

The Greatest Shortcoming

Quantitative Chauvinism and Lebensraum Imaginaries

First Edition

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“The greatest shortcoming of the human race is our inability to understand the exponential function.
—Al Bartlett, 1972

“Immigration is the main driver of population growth in the U.S. ...therefore any discussion of sustainability in the U.S. must address the need to reduce or eliminate immigration, both legal and illegal, into the U.S.
—Al Bartlett, 1992

“Anyone who thinks that you can stop population growth to save the environment without reducing immigration is innumerate.
—Al Bartlett, 1998

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Preface

I AM NOT A DEMOGRAPHER. I trained as a mechanical engineer, then in quantitative communication studies, with detours through political science and business. I now work as a computational social scientist in the discipline of misfit toys known as information science. Statistical modeling and data science methods remain central to my professional identity. I am most at home when studying the digital traces of social behavior from online communities like Wikipedia, Reddit, or Twitter. Readers looking for a jeremiad about the limits of numbers or settling internecine debates between “quantitative” and “qualitative” epistemologies will be disappointed.

What this book does do is to *reclaim* quantitative methods from reactionary political projects that have, for decades and centuries, abused quantitative data to advance cruel visions for protecting the future from the present. Collecting, analyzing, and interpreting numerical data has never been neutral and this tradition has much to answer for. Quantitative data and methods will play a crucial role in the contemporary polycrisis of metastasizing democratic backsliding, accelerating climate crisis, and compounding technological disruptions. A more reflexive post-positivism that grapples with the social construction and political economies of data will not rescue us from this polycrisis, but it could help clarify the consequences of data analysis choices. Immunological metaphors (and their limits) are central to this book’s argument. Seen through that lens, how do we inoculate ourselves to recognize and resist uncritical appeals to numbers?

There is one key bit I omitted from my biography: my exposure to science and technology studies (STS) as an undergraduate student transformed my educational and professional trajectories. STS as a discipline has a genealogy that coalesced alongside the Cold War history of demography that later chapters trace. STS scholars study how scientific methods and technological tools are constructed, how they circulate, and what consequences they produce within and between social groups. This project is therefore a homecoming to my STS roots after almost two decades in the theories and methods of other disciplines. As an STS project, it combines historical, rhetorical, and ethnographic modes of analysis. What I can evaluate is the rhetorical and orga-

nizational structure of how Bartlett's models travel: from the lecture hall to the policy document, from the physics classroom to the advisory board of an immigration restriction organization, from the pedagogy of exponential arithmetic to the production of moral permission for coercive governance. This is a theme that the United States has very much been grappling with in 2025 and 2026. Three additional motivations converge in this project.

The first motivation is professional. Since 2018, access to the social data that quantitative researchers rely on for scholarship, journalism, and civic accountability has retreated behind rationales of privacy, monetization, and artificial intelligence training. Scientists like myself build their research programs around access to the high-resolution and archived digital traces of social behavior on platforms like Facebook, Twitter, and Reddit that have unilaterally closed off data access. This retreat selectively forecloses the kinds of empirical inquiry that might contest quantitative claims made by organizations with access to privileged data. The kinds of science that can be imagined and accomplished are very different if the Large Hadron Collider implemented paywalls or NASA shut down its space telescopes.

The second is political. The kinds of reactionary politics that have taken root not just in the United States are inheritors of architectures with a longer history of combining demographic projections, environmental scarcity, and immigration restriction.

The third is the recognition that the infrastructure consecrating Bartlett's authority continues to confer legitimacy on the arguments he developed with it, long after his death and long after demographic science has moved on.

These converging backgrounds make sense when read through a frame scholars call *ecofascism*. This book uses that term deliberately and early because the concept has a history of re-appearing. Ecofascism is not a fringe movement far removed from mainstream environmentalism. Neither is it a rhetorical excess to be carefully distinguished from the legitimate environmental concern it resembles. Ecofascism's fusion of ecological emergency with racialized demographic hierarchy, the mobilization of environmental scarcity arguments to justify coercive governance of populations defined by race and national origin, the inscription of exclusionary political projects in the language of natural limits and arithmetic are central to its durability.

"Ecofascism" is not meant to be a verdict on individuals either. There is little to be gained from "canceling" a scientist who died in 2013. What the book does instead is trace *Lebensraum imaginaries* as a recurring epistemic architecture within ecofascism that describes an ensemble of future narratives, organizational practices, and rhetorical repertoires through which ecological emergency is converted into demographic hierarchy and coercive governance is presented as reluctant but responsible management of natural limits. *Lebensraum* is a deliberate reference to a concept crossing biological, ecological, social, and political boundaries that not only animated the calamities

of the Third Reich but has a longer history and more unsettling afterlives. An imaginary is a collectively held structure that makes certain actions seem necessary, certain populations seem threatening, and certain policies seem like responsible concessions. Individuals, organizations, data, and arguments enter and exit these structures even as the structures of the imaginary persist and adapt over time.

This book's examination of Bartlett's lectures, writing, and archive does not attempt to definitively classify his beliefs and behaviors as being "ecofascist." Any honest accounting of Bartlett's legacy should also examine why he spent two decades in his retirement lending his credibility to and ultimately being eulogized by the post-Cold War white nationalist anti-immigration movement that has only metastasized since 2016. His institutional authority as a gifted lecturer who spent decades evangelizing the environmental implications of his oversimplified math was important to organizations that operated through *lebensraum* imaginaries that remain recognizable through the Cold War, in contemporary political formations, and the climate polycrises that will unfold in the decades to come.

The other name this book requires is *quantitative chauvinism*: a style of reasoning that privileges quantitative metrics as having jurisdiction over political judgments, collapses heterogeneous social complexity into a single aggregated trajectory, and then inscribes the result in portable artifacts that travel as if they were neutral facts. Quantitative chauvinism is the use of numerical abstractions to close political questions before they can be opened, to present simplifications as comprehensiveness, and to frame dissent as a failure to understand the data. Bartlett's lecture and its circulation through both mainstream and fringe environmental movements is just one of its clearest pedagogical expression. How does a set of graphs, a doubling-time calculation, and a rhetorical performance delivered by overhead transparency transform an arithmetic thought experiment into moral mandate for coercion and sacrifice that evades our cultural antibodies for recognizing eugenic and genocidal arguments. The lecture's portability — 1,742 performances across four decades — contributed to a movement that abstracted the overpopulation anxieties of the late 1960s into the conviction that the ability to calculate exponential doubling time justifies urgent and coercive policies.

The concepts of *lebensraum* imaginaries and quantitative chauvinism are distinct but co-evolving. Quantitative chauvinism provides the epistemic infrastructure that *lebensraum* imaginaries require: without simplified models that suppress feedback and alternative trajectories, naturalizing demographic hierarchy loses its causal warrant; without expertise that confers authority on simplification, hierarchizing populations loses its scientific alibi; without the foreclosure of alternatives that quantitative inevitability performs, instrumentalizing coercive governance loses its claim to reluctant necessity. The chapters that follow trace this interaction across four historical eras and two paths through a projected future. The goal is not to discredit quantification but to show

what happens when quantitative form is separated from the epistemic commitments that give them appropriate claims to legitimacy and authority.

This book addresses four scholarly communities whose debates converge on the problem it analyzes. For scholars in science and technology studies, the manuscript contributes a case study in how quantitative authority is constructed, made portable, and recruited for political ends that extends Porter's account of mechanical objectivity as a technology of trust into the specific domain of demographic forecasting and its troubling tendency to be captured for reactionary political projects. For scholars in critical and historical demography, it extends the well-documented genealogy of the population concern community's institutional entanglement with exclusionary politics from pre-World War II eugenic organizational forms and its post-World War II institutions infrastructure into contemporary political assemblages. For scholars in environmental sociology, it contributes an account of how the greening of nativist politics operates through scarcity arguments, carrying capacity frameworks, and environmental impact metrics that justify racial hierarchies. And for scholars in extremism studies, it contributes a mechanism for understanding how mainstream scientific and institutional authority is recruited by extremist organizational networks without those networks having to capture the institutions outright. How does a physicist's celebrated lecture become one of the crown jewels for white supremacist networks and continues to do important work for the network decades after his death?

The methods of quantitative chauvinism and their role within *lebensraum* imaginaries are not legible from inside any single disciplinary framework. They require the tools of STS to trace how quantitative authority is built and made portable; the tools of critical demography to situate those authority claims within the history of population knowledge as governance knowledge; the tools of environmental sociology to explain how ecological emergency becomes a warrant for demographic hierarchy; and the tools of extremism studies to analyze the organizational forms through which those warrants travel from scientific institutions to political movements.

A final motivation is to weigh how immunological metaphors like inoculation are helpful as a response of anticipatory recognition: learning the recurring rhetorical and technical moves by which numerical claims are converted into moral permission structures and rehearsing how to identify and interrupt those moves before their anxieties harden into policies. Quantitative chauvinism rarely announces itself as domination but as responsibility: a small set of numbers, a graph, a trajectory too steep to ignore. *Lebensraum* imaginaries intensify the effect by binding those projections to narratives of ecological emergency, scarcity, invasion, and triage. The chapters return to this coupling through four historical cases so that its signature becomes recognizable. The last third of the book speculate about might this structure look like under likely climate

trajectories, the traditions that might help us resist the siren's call of these imaginaries, and the limits of these sociobiological analogies.

A Politically Direct Closing

The organizations this book analyzes are not historical curiosities and very much remain active and influential. The Federation for American Immigration Reform, the Center for Immigration Studies, NumbersUSA, and the network of affiliated outlets and advocacy organizations that John Tanton, Cordelia Scaife May, and other white supremacists assembled from the late 1970s onward are active, well-funded, and institutionally connected to federal immigration enforcement policy after 2016. The arguments they deploy about demographic projections, environmental carrying capacity, fiscal impact assessments, continue to circulate through media environments that treat numerical form as objective and serious evidence. Under the second Trump administration, those arguments have moved from fringe movements to a center of a trans-national project of immigration enforcement, selective reading of demographic data, and the application of environmental emergency to deeply reactionary political projects. The Tanton network did not create these conditions, but it built the infrastructure of white papers, think tanks, op-eds, journals, and lexicons through which *Lebensraum* imaginaries could be rationalized.

This book does not pretend to offer a “neutral” account of these entanglements: fascism and chauvinism are bad and less of them are unequivocally good for society. The distinction this book draws is between the legitimate practice of quantitative inquiry and the organized project of converting quantitative claims into justifications for coercive governance. I refuse the choice that quantitative chauvinism always tries to impose: between accepting its simplified projections as mandates and being dismissed as innumerate if you disagree.

The book's counterproposal is *quantitative egalitarianism*, grounded in three commitments. First: treat models as conditional representations, not moral verdicts. Require explicit assumptions, uncertainty quantification, and validation. Make visible what the model suppresses. Second: treat data as civic infrastructure. Protect access, provenance, and accountability through governance that serves democratic publics rather than private enclosure or institutional self-interest. Third: treat measurement as an arena of democratic contestation. Make categories, metrics, and trade-offs available for challenge, especially when they govern mobility, belonging, and risk exposure. These demands are not dramatic: that the assumptions behind projections be available for scrutiny, that the populations governed by those projections have standing to contest them, and that the numerical claims made in the name of environmental necessity be answerable to the pluralistic life they claim to protect.

This book opens and closes in Boulder, Colorado — which is only a little bit of navel-gazing. Boulder is useful because its trajectory mirrors so many other progressive communities where the slide from conservation into exclusion has been accomplished gradually enough that each step looked like “common sense” responsible stewardship in the name of conservation and sustainability. Its hazards expose the limits of imaginaries that promise to provide long-term safety in exchange for coercive and dehumanizing control now. Boulder’s environmental self-image exposes how easily conservation slides into coercion when the categories used to define the community worth conserving are treated as natural rather than political. But Boulder’s institutional history also contains the resources for a different kind of quantitative governance: one that treats the city’s environmental future as a question that belongs to everyone who lives within its picturesque wildland-urban interfaces.

The measure of preserving a livable future in the face of climate polycrises will not be the precision of any quantitative forecast. It will be around our arrangements for re-assembling our institutions to survive destabilizing polycrises to ensure that who bears risk, where space is shared, how ecosystems are stewarded, and the numbers that shape these discussions do not end up being captured, one again, by odious political projects and narrow social imaginations.

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1 Boulder

BOULDER IS NOT A SAFE PLACE. Beneath the endless blue vault of Colorado's mile-high skies, the Flatirons stand as perilously perched reminders that this valley is never at rest. This is where the sea of grass of the North American interior abruptly collides with the Rocky Mountains. It is a landscape of accumulated catastrophes: seas that dried up, mountains erupting from the crust, ice grinding those mountains down, fires incinerating forests and prairies. This valley has for millions of years been in some state of convulsion, punctuated by only brief respites of stability. The aggregations of geology, ecology, and human settlement in this place encode many memories of floods, fires, and churning earth that have erased prior claims on the valley.

The Rocky Mountains themselves are geological adolescents, the product of a dying tectonic plate plunging beneath the North American continent a thousand miles away, cracking the Earth's crust and forcing ancient seafloor skyward. These slabs of rock were raised higher than the modern Tibetan plateau, and a vertical mile of sandstone and granite has since eroded into "mere" 14,000-foot peaks. Glaciers repeatedly descended from the high country without quite reaching the valley floor, yet they released massive pulses of meltwater that carved canyons, cut creek beds, and washed gravel into the plains below. What looks like solid ground is an alluvial fan, a delta of everything the mountains could no longer hold. The city of Boulder — and every settlement that precedes and follows it — sits on this debris.

Life in all its opportunism was attracted to the fresh water coursing through these rich sediments. Plant and animal ecologies climbed Boulder's mountainsides from the plains as prior climates warmed and glaciers retreated through the Cenozoic era. Cottonwoods and willows threaded along Boulder Creek and ponderosa pine and juniper claimed the drier ridges above. The ecological logic of this place has always been one of sharp transitions: mountain gives way to plain, wet to dry, cold to hot within the span of a single afternoon. Those transitions attracted everything that could exploit the gradients and survive their stresses. The flora and fauna that established themselves here treated catastrophe as a condition to endure rather than an aberration: the ponderosa

pine requires fire to open its cones, and the cottonwood requires flood to disperse its seed. Calamity and rebirth were the terms of survival.

The first humans to arrive in the region already understood this bargain based on the conditions they survived and remembered. They likely came south through thawing Ice Age corridors some thirteen thousand years ago, following megafaunal game. Clovis and Folsom cultures left bone fragments and projectile points in creek sediments as they moved between mountain and plain with the seasons. When the climate warmed and rendered age-old hunting and settling strategies obsolete, their successors adopted altered mobilities, developed specialized tools, and managed the landscape more deliberately through controlled burns that cleared meadows and concentrated game. Boulder Valley was a seasonal crossroads rather than a permanent home. What looked like the absence of permanent settlement was in fact a different theory of it: that a valley this volatile rewards those who move with its cycles rather than those who attempt to hold their position against them. The governance these peoples practiced for millennia was sophisticated enough to match the landscape's own terms.

Those terms did not survive contact with European settlers, who arrived in the Front Range in the middle of the Christian nineteenth century. By the time horses introduced by Spanish conquistadors spread north through the Plains in the seventeenth century, the Arapaho and Cheyenne, whose oral, genetic, and linguistic histories point to a departure from an ancestral home along the Great Lakes in the upper Midwest, had learned to ride them to follow the bison. These peoples were survivors of a pandemic that killed as much as 90% of the population, but they nevertheless carried beliefs about what the land was for and strategies for how it could be managed.

The tectonic convulsions that raised the Rockies also brought minerals containing gold, silver, and copper from deep within the Earth's crust. The immense value Europeans ascribed to these minerals brought abrupt and profound change to the landscape. Miners reached Boulder Valley in 1858, and what followed was swift: the displacement and slaughter of indigenous inhabitants, the privatization of land that indigenous peoples had treated as shared, and the imposition of infrastructures designed to extract whatever the mountains could yield. Forests were razed, streams dammed, ditches dug, railroads laid, and properties gridded along routes that treated the landscape as an obstacle to be engineered rather than a condition to be inhabited. Within a decade of the first mining camps, a city called "Boulder" had been incorporated in a territory called "Colorado."

What happened in Boulder over the next century was a transformation not only of the landscape but of the ideology governing it. The extractive violence of the mining era did not give way to something gentler so much as to something more capable of obscuring its coercions. A land-grant university arrived in 1876, charged with transforming Colorado's miners and farmers into good American citizens. In 1898, a

coalition of Texas educators and Boulder civic boosters established the Colorado Chautauqua on a barren mesa at the foot of the Flatirons. It was the first parkland the city ever purchased, and it was part of a national movement that by its peak in the 1910s attracted thousands of upwardly mobile Protestants to programs of lectures, concerts, and coursework framed as moral self-improvement. The Ku Klux Klan's parades on Main Street in the 1920s are carefully bracketed as regrettable ideological excesses rather than the product of decades-old movements. The federal government's investments in laboratories for building weapons, standards, and spacecraft during the Cold War filled the city with physicists and engineers who brought with them a faith in technical solutions whose afterlife would prove consequential. A blue line was drawn to stop the city from spreading into the mountains, and a greenbelt was established to stop it from spreading into the plains. Population growth was limited in the name of environmental protection.

The subject of this book is how a community abstracts its anxieties about disorder, contamination, and change into technical problems, how those problems become subject to "objective" numerical analysis, and where those quantitative logics travel as justifications for who has access to community, beauty, or safety. These mechanisms unfold in different ways in every American community that has scarcely grappled with how its contemporary urban forms and their consequences follow from colonial dispossession and white supremacist justification. But they are unusually easy to see in Boulder because these inheritances are absorbed into a civic identity that has learned to present exclusion as progressive environmental stewardship and has implemented this stewardship through the "neutral" application of "common sense" arithmetic. Calculations about population growth, density, and carrying capacity become judgments about who belongs here without ever requiring any justification beyond the numbers themselves.

Olmsted and Chautauqua

The institutional foundations of that identity were laid earlier than most Boulder histories acknowledge, and more deliberately than the city's popular self-image would suggest. In the autumn of 1908, the Boulder City Improvement Association retained Olmsted Brothers, the Brookline, Massachusetts landscape architecture firm founded by the sons of the designer of Central Park, to survey the city and recommend a program of municipal improvements. The resulting document, issued in 1910 under the authorship of Frederick Law Olmsted Jr., is a genuinely reformist work in many respects: it proposes parks, defends wages against land speculators, and advocates for systematic management of Boulder Creek's dangerous floodplain [1]. But it is also one of the earliest codifications of a vision of environmental preservation that specifies who

belongs in the community it protects and what must be kept out to preserve the community's character.

The Olmsted report's organizing ambition is captured in a phrase repeated across its hundred pages: Boulder should become "a city of homes." Olmsted distinguishes Boulder and its proximity to nature as the agreeable destination to escape detestable cities. Its proper residents are people of cultivated taste who have chosen to settle permanently in a place whose value lies in climate, scenery, and a quality of life available only where those assets have been protected against uncomfortable encounters with the deprivations of capitalism. The desirable newcomers for this new community will contribute to its character as responsible citizens who own property and pay taxes. Olmsted's imagination of a city of homes was grounded in anxieties about protecting Boulder's purity from the changes washing over other urban communities.

Olmsted was not inventing this idea for Boulder. Civic reformers had been cultivating Boulder as a destination for a decade, most visibly through the Colorado Chautauqua. The Chautauqua, which occupied land the city had purchased and leased back to a nonprofit association, was organized expressly for schoolteachers and their families. Its intended audience was educated, Protestant, upwardly mobile, and socially removed from working-class industrial labor. It charged admission, required transportation from the downtown railroad depot, and housed its summer residents in a cottage grid governed by design guidelines that enforced a "feeling of a large camp rather than a collection of individual properties" [2]. A perimeter fence and a ten-o'clock curfew policed its borders in the early years. It was, from its founding, a bounded and selective community on public land, defining its character against the commercial entertainments it refused to host [3]. The Chautauqua had already demonstrated the physical as well as cultural infrastructure that the Olmsted report would extend into an administrative program for the rest of the city.

The most enduring element of that plumbing was a grammar of scarcity demanding urgent intervention. The Olmsted report emphasized that Boulder's mountain views were a finite common resource subject to irreversible privatization. The public authority it envisioned over the density and character of residential development, which subordinated private property rights to a collectively defined vision of what the community should look like, would become indistinguishable from the policy tools Boulder deployed in later decades to suppress new housing and restrict population growth in the name of environmental sustainability. The progressive intent and the exclusionary outcome ran through the same institutional plumbing. The subheading "The Cost of Delay" recurs through the report like a drumbeat, marking each moment where inaction is characterized not as a value-neutral choice but as responsible stewardship deferred at collective expense [4]. The resource Olmsted identified was aesthetic and spatial rather than ecological or cultural, but the argumentative structure is identical

to the carrying capacity arguments that would appear in Boulder's planning debates decades later: a bounded endowment, a threat of irreversible loss, expert authority, and coercive intervention on behalf of the future presented as the only responsible response.

Neither the Olmsted report nor the Colorado Chautauqua makes explicit demographic arguments. Neither invokes the categories of race, national origin, or biological stock that Chapter 2 documents about Progressive Era reforms when "eugenics" had overwhelmingly positive popular connotations. Their logic is grounded in aesthetics, access, and the governance of taste: the question of what kinds of use, what kinds of resident, and what kinds of development belong in a community organized around the cultivation of natural scenery. This displacement of political argument into aesthetic and environmental categories is precisely what makes these documents important for the history that follows. They demonstrate that the components of a preservation imaginary can be assembled and made institutionally durable without recourse to the now-tawdry rationales of biological hierarchy or genetic inevitability. These components include a bounded natural endowment treated as a finite resource, a selective community defined against undesirable uses, administrative instruments for exclusion, and expert authority naturalizing that exclusion as stewardship.

Forgotten reactionaries

The 1920s brought Boulder its first explicitly demographic politics. A Boulder County chapter of the Ku Klux Klan was the third chartered in the state. It paraded down Pearl Street in December 1922, burned a cross on Flagstaff Mountain in May 1924, and by 1925 claimed over a thousand members in a town of just eleven thousand [5]. Colorado's Klan in those years was one of the strongest in the country, controlling the governorship, a U.S. Senate seat, the secretary of state's office, the Denver mayoralty, and a majority of the state House [6]. The movement's high point was the Johnson-Reed Immigration Act of May 1924, which set annual quotas on immigration to two percent of its 1890 foreign-born population, a baseline year purposefully selected to throttle southern and eastern European immigration and to bar Asian immigration almost entirely [7]. The Boulder Klan's weekly newspaper, *The Rocky Mountain American*, published from a Walnut Street office in the first half of 1925, ran cartoons celebrating the new restriction. Figure 1.1 shows a hooded Klansman standing before a padlocked door labeled "American Ban on Immigration," holding the key, while caricatured Italian and Asian figures plead for entry from outside in mock dialect. Boulder's klavern and its newspaper dissolved in July 1925.

While the organization collapsed, its supporters did not evaporate. Three years later, Boulder adopted a new administrative instrument that performed the sorting the Klan had attempted using a technocratic vocabulary that required neither quotas on nationalities nor burning crosses. In the aftermath of the Olmsted report, Boulder had already

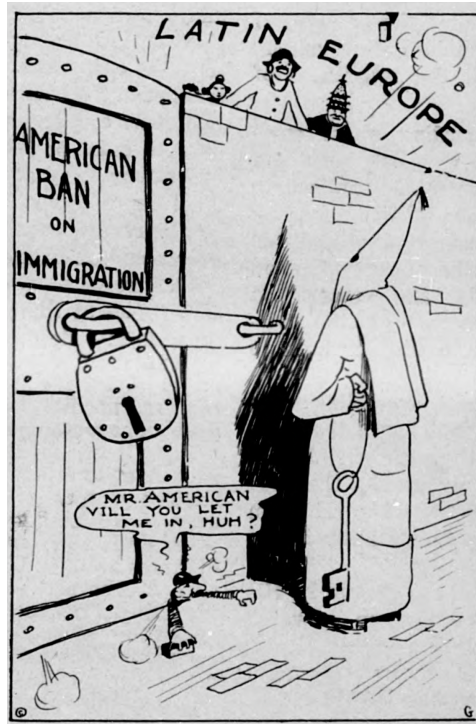


Figure 1.1

Anti-immigration cartoon from *The Rocky Mountain American*, Vol. 1, No. 11, April 1925, the short-lived weekly newspaper of Boulder County's Ku Klux Klan. A hooded Klansman holds an over-sized key next to a brick wall and a padlocked door labeled "American Ban on Immigration". Three caricatured figures labeled "Latin Europe" peer over the wall and another figure trapped by the door pleads in mock-foreign dialect, "Mr. American, vill you let me in, huh?"

chosen to not expand its industrial base and to remain "quiet and attractive" rather than to court the factories and their laborers [8]. That choice made the city's residential character its principal asset. The first official conversation about zoning, at a Planning and Parks Commission meeting in September 1922, framed the instrument as safeguarding residences from the encroachments of business buildings, apartment houses, and their associated characters. The proximate conflicts were fights over commercial creep on University Hill, where gas stations, garages, and small business houses were appearing among the fraternity houses and faculty homes across from the campus.

University president George Norlin, who would later refuse the Klan's demands to purge Catholic and Jewish faculty, protested a proposed garage opposite the campus in 1923 on the ground that no "sensible program to keep or enhance the beauty of

Boulder” could permit it [9]. The Planning and Parks Commission hired the Denver landscape architect and city planner Saco R. DeBoer, the author of Denver’s 1925 ordinance and an emerging expert, to design its zoning. After the Supreme Court’s November 1926 decision in *Village of Euclid v. Ambler Realty* removed the lingering doubt about the constitutionality of zoning policies, the city enacted DeBoer’s code in January 1928.

The ordinance sorted Boulder into seven districts in a strict cumulative hierarchy with single-family residential at the apex, a hierarchy the *Euclid* court had ratified in famously unguarded language describing apartment houses as “a mere parasite” that “come very near to being nuisances” when set among detached single-family homes [10]. Boulder’s new zoning was formally race-neutral. Boulder’s 1920 population of eleven thousand included only 128 Black residents, alongside a small Hispanic and Italian working class concentrated in the coal-mining and sugar-beet districts to the east, the same populations the Klan had named as threats just three years earlier [11]. The code that prioritized detached single-family construction and pushed apartments, businesses, and renters down its hierarchy of permitted uses did not have to name those populations who inhabited them to sort and exclude them; questions of land use short-circuited the possibility of debate and naturalized it as planning expertise.

The architecture had its champions among the city’s professional class. Ira DeLong, the retired head of the University of Colorado’s mathematics department, was one of them. He sat on the city’s Planning and Parks Commission and defended the ordinance against the property owners contesting its district lines, dismissing one such fight as a “fuss in a teakettle” [12]. What the 1928 code installed was the administrative machinery for sorting the city by use and therefore selecting the kinds of occupants who deserved to reside within the city of homes. Eighty percent of Boulder remains zoned exclusively for detached single-family construction a century later, and between 1981 and 2024 the city enforced occupancy limits restricting unrelated cohabitants in those zones to three or four persons per household [13]. While the Klan’s politics had required hoods, parades, and a weekly paper, the zoning code required none of those things and outlasted them all.

PLANning for the future

In 1958, a passing conversation between two hiking CU Boulder faculty members changed the course of the city again: math professor Bob McKelvey declared that “something had to be done about those foothills” or they would be covered in houses. The other half of that celebrated 1958 conversation with McKelvey was Professor Albert Bartlett. A nuclear physicist who had joined the University of Colorado faculty in 1950, Bartlett had completed a physics doctorate at Harvard following his time measuring plutonium isotopes for the Manhattan Project at Los Alamos. To use the language

of a later era, McKelvey and Bartlett were community organizers: scientists who understood that technical authority could be translated into policies, and that charisma could convert technical arguments into a social movement.

In 1959, a coalition of Boulder citizens alarmed by proposed real estate development on the mountain backdrop organized under the People's League for Action Now (PLAN-Boulder) and successfully campaigned for what became known as the Blue Line Amendment: a provision prohibiting the city from extending water service above a fixed elevation contour, thereby making hillside development administratively impossible without annexation and infrastructure investment the city could simply refuse to provide [14]. The views Olmsted had designated Boulder's "priceless possession" were protected not by scenic easement or preservation ordinance but by the bureaucratic logic of access to municipal water. Bartlett would spend the next half century translating his expertise with the exponential mathematics of radioactive elements into a persuasive template for grappling with the consequences of growth.

The Blue Line was an early demonstration of this translation. It took the Olmsted report's scarcity frame, a finite natural endowment subject to irreversible privatization, and grounded it not in aesthetic judgment but in arithmetic: above a certain elevation, the physics of water pressure and the economics of infrastructure could be mustered to defeat development without confronting the stickier implications of settler entitlement or capitalist accumulation. The city needed only to define a line on a map to achieve an aesthetic policy goal. What Olmsted had argued through the cultivated authority of the landscape architect, McKelvey and Bartlett could argue through the harder authority of mathematics and physics. Bartlett's lecture and arguments at the end of his career and the political uses to which his arithmetic would eventually be put are the subject of the chapters that follow. But the origins of Bartlett's conviction that the right number, placed in the right instrument, could make exclusion feel like stewardship lay here in the same foothills Olmsted had designated Boulder's "priceless possession."

Bartlett and his colleagues at PLAN-Boulder went on to anchor a cascade of instruments through Boulder's mid-century, each of which extended the original Olmsted grammar of bounded endowment, selective access, and expert stewardship into new administrative domains. Within Boulder's enlightened self-conception as "twenty-five square miles surrounded by reality" and its non-partisan local political culture, PLAN-Boulder essentially ruled Boulder as one-party local government for decades. The open space program, launched in 1967, authorized the city to purchase and permanently retire land on its mountain and plains perimeter, removing it from the development market and protecting the views that defined Boulder's civic identity. The Danish Plan capped residential building permits at 450 units annually, a figure derived from calculations about infrastructure capacity and environmental carrying capacity that arrived at the city council bearing the authority of technical necessity rather than the vul-

nerability of political choice [15]. The Boulder Valley Comprehensive Plan of 1977 formalized a population ceiling for the planning area, translating an intuition about the city's proper scale into an administrative target. Height limitations, floor-to-area ratios, service boundaries, and growth allocation systems accumulated around these anchor instruments across the 1970s and 1980s, each adding one more quantitative threshold to a planning architecture that was, by 1985, among the most elaborate in any American city of comparable size.

Horseshoe theory

None of Boulder's Blue Line, open space program, or growth management instruments required the invention of new political logic. Each extended a template that had already been in place for decades and further rationalized its claims to control through the emerging vocabularies of environmental protection and technocratic governance. Flood control levees, fire risk maps, traffic models, demographic forecasts, zoning ordinances, and permitting regulations all drew on a shared faith in numbers as instruments of order [16]. Theodore Porter's account of how quantification functions as a "technology of trust" describes the mechanisms at work in Boulder's governance: the impersonal authority of calculation substituted for the messy, contested, interest-laden process of democratic negotiation, making exclusionary outcomes appear as objective responses to ecological constraint [17]. Numbers became the infrastructure of safety for a city organized around the preservation of scenery and the cultivation of "cultivated" residents. How those same tools of quantitative rationalization became embedded within ideologies committed to exclusion and dominance motivates the remainder of the book. While Boulder is not the origin of those ideologies, it became one of their most legible proving grounds: a nominally progressive community where the slide from conservation into coercion was accomplished so gradually and with such apparent good faith that it is difficult to know when it crossed the line into embracing values that should have remained outside of polite society.

Boulder's planners and civic advocates experienced this apparatus as environmental protection. The Blue Line kept houses from dotting Boulder's dramatic foothills. The open space system has preserved farms and grassland from becoming suburban sprawl. The environmental commitments behind them were, in most cases, genuine. But the apparatus also produced a city whose demographics reproduced the structural prerequisites of mid-century conservation: Boulder became white, educated, and prosperous at rates that exceeded what its research university alone would predict. In 2020, Boulder was 83 percent non-Hispanic white in a state that was 66 percent and in a nation whose demographic composition was rapidly diversifying. Boulder's housing costs rose at multiples of the regional average, pricing families, recent immigrants, and blue-collar households out of the city's housing market. Car-centered sprawl exploded

in the exurbs of the “L”-towns surrounding Boulder (Lyons, Longmont, Louisville, Lafayette — and Erie and Superior) whose commitments to conservation were more flexible. The infrastructure of environmental conservation was also convenient for demographic selection of “deserving” residents, and the two functions ran through the same policy instruments, the quantitative calculations, and technical invocations of ecological necessity.

How did that happen? The question this book pursues is not why Boulder produced exclusion: exclusion is the ordinary product of American land-use governance and the longer history of settler dispossession, white supremacist red-lining, exclusionary zoning, and neoliberal infrastructural (dis)investment that has structured most American urban geographies. The question this book presents is more specific: how did the quantitative tools of environmental stewardship come to share the grammar of far-right nativist politics? How did carrying capacity calculations, exponential growth curves, and population impact assessments travel from progressive activists in Boulder’s city council chambers to the pseudo-scientific journals of white nationalists? What about Bartlett’s exponential lectures caused him to spend his retirement advocating for draconian immigration restrictions that would be eulogized by virulent white supremacists? What does it mean that the quantitative tools Bartlett used to justify protecting Boulder’s open spaces from development continue to be invoked since 2016 to justify reprehensible policies pretending to protect the nation’s European heritage from “replacement”? How might these quantitative arguments re-appear in the future as justifications for excluding and abandoning migrants fleeing climate catastrophes?

The archival record provides a partial answer. Bartlett served on the advisory board of the Federation for American Immigration Reform (FAIR), an organization founded by John Tanton, a Michigan ophthalmologist who had chaired the Sierra Club’s population committee before concluding that mainstream environmentalism would never embrace demographic restriction. He contributed to *Population and Environment*, a journal edited by Virginia Abernethy, who would later accept the endorsement of the white nationalist Council of Conservative Citizens in her 2004 campaign for Arizona’s Prop 200. He sat on the board of Carrying Capacity Network, which argued that the United States had already exceeded its ecological carrying capacity and that immigration was the primary driver of the excess. None of these affiliations were secret. They appeared on his curriculum vitae, in his published correspondence, and in the organizational records of the groups themselves. The archive does not reveal a hidden life. It reveals a public one whose connections were legible all along to anyone who cared to trace them. When Bartlett died in 2013, that public life received its plainest acknowledgment: *The Social Contract*, the quarterly Tanton had founded to circulate restrictionist argument, marked his death with memorial tributes that claimed him without hesitation as one of the movement’s own [18]. Chapter 3 returns to those eulogies. What

matters here is that they were offered in earnest, and that the same arithmetic Bartlett used to protect Boulder's foothills was, by the community that knew his work best, understood as continuous with the politics of exclusion.

The answer lies not in Boulder's aesthetic peculiarities or a blinkered physicist's celebrated lecture but in a mechanism that operates wherever quantitative authority meets social anxiety: the recruitment of numerical instruments and naturalistic arguments to perform political work that their form disguises as technical and biological necessities. This book traces that relationship across a hundred years of American demographic and environmental governance. It follows how certain kinds of numbers, simple, visual, alarming, and portable, became infrastructure for reactionary politics. It shows how exponential arithmetic became environmental "common sense," how immigration impact statements became policy expertise, how demographic projections became racial countdowns, and how fertility rates became investments requiring the tools of portfolio management. It identifies the actors who built these instruments, the organizations that circulated them, the metaphors that made them persuasive, and the political projects they served. The book asks whether the same quantitative tools that narrowed the circle of concern can be resisted or redirected to expand it to meet the compounding challenges of a rapidly warming world in the decades to come.

The Road Ahead

The book's argument unfolds across an inheritance chapter and five historical eras between 1968 and 2068. The inheritance is the architecture Bartlett's lecture activated when it began its long career. The four chronological eras that follow trace the architecture's portability across instruments, institutions, and political projects; the fifth extends the established patterns into plausible futures, followed by the reclamation of the long term and a return to Boulder.

The Inheritance Bartlett's lecture did not arrive in 1969 as an original act of intellectual creation. It entered a formation whose intellectual coordinates had been assembled across the first half of the twentieth century. Chapter 2 traces four currents that converged into the population concern community Bartlett joined. The first is the institutional eugenics apparatus assembled at the Eugenics Record Office at Cold Spring Harbor between 1910 and 1940, codified through state sterilization regimes after *Buck v. Bell*, and reorganized after the postwar repudiation of explicit eugenics into the demographic and population-science institutions that absorbed its categories, its personnel, and its donor networks. The second is the German biogeographical tradition that Friedrich Ratzel coined as *Lebensraum*, that Johan Rudolf Kjellén translated into the administrative apparatus of the modern state, and that Karl Haushofer fused into the territorial doctrine the Nazi state administered. The third is the American conservation movement that consolidated between 1900 and 1940 as a parallel expression of the same

intellectual project, sharing personnel and funding networks with the eugenics movement through figures like Madison Grant and Henry Fairfield Osborn. The fourth is the postwar population concern community that the Population Council assembled in 1952, that Paul Ehrlich popularized in 1968, that Garrett Hardin radicalized between 1968 and 1974, and that the Club of Rome credentialed with computational authority in 1972. By the time Bartlett walked to the chalkboard in September 1969, the apparatus, the vocabulary, and the audience were already in place. The chapter's question: what did Bartlett's lecture inherit?

The Bottle (1968–1994) In the late 1960s, the Cold War's ambient dread found a new target. Paul Ehrlich's *Population Bomb* opened with a scene of a Delhi taxi ride and concluded that hundreds of millions of people would starve to death in the coming decade. The book sold three million copies. Ehrlich appeared on *The Tonight Show* more than twenty times. Zero Population Growth recruited hundreds of thousands of members. Garrett Hardin published "The Tragedy of the Commons" and later "Lifeboat Ethics," arguing that some must drown so that others might survive. In Boulder, Bartlett was giving his lecture. While he was not the formation's most famous figure, he was its most persistent, repeating his thought experiment year after year, 1,742 times over four decades. His lecture worked at a different register. It did not predict famine or propose letting nations starve. It taught arithmetic, and it made the mathematics feel like insight and the insight feel like obligation [19]. Chapter 3 traces how Bartlett's tutorial converted exponential arithmetic into environmental common sense, how a process this book calls *domesticated scientism* operated as the mechanism through which the arithmetic was naturalized as ecology, and how the 1995 *Population and Environment* exchange between Bartlett and the Ehrlichs first staged the formation's exclusionary logic in the form of a dispute. The era's question: how did exponential arithmetic become environmental common sense?

The Quarantine (1995–2008) In September 1994, delegates from 179 countries gathered in Cairo for the International Conference on Population and Development. The conference produced a Programme of Action that replaced coercive population control with a rights-based framework centered on women's reproductive autonomy. For the population concern community, Cairo was a catastrophe [20]. The institutional consensus that had sustained their work for three decades was shattered. But population anxieties did not disappear after Cairo; they were displaced into new and troubling movements. With fertility control delegitimized, immigration restriction became the residual lever. John Tanton, the ophthalmologist from northern Michigan who had chaired the Sierra Club's population committee, had already anticipated this displacement. Beginning in 1979, he built a network of organizations that would launder nativist conclusions through the aesthetics of neutral policy expertise [21]. The network produced fiscal impact statements that converted the political question "should

we admit these people?” into the accounting question “what do these people cost?” It constructed a parallel credentialing infrastructure that could circulate restriction as research. Chapter 4 traces how the Tanton network survived Cairo, built its institutional architecture, captured and lost the contest for the Sierra Club, and operated through the twin mechanisms of colorblind racism and what this book calls *agnetological infrastructuring*: the production of ignorance through parallel institutions and manufactured uncertainty. The era’s question: how did these ideas survive their own delegitimation, and lose their association with racism in the process?

The Swarm (2009–2024) On August 14, 2008, the U.S. Census Bureau issued a press release announcing that non-Hispanic whites would become a minority of the American population by 2042. The projection was accompanied by a chart showing racial and ethnic composition lines converging and crossing. Within months, the crossover chart was circulating through restrictionist networks and nationalist forums, and within years it surfaced in the chan-board cultures that incubated the “great replacement” narrative [22]. The Census Bureau had not intended to produce replacement anxiety. Federal demographers were applying standard methods to standard data. But the decomposition technique provided something that no restrictionist think tank could have produced with comparable authority: a federal statistical agency confirming that the racial composition of America was changing. The data entered the world from a legitimate source and was picked up by actors with no institutional connection to its producers. Screenshots stripped the projection of its caveats. Memes stripped it of its conditionality. Manifestos cited it as an epistemic warrant for violence. Chapter 5 traces the path from Census data to manifesto citation, from federal press release to eliminationist politics, and names the operation that connects them: a process this book calls *demographic engineering*, in which decomposition and risk-tranching logics convert heterogeneous populations into a portfolio whose racial composition can be projected, ranked against a benchmark, and managed through policy levers. The era’s question: how did a federal projection become the epistemic infrastructure of replacement politics once it traveled beyond its makers?

The Portfolio (2025–2048) Something strange happened as the replacement anxiety of the Swarm era crested: anxieties about *over*-population were replaced with anxieties about *under*-population. Birth rates, already in decline across the developed world when Bartlett and his collaborators were cultivating overpopulation panics, shifted headlines from “minorities will be the majority” to “where have all the babies gone?” Pronatalist politics appeared as the mirror image of overpopulation anxieties and it recruited the same quantitative infrastructure. The fiscal burden analysis became the demographic return-on-investment calculation and wombs became policy levers. In 2024, a legal complaint challenging the availability of mifepristone included arguments about the fiscal and political consequences of declining birth rates. The complaint treated repro-

duction as a variable to be optimized, each prevented birth carrying an implied economic and political cost, each additional birth carrying an implied fiscal return. The same logic that governed portfolio management in finance was being applied to human reproduction. Chapter 6 develops two concepts: *fine-tuning*, the application of financial optimization to demographic governance, and *temporal gerrymandering*, the manipulation of demographic time horizons to manufacture urgency. Despite the superficial tension, zero population growth and pronatalism are structurally the same project of coercive interventions in the present on behalf of the future. The era's question: why are zero population growth and pronatalism the same project, and what kind of governance does the same-project structure produce?

The Evacuation (2049–2068) The final era is speculative, but its mechanisms are not. Cascading climate catastrophe shifts the governing question from who gets to stay to who gets rescued, and the instrument that answers it is no longer a curve or a chart but a model of the shocks themselves: across the 2020s the science of climate migration becomes a science of shock spillover, in which a disruption in one region is modeled as a cascade of secondary shocks carried by the displaced into the regions that receive them. Chapter 7 follows that apparatus to its speculative destination — an autonomous decision system that processes its outputs and issues evacuation and resettlement determinations faster than any human review can keep pace with — and shows how a governance regime this book calls *regenerative sovereignty* (bioregional, restorative, participatory) becomes the progressive vocabulary in which Garrett Hardin's lifeboat returns. The architecture's instrumentalization is complete when the lifeboat is delegated to a machine and selective rescue is made to feel like the responsible operation of complex systems rather than the abandonment of vulnerable people. The era's question: when emergency replaces planning, and the decision is handed to a system no one in the water can argue with, whose survival counts?

The Long Term Chapter 8 turns from trajectory to refusal, and it begins by reclaiming the ground the architecture has fought on all along: the long term. Long-term thinking has been ceded, by default, to longtermisms that treat the future as a warrant for present coercion — the existential-risk calculus of effective altruism, the discount rates of climate economics, the captured regenerative sovereignty of the previous chapter, the abstract deep-time of authoritarian futurism. Against them the chapter develops what it calls *intersectional longtermism*: long-term thinking drawn from the communities that have already survived long-term projects of erasure and have had to think in generations to do it. Five traditions are developed on their own terms — Black ecologies, Indigenous continuance, disability justice, reproductive justice, and antifascist organizing — and from them the chapter distills a constructive program: *quantitative egalitarianism*, which demotes the number from verdict to evidence, and *solidarity infrastructuring*, the institutional work of carrying these traditions across generations.

The chapter's question: who owns the long term, and what does it look like to wield quantitative tools without recruiting them for exclusion?

Boulder Again Chapter 9 returns to Boulder, where the book opened because this is where the arithmetic of concern was taught, repeated, and absorbed into civic identity. The coda reads Bartlett's bacteria-in-a-bottle thought experiment as a micro-macro isomorphism: what the model strips from real bacteria — feedback, coordination, memory, survival — is what the *lebensraum* imaginaries strip from human communities, and the five traditions of the previous chapter are the macro counter-case, paired as deliberate provocations with the biological mechanisms of recombination and immune memory. From there the chapter brings the spectrum of recombinatory futures down to Boulder's own ground — the Cheyenne and Arapaho arrivals, the Sand Creek Massacre as the local precedent for what the architecture produces under climatic stress, and the city's deep-time ecological and habitability future of water, fire, and species under the long heat — and turns the nativist question of who belongs into the question of what the valley owes the people in a rapidly warming world.

The book gives this analysis two names it will use throughout. The first is *quantitative chauvinism*, which extends Porter's account of mechanical objectivity as a technology of trust by specifying the three orientations — the will to simplify, the rule of expertise, the politics of inevitability — through which that technology operates in demographic governance [23]. The second is *lebensraum imaginaries*, which extends Griffin's account of palingenetic ultranationalism by specifying how ecological crisis supplies the rebirth myth and quantitative instruments supply the epistemic infrastructure [24]. The book uses *lebensraum imaginaries* for the recurring architecture and reserves *ecofascism* for the verdict, on the reasoning Chapter 2 sets out. The two concepts co-evolve: quantitative chauvinism is the epistemic infrastructure that *lebensraum* imaginaries require, and the chapters that follow trace their interaction from one instrument to the next.

Infrastructures of Safety

Boulder is not safe. It built governance systems to protect the community's quality of life but also systems to protect a specific demographic configuration that leverage "neutral" quantitative instruments to embed exclusionary assumptions. A doubling-time calculation, the planner's growth cap, the budget analyst's fiscal impact assessment, and the climate plan's emissions target each present a governance decision as a technical finding, and the accumulation of these technical findings produces a community that is overwhelmingly white, overwhelmingly affluent, and overwhelmingly insulated from the demographic changes reshaping the nation around it. None of these instruments were explicitly designed as tools of demographic exclusion but all of them function to filter out classes of people who lack the historical roots, social connections, and fi-

nancial resources that reproduce historic systems of marginalization. Were these filters encoded as legislative language barring new residents on the basis of their tenure in the community, success of their businesses, or their outright wealth, it would be easier to connect legislative intent and substantive impact. But the quantitative form in which these programs are administered ensures that numeric thresholds about ostensible ecological constraints conveniently enact political preferences as well.

The chapters ahead show the same operation working at larger scales and in different vocabularies. A doubling-time tutorial that warns of ecological limits also forecloses structural alternatives to demographic restriction. An impact statement that quantifies the fiscal cost of immigration also launders a racial hierarchy through administrative aesthetics. A population projection that decomposes the future by race also manufactures a countdown that demands eliminationist politics. A fertility model that optimizes the dependency ratio also treats women's reproductive capacity as a portfolio subject to financialized optimization. In each case the instrument's form — its apparent objectivity, its institutional credentials, its capacity to convert political claims into technical findings — is what enables the slide.

The book traces this slide not to demonstrate that quantification is inherently exclusionary, which is both false and useless, but to show how quantitative instruments, wielded by specific actors through specific institutional channels, have been recruited to serve projects of demographic exclusion that their form disguises as ecological necessity. Before that lecture could enter the world, however, the world had to be prepared to receive it. Chapter 2 traces that preparation.

8 The Long Term

“THE CHIEF CAUSE OF PROBLEMS IS SOLUTIONS” is a popular quote attributed to Eric Sevareid, the CBS News journalist whose two-minute commentaries accompanied Walter Cronkite’s celebrated broadcasts. Sevareid’s quote was one of Bartlett’s favorites and is cited in his lecture and writings as a justification for his brand of lifeboat ethics; public health, agricultural innovation were solutions that simply delayed reckoning with the longer-term problem of overpopulation his exponential math proved. [1] The diagnosis the previous chapters have built generates an impulse to *do something*. The futures the *Lebensraum* imaginaries are organizing toward, the policies the chauvinist instruments are already operationalizing, the captures already under way all press toward a constructive response. But the form the response must take is fixed by something the diagnosis has also disclosed. The operations the book has tracked are not the work of a news cycle, an election, or a single administration. They are intergenerational projects. The population-concern community trained the population-bomb generation; the population-bomb generation trained the Tanton network; the Tanton network trained the contemporary restrictionist infrastructure; and the pronatalist economists trained the longtermist research community whose framework now structures the regenerative-sovereignty imaginary that Chapter 7’s autonomous decision system was designed to operationalize. Quantitative chauvinism reproduces through journals, departments, funding networks, and curricula. It thinks in centuries. The reflex is to answer urgency with urgency, but the adversary is patient, and a response pitched to the electoral cycle will lose to a project pitched to the generational one. The terrain on which the contest is actually conducted is the long term.

The long term, however, is already occupied. “Longtermism” is an intellectual program that belongs to a specific and well-funded constellation, and the broader business of imagining what comes after the present has been ceded, by default, to projects that treat the future as a warrant for present coercion. However, long-term thinking does not belong to them. Communities have been practicing it, under names other than “longtermism,” for as long as the operations this book has documented have been op-

erating against them. They have had to because the marginalized people targeted for erasure that did not think in generations would not survive to be studied. The resources for more just long-term thinking already exist, fully developed, in the traditions that built them under the worst conditions the modern world has produced.

The chapter names this inheritance *intersectional longtermism*. It is long-term thinking drawn from communities that have already survived long-term projects of erasure: a way of treating the future as multiple rather than singular, and the present as the moment at which a choice among possible inheritances is made, rather than as a toll to be paid for a future that someone else has already specified. It refuses the trade at the heart of the dominant longtermisms, the exchange of present coercion for future security, because the communities it draws on have been on the receiving end of that exchange every time it has been offered. Its sources are five traditions of surviving and prevailing: Black abolition and environmental justice, Indigenous continuance and decolonial ecology, disability justice, feminist political economy and reproductive justice, and antifascist-cooperative organizing. Its horizon is what comes after the speculative threshold of 2068 and into the long heat [2], the period of compounding climate stress in which the question of who is allowed to persist will be asked with new force.

The chapter proceeds in three movements. It first examines the prevailing conceptions of the long term, the claimants who have made longtermism their own, and the single structural move they share. It then develops the five traditions on their own terms, each with its own thinkers, its own conception of time, and its own account of what is owed across generations. It closes by drawing from them a compressed constructive program, *quantitative egalitarianism* and *solidarity infrastructuring*, and returning the whole apparatus to the diagnostic the book has built.

The Long Term and Its Claimants

Begin with the difficulty of thinking about the long term at all. Amitav Ghosh has named it the great derangement: the strange incapacity of the dominant cultural forms to represent climate change, to hold in imagination a process whose scale and duration exceed the conventions through which the modern West learned to narrate the world [3]. The future arrives as either statistical abstractions or apocalyptic collapse, never as time inhabited by people. Nathaniel Rich's account of the decade between 1979 and 1989, when the science was settled and the political will briefly seemed to exist, documents the institutional version of the same failure: a long-term problem repeatedly deferred because no institution was built to hold it across the electoral and quarterly horizons that organize political and economic life [4]. The long term is hard to think and this difficulty clears the field for those who arrive with apparatuses to fill the vacuum with futures of their own design.

The most prominent brand of contemporary longtermism grew from a movement called “effective altruism.” In the work of William MacAskill and Toby Ord, the moral weight of the vast number of people who might exist in the future is made to bear on present priorities: if the future could contain astronomical numbers of worthwhile lives, then safeguarding their possibility dominates almost every other consideration, and the catastrophes that would foreclose them, the existential risks, become the supreme objects of concern in the present [5]. The framework includes its own deferral mechanism, the “long reflection,” in which humanity is to postpone irreversible commitments until it has had time to deliberate. Timnit Gebru and Émile Torres have traced the genealogy of this constellation, the bundle of ideologies they label TESCREAL, back to the eugenic traditions the present book has been documenting from another direction [6]. The structural feature worth naming here is the conversion of the future into a quantity to be maximized. Once the future is a number of potential lives, present people, and present vulnerable people in particular, become small relative to it, and their claims are discounted against the astronomical stakes that the framework has discovered. The will to simplify, encountered in the Test Tube era as a single curve, reappears as an incomprehensibly large count of lives for the remaining lifetime of the universe.

The second claimant is technocratic and wears the dress of sober realism. It is the long term of integrated assessment modeling, the social cost of carbon, and financial discount rates. The long-running quarrel between Nicholas Stern and William Nordhaus over how steeply to discount future welfare is, beneath its technical surface, an argument about how much a future person is worth in present accountings of costs and benefits, and the choice of a parameter decides in advance how much present sacrifice the future can command, or how much future suffering present comfort may impose [7]. The Shared Socioeconomic Pathways supply the official scenarios within which this accounting is conducted [8]. Andreas Malm and Wim Carton have exposed the temporal sleight at the center of the contemporary version, the logic of overshoot: the promise that the world may exceed its carbon budget now and draw the excess back down later licenses present emission against a future removal that the models assume and the technologies do not deliver, so that the future is mortgaged to underwrite the present in the precise inverse of the sacrifice the philosophical longtermists demand [9]. James Hansen’s appeal to his grandchildren belongs to this register too, in its more honorable form, the climate scientist staking his authority on an intergenerational claim [10]. Dan Wang’s account of the engineering state, the polity that plans in decades and builds at scale, names the geopolitical version of technocratic long-termism, admirable in its capacity and indifferent, in itself, to whose future it engineers [11]. What the technocratic long term shares with the philosophical one is the rendering of the future as an object of calculation, and the smuggling of the decisive political choices into parameters that present themselves as technical.

The third claimant is the one the book has already named. *Regenerative sovereignty*, the contemporary synthesis Chapter 7 watched assemble itself, is a speculative flavor of longtermism already captured by ecofascist politics. “Regenerative” agriculture, economics, and culture were developed by communities this chapter will return to, and their founding politics were anti-colonial, anti-plantation, and anti-extractive; “sovereignty” over food, body, data, and land was developed by Indigenous nations, reproductive-justice organizers, and Global South movements as a refusal of imperial governance. Paul Hawken’s call to end the climate crisis in a single generation gives the benign mainstream version of the regenerative vocabulary, earnest and widely read [12]; Laura Martin’s history of ecological restoration supplies the genealogy that explains how such vocabulary becomes available for capture, tracing how the project of restoring nature was, from the beginning, entangled with decisions about which nature, and whose, was to be restored [13]. When the regenerative and sovereign vocabularies are detached from their core commitments and reassembled around the spatial-biological governance that *Lebensraum* has always specified, the result are imaginaries of the long term that may profess repair while performing selection.

The fourth claimant is the gentlest and, for that reason, the most insidiously available. It is the civic deep-time culture of the Long Now, the ten-thousand-year clock, the invitation to lengthen our horizons and think like good ancestors [14]. It includes the international law of intergenerational equity, Edith Brown Weiss’s effort to give future generations standing in present jurisprudence [15], and the holistic civilizational visions, like Jeremy Lent’s ecological civilization, that ask for a wholesale reorientation of worldview [16]. There is much of value here, but its long term is an abstract horizon detached from any specific obligation to any specific people. The most telling symptom is the free circulation of “seven generations” as a civic slogan, lifted from Haudenosaunee governance and emptied of the specific nation, the specific land, and the specific obligations that gave it meaning, until it can ornament an NGO’s annual sustainability report. A long term owed to everyone in general is owed to no one in particular, and can be redirected to whatever project supplies the next sentence.

Around these four runs a fifth register, neither program nor policy but mood: the doomerism and catastrophism that imagines the long term as already locked-in and foreclosed. David Wallace-Wells’s cascade of worst cases and Roy Scranton’s counsel that we learn to die as a civilization both grasp the gravity of the situation and, in grasping it, evacuate the future of agency [17]. The collapse literature gives this mood an apparatus. Luke Kemp’s history of societal collapse and Peter Turchin’s complexity science of the rise and fall of complex societies offer to make the future legible as recurrence [18]; and Turchin’s cliodynamics, in particular, must be received with the book’s standing caution, because a quantitative science that resolves the human past into a small number of structural variables and projects their cycles forward is performing

the very will to simplify the book has tracked through demography. Collapse is not a mechanistic law but one outcome among many.

The claimants differ in idiom, constituency, and good faith. What they share are commitments to the future as singular destination that is a destiny to fulfill, an optimal path to pursue, an aggregate to maximize, or a collapse to await. Each uses that singular future to license a persistent arrangement: coercion in the present on behalf of the future. Intersectional longtermism recognizes and refuses this move by treating the future as multiple and the present as the site of choice among the inheritances to pass along to those futures.

Five Traditions of the Long Term

The traditions that follow are not five instances of a single thing. Each developed its own thinkers, its own conceptions of time, its own institutions of transmission, and its own accounts of what one generation owes the next. What they share are intergenerational durabilities in spite of active efforts to erase them. The chapter sets them alongside one another, but it should not flatten them into manifestations of a latent common doctrine or subordinate them to facile comparisons within an organizing grid. The comparison runs along three questions, which each tradition answers in its own register: what is the long term *for*; how is the capacity to think and act across it *transmitted*; and to whom is the future *owed*. Read together, the answers compose something the dominant longtermisms cannot: a long term that is plural, situated, and accountable to the living. The traditions are credited where their analytical work appears, and the historical and archaeological cases the chapter mobilizes are presented as what they are, the documented long-term survival of specific peoples, rather than as illustrations of a thesis they did not author.

Black Ecologies and the Long Memory of Disposability

The Black radical tradition has been thinking in centuries because it has had to survive in centuries, through the Middle Passage, plantation slavery, the barbarity of the Civil War, the overthrow of Reconstruction, Jim Crow, mass incarceration, and the ongoing condition Ruth Wilson Gilmore names organized abandonment. Its long term is a long memory: the accumulated recognition of disposability as a recurring structure rather than a series of discrete misfortunes and of survival as an achievement that must be actively produced and transmitted. Clyde Woods uses the blues, developed through the political economy of the Mississippi Delta, to describe a sustained cultural practice in which the conditions of Black life under racialized capitalism became a medium of a counter-temporality, one that refused the catastrophist countdown and held open a horizon of endurance and reconstruction where the dominant register saw only collapse and abandonment [19]. Blues epistemology is long-term thinking because it en-

codes the memory of what the apparatus has done before and the expectation that it will try again under a new name.

The tradition's conception of the future is most precisely stated in its spatial imagination. Katherine McKittrick's plantation futures names the persistence of plantation logic in contemporary geography and names the forms through which Black life has refused enclosure: the plot worked against the plantation, the marronage geographies beyond the reach of survey, the Black ecologies of swamp and bayou and island that the surveillance apparatus could map institutionally but could not see politically [20]. J. T. Roane and Justin Hosbey extend this into a method for documenting the specific places where Black communities have built ecological relations the managerial frameworks of environmental governance cannot comprehend [21]. This is environmental thought with a long memory and a long horizon, and it has an institutional spine. The Black agrarian line runs from Booker T. Washington's Tuskegee through Fannie Lou Hamer's Freedom Farm Cooperative to Leah Penniman's Soul Fire Farm and the contemporary Black land trust movement, a continuous practice of working land on terms set against the plantation. Soul Fire Farm is the decisive contemporary instance because its surface, Black people growing food on rural land, can be confused with the trad-wife homesteader aesthetic only until one attends to its abolitionist and anti-plantation commitments.

If the blues encode the tradition's long memory, Black speculative fiction encodes its long future, and it does so in direct competition with the longtermisms of the first movement for the authority to imagine what comes after. Octavia Butler's *Parable* novels, written in the early 1990s and set in a 2020s California unraveling under climate breakdown, corporate enclosure, and a resurgent authoritarianism that campaigns on making America great again, are climate fiction a generation ahead of the genre's later prestige, and they imagine survival not as a lifeboat to be defended but as a dispersed, deliberately heterogeneous community carrying a transmissible discipline across catastrophe [22]. The wider Afrofuturist tradition, from Sun Ra through Samuel Delany to the present [23], treats the far future as territory where Black people not only exist but thrive rather than as objects to be managed by existential-risk frameworks. A people whose place in the present was always contested has had to claim the future in imagination. This is all long-term thinking, older and more durable than the latter-day discipline that has claims the name.

The tradition's account of the present is correspondingly long. Ruth Wilson Gilmore's concept of organized abandonment names the deliberate, infrastructural withdrawal of the resources a community needs to live, the slow violence that the catastrophist register, fixed on the sudden event, cannot see; and her abolition geography insists that the alternative is not the reform of the carceral apparatus but the construction, across generations, of the institutions that make it unnecessary [24]. Saidiya Hartman's afterlife of

slavery [25] and Christina Sharpe's figure of the wake [26] name the same long duration from the side of those who bear it: the present as a time still structured by a past the official history declared closed, and survival within it as sustained practice rather than settled fact. What these analyses share with the agrarian and environmental-justice work already named is the refusal of the short horizon. The harm is centuries deep; the repair must be centuries long; and the tradition has built strategies and institutions to hold that horizon open.

That defining content is the tradition's account of what is owed. The environmental justice scholarship of Robert Bullard, Laura Pulido, and David Pellow established that environmental harm to Black communities is the product of racialized siting, zoning, and disinvestment, not of any attribute of the communities themselves, and so refused at the level of structural analysis the threat-object construction that every longtermism of selection requires [27]. Abolition supplies the warrant beneath the analysis: the founding refusal of the plantation's disposability logic, carried forward into the contemporary movement against carceral and immigration-enforcement regimes. The Black church, the historically Black colleges and universities, the mutual-benefit societies, the NAACP, the Black Panther Party's survival programs and their descendants, and the agrarian network already named are not a list of organizations but a demonstration that the transmission problem has been solved, repeatedly and deliberately, across changes in political condition, funding, and leadership. The long term, in this tradition, is the time of abolition: not a deadline by which a catastrophe must be averted, but a horizon toward which a centuries-long labor of refusal and reconstruction is sustained.

Indigenous Continuance and the Politics of Persistence

Indigenous environmental thought has been developed by peoples who have already survived an apocalypse, and it carries the conceptual resources that survival required. Kyle Whyte's framework of collective continuance names the capacity of an Indigenous nation to adapt to change without ceasing to be itself, to maintain the relationships, responsibilities, and self-determination that constitute it as a people across transformations that the settler imagination can only read as loss [28]. This is a theory of the long term built from the inside of long-term survival, and it reframes the seven-generations principle that the civic deep-time culture has appropriated. In its place of origin the principle is not a slogan about lengthening horizons but a specific structure of obligation binding a specific people to specific relatives, human and nonhuman, across descending and ascending generations. Robin Wall Kimmerer develops its ecological grammar as reciprocity, an exchange of gifts and responsibilities between human and other-than-human kin that refuses the extractive calculus converting webs of relations into commoditized resource flows [29]. Vanessa Watts presses further with place-thought, the proposition that land itself knows and that agency is not separable

from territory, an ontological commitment that the visual and rhetorical conventions of Western environmental communication cannot register [30]. Nick Estes shows the principle in motion as politics, tracing Oceti Sakowin resistance from the nineteenth century to the Standing Rock encampments as a single long tradition in which the obligation to the Missouri operated as a warrant for collective action that the language of water rights could describe but not comprehend [31]. In the contemporary governance register the same commitments appear as Indigenous data sovereignty, the CARE Principles and the protocols for Indigenous artificial intelligence that insist on authority, accountability, and benefit prior to the technical question of measurement [32].

What the indigenous tradition's long term looks like in practice is documented in the archaeological and historical record of the North American Southwest, which supplies two of its clearest demonstrations. The first is the dispersal from Mesa Verde. In the late thirteenth century, across a sustained megadrought, the Ancestral Puebloan communities of the Four Corners left the cliff dwellings and the mesa-top fields that had sustained them for centuries. The standard narrative, popularized by Jared Diamond and the broader collapse literature, reads this departure as civilizational failure, demographic overshoot meeting ecological limit, and enrolls the empty dwellings as evidence for the inevitability of carrying-capacity collapse [33]. Indigenous scholars have refused the reading, and the evidence is on their side [34]. Their dispersal was not collapse, but reorganization. The communities did not scatter at random but moved to specific receiving communities along the Rio Grande, organized by kin and ceremonial relationship, carrying with them the kiva traditions, the calendar systems, the agricultural knowledge, and the kinship structures that constituted them as peoples. The receiving sites show the integration of Mesa Verde architectural and ceramic forms alongside existing local traditions, the material signature of a people transmitting itself into a new place rather than ending. The Pueblo nations of the Rio Grande, the Hopi mesas, and the Zuni River are the descendants and the inheritors of that movement. What the collapse frame reads as a tale of limits and overshoot is actually an instance of persistence: the survival of peoples through a transformation severe enough to empty a landscape, accomplished by carrying forward knowledges and practices by re-configuring relationships in the face of changing ecologies.

The second demonstration is what that persistence made possible. The communities the dispersal seeded were, within two centuries, surviving a second and far greater catastrophe, the pandemics that followed European contact and reduced the Indigenous population of the continent by something on the order of ninety percent across the first century and a half after 1492, a depopulation severe enough to register in the atmospheric carbon record [35]. The Pueblo peoples who rose against Spanish rule in 1680 were survivors of that catastrophe and of eight decades of colonization, the *encomienda*

system, and the suppression of their ceremonial life. Under the leadership of Popay of Ohkay Owingeh, communities speaking Tewa, Tiwa, Towa, Keres, Zuni, and Hopi, languages of limited mutual intelligibility, coordinated a synchronized uprising across hundreds of miles, timed by knotted cords whose untying counted down the days, and drove the Spanish colonial regime from New Mexico for twelve years [36]. The Pueblo Revolt is the political achievement the Mesa Verde dispersal made possible, and the connection between them is the heart of the matter. Four centuries separated the departure from Mesa Verde and the coordination of 1680, and across those centuries the communities did not preserve their knowledge and practices in proverbial amber but preserved themselves *through* change, transmitting the practices, the memory of suppression, and the recognition of the colonial pattern that made a coordinated refusal possible when the moment came. The contemporary Pueblo nations carry that transmission forward across nine centuries now, through further impositions and further recoveries, organized today in the All Pueblo Council of Governors. They are not the preserved remnant of a purer predecessor, they are representations of intergenerational persistence by carrying forward relationships across catastrophe and reorganization.

The contrast that clarifies the achievement is the case where transmission failed. The Late Bronze Age collapse around 1200 BCE is the canonical instance of systemic breakdown across an interconnected world: drought and disruption across the eastern Mediterranean, the fragmentation of the Hittite empire, the burning of the Mycenaean palaces, the loss of literacy, and a demographic disruption that took centuries to repair [37]. Even there the lesson is not the inevitability of collapse but the difference transmission makes, for the communities that retained the capacity to reorganize, the Phoenician city-states that became the dominant traders of the Iron Age, the upland communities that emerged where the palace economies had been weakest, passed through the breakdown into altered but viable forms, while those that did not, ended. Collapse is a real pole of the spectrum of outcomes. What determines a people's position on that spectrum is not the magnitude of the shock but whether the practices of persistence have been built and carried forward.

This tradition operates in close dialogue with a broader decolonial and Global South body of thought that the chapter names as adjacent rather than identical, and declines to fold in. The Andean philosophy of *Buen Vivir*, constitutionalized in Ecuador and Bolivia, replaces growth-based development with frameworks of reciprocity, sufficiency, and the rights of nature [38]; Sabelo Mhlambi develops from *ubuntu* an account of relational personhood that refuses the individual-utility calculus on which the fitness warrants depend [39]; Arturo Escobar's pluriverse, in dialogue with Zapatista practice, names the ontological plurality that decolonial politics defends [40]. These are distinct traditions with distinct genealogies, and the resonances among them should not be collapsed into a single "indigenous" lens. What they share with the North Amer-

ican traditions, and what matters for intersectional longtermism, is an account of the future as continuance and persistence of people and their relationships across time as obligations without fantasizing about the future welfare of interstellar civilizations.

Disability Justice and the Indispensability of All

Disability justice contributes the conception of the long term that the dominant longtermisms are least able to absorb, because it begins from the refusal of the fitness logic on which they all depend. The principle that Patty Berne, Mia Mingus, Stacey Milbern, and their collaborators named in the mid-2000s, and that Sins Invalid articulated as the ten principles of disability justice, is the indispensability of all bodies and minds: the categorical claim that every body and mind has equal standing to the resources necessary for life, asserted not as a value to be weighed against others but as a refusal that renders incoherent any operation that sorts bodies into the expendable and the indispensable [41]. Set this beside the longtermisms of the first movement and its force is plain. The effective altruism's longtermism ranks lives by their contribution to aggregate future welfare; the technocratic version ranks them by the discount rate; the regenerative and triage versions rank them by their fit to ecological or civilizational capacity. Each is a fitness logic, and the eugenic continuity the book has traced across its eras runs through through them all. Disability justice does not propose a "fairer" ranking but refuses the ranking altogether. David Pellow's fourth pillar of critical environmental justice is grounded substantially in this refusal, which is what allows the framework to register as indispensable the populations the dominant framework has already classified as expendable [42].

The tradition also contributes a distinct temporality. Alison Kafer's crip time names time as relationally produced rather than uniformly distributed, the rhythms of care, fatigue, recovery, and access that the productivity calculus cannot accommodate and that the single curve of exponential growth was built to suppress [43]. Crip time is long-term thinking of a particular kind: it measures duration in the labor of sustaining a life rather than in units of output, and it knows that a future built to exclude the slow, the dependent, and the unwell is a future built on the same logic as the past the tradition has survived, the eugenic institutions, the sterilization regimes, the asylums, the neoliberal disinvestment, and the pandemic abandonment in which elite discourse openly named the deaths of the disabled, the chronically ill, and the elderly as acceptable losses for the resumption of economic life. Against that abandonment the tradition has built care webs, the relational infrastructures of mutual support that disabled and chronically ill people maintain outside the formal medical and emergency management systems, which Leah Lakshmi Piepzna-Samarasinha has documented as a political practice with worked-out forms, expanded sharply during the pandemic-era withdrawal of collective protection [44]. Alice Wong's curatorial work made disabled lives visible on disabled

people's terms [45], and Mia Mingus's access intimacy named the relation in which access is produced together rather than granted.

The memory this tradition carries is specifically the memory of the eugenic century, the same century the book has tracked from the demographers' side. The institution, the colony, the sterilization law upheld in *Buck v. Bell*, and the long medical project of curing disability out of existence are the operational face of the fitness logic, and disability justice was built in their shadow. Eli Clare's critique of the ideology of cure distinguishes the wish to relieve suffering from the wish to eliminate disabled existence, and names the second as continuous with the eugenic project rather than opposed to it [46]. Sunaura Taylor draws the connection the environmental frame requires, linking disability and animal liberation to show how the category of the defective organizes both the treatment of disabled people and the treatment of the more-than-human world, so that an ecological politics indifferent to the fitness logic will reproduce it [47]. Aimi Hamraie's history of accessible design [48] demonstrates that the built environment is itself a long-term inscription of assumptions about which bodies will exist and matter, and that access is the slow material labor of building a world on different assumptions.

This bears directly on the long heat. Disabled and chronically ill people are the population that disaster most reliably abandons, left behind in the evacuation, cut off when the power fails, sorted out when the modeling assigns survival probabilities, and the disability-justice tradition has produced the most rigorous available account of how that abandonment is engineered and how it is refused. The care web is the refusal in institutional form, a distributed and intergenerational infrastructure of mutual provision that does not wait for the formal system to certify a life as worth sustaining. Where the dominant longtermisms imagine the future as a frontier secured by the fit, disability justice imagines it as a web of interdependence maintained by everyone, and it knows from a century of practice that interdependence rather than independence is the actual condition of survival across time.

The transmission of this tradition is the youngest of the five and, for that reason, the most self-conscious about transmission as such. The Independent Living Movement that Ed Roberts and the Berkeley Rolling Quads built into the first Center for Independent Living in 1972 won the legal scaffolding that culminated in the Americans with Disabilities Act of 1990; the disability justice framework that succeeded the rights frame pushed past legal accommodation toward indispensability; and because the tradition is young, it has documented its own principles and archived its own history deliberately, building the counter-archive even as it builds the movement. Its account of what is owed is the simplest and the most radical of the five: nothing in the future may be purchased at the cost of sorting bodies in the present, because the sorting is the harm the tradition exists to refuse.

Reproductive Justice and the Feminist Long Term

The feminist tradition holds the longest memory of the specific operation this book tracks, because the government of populations has always run through the governance of reproduction. Silvia Federici's account of the European witch hunts as a moment in primitive accumulation, a violent enclosure of women's bodies and reproductive capacity that accompanied the enclosure of the commons, supplies the deep history: the demographic apparatus did not begin with the census, and the feminist refusal of it did not begin with the clinic [49]. The contemporary articulation is reproductive justice, the framework that Loretta Ross and a collective of Black women named in 1994 and that SisterSong institutionalized after 1997, built on three rights held together: the right to have children, the right not to have children, and the right to parent children in safe and supportive environments [50]. The structure of the framework is itself an argument about the long term. It refuses both the anti-natalist coercion of population control and the pronatalist coercion of demographic vitality by locating reproductive futurity in the autonomy of the people who bear it, rather than in any planner's objective to engineer the composition of future generations. Betsy Hartmann's documentation of population control coercion in the Global South and Anne Hendrixson's renewed call for feminist resistance to populationism carry the analysis into the present, where both poles of the coercion operate at once [51].

The feminist long term has an explicitly ecological lineage. Carolyn Merchant's history of the scientific revolution traced the moment at which a living, female-figured nature was reconceived as inert matter available for domination, and located in that reconception the joint subordination of women and the natural world that ecofeminism exists to refuse [52]. Vandana Shiva's defense of subsistence and seed sovereignty [53], Maria Mies's subsistence perspective [54], and Val Plumwood's dismantling of the reason-versus-nature dualism [55] compose a tradition that has, for half a century, named the conceptual machinery the *Lebensraum* imaginaries depend on: the rendering of certain people and certain natures as background, resource, and substrate for political projects of purification and rebirth. Ecofeminism is long-term thinking because it attends to precisely what the productive economy treats as free and inexhaustible, the reproductive and regenerative work, human and ecological, on which the visible economy silently draws.

In contemporary theory, social reproduction is where the feminist tradition meets the structural analysis the later sections develop. Nancy Fraser's account of cannibal capitalism describes a system that consumes the social and ecological conditions of its own possibility faster than it replenishes them, devouring the care work that sustains life in pursuit of the value that accumulates capital [56]; the social-reproduction theorists locate the making and remaking of human life, daily and generationally, as the hidden ground of the whole economy [57]; and the ethics of care proposes care itself

as the organizing value of a just politics [58]. The long term, in the register of feminist resistance, is simply the time across which life is reproduced and any politics organized around accumulation will always raid that time until commitments to care reverses the priority. Reproductive justice's third right, the right to parent children in safe and sustainable environments, is already a climate claim: it asserts that the conditions for raising a next generation are themselves owed, and may not be spent.

This tradition also supplies the epistemology and the transmission technology that intersectional longtermism most directly inherits. Donna Haraway's situated knowledges named and refused the view from nowhere, the god-trick of a disembodied objectivity [59], which is precisely the position from which the single official projection and the catastrophist countdown are issued. Long-term thinking that knows itself to be situated, partial, and accountable is feminist before it is anything else. The "intersectional" in intersectional longtermism is a direct debt to Patricia Hill Collins and Kimberlé Crenshaw [60], whose insistence that systems of domination are co-constituted and cannot be analyzed one axis at a time is the methodological core. The consciousness-raising group, the signature form of 1970s women's liberation, was itself a transmission technology, converting private grievance into shared analysis and solidarity and passing the method from one cohort to the next. The institutional infrastructure that followed, the reproductive justice organizing, the clinic networks and abortion funds that expanded sharply after the 2022 *Dobbs* decision, the feminist publishing of the Kitchen Table: Women of Color Press that Barbara Smith and Audre Lorde helped found, and the women's and gender studies programs demonstrate a transmission sustained across more than a century. The contemporary feminist climate writing, the moral clarity of Kathleen Dean Moore, Ayana Elizabeth Johnson, and Katharine Wilkinson carries the tradition to the edge of the long heat [61]. The feminist long term is reproductive futurity held in autonomous hands, and its account of what is owed is that no generation may mortgage the bodily self-determination of the next to secure a future of its own design.

Antifascism and the Transmission of Pattern Recognition

The antifascist tradition contributes the long term as sustained organized refusal, and a transmission mechanism unlike any of the others: the deliberate teaching of pattern recognition across generations, so that each cohort can identify the fascist formation its predecessors fought when it returns under a new name. Mark Bray's history locates contemporary antifascism in a century of documented organizing, the Italian *Arditi del Popolo* against Mussolini's Blackshirts in 1921, the German *Antifaschistische Aktion* of 1932 whose twin-flag emblem the movement still carries, the International Brigades, the partisan resistance across occupied Europe, the 43 Group of Jewish veterans against Mosley's reconstituted fascists in 1940s London, and the Anti-Racist Action networks

of late-twentieth-century North America [62]. This is a century-old longtermism that recognizes that the adversary is also engaged in a long-term project with intergenerational institutions and recognizes that effective opposition must therefore be pitched at the same temporal scale of transmitting capacities to recognize the fascisms chameleon character.

The Mondragón federation, the Emilia-Romagna cooperative network, the U.S. Federation of Worker Cooperatives, and the platform-cooperativism movement theorized by Trebor Scholz and Nathan Schneider supply working alternatives to shareholder-governed production in which the objective function is set by workers and members rather than external investors [63]; and the mutual-aid practice that Dean Spade and Rebecca Solnit have documented across Hurricane Maria, the Covid-19 pandemic, and the fires of the past decade demonstrates that horizontal solidarity is a tested mode of holding populations through disruption, not a sentiment [64]. Here the tradition meets the structural analysis it requires. Jason Moore's account of capitalism in the web of life names the accumulation logic, the organization of cheap nature and cheap labor across a five-century world-ecology, that the antifascist and cooperative traditions confront, and locates the present crisis in the long history of that accumulation rather than in a recent deviation from it [65]. Kohei Saito's reconstruction of Marx's late ecological thought, and his case for a degrowth communism, supply the most developed contemporary statement of an emancipatory long term built on the deliberate reduction of throughput under collective control rather than its coerced imposition on the vulnerable [66].

The tradition carries a constructive ecological vision as well as a defensive posture because it was built in direct opposition to the Malthusianism the early chapters tracked. Murray Bookchin's social ecology argued, against the population alarmism of his own environmental moment, that ecological crisis originates in social domination rather than in human numbers, and that its resolution is the reconstruction of society along confederal, directly democratic, and ecological lines [67]. The democratic confederalism of the Kurdish freedom movement in Rojava, drawing explicitly on Bookchin, has organized communities around women's liberation, ecological stewardship, and decentralized self-government in the middle of a war, and the Zapatista communities of Chiapas have sustained an autonomous, ecological, Indigenous-led political order for three decades. These are not models to be directly copied into Boulder, but they are existence proofs of ecological politics grounded in solidarity that have survived longer than French revolutions and communes that so often are used to disqualify the possibilities of sustained solidarity.

The tradition has also named the specific adversary the long heat will produce. Andreas Malm and the Zetkin Collective have documented the deep affinity between fossil-capital defense and the far right, and the emerging formation in which climate

denial gives way to a climate nationalism that concedes the warming and proposes to meet it by fortifying the borders and sorting the displaced [68]. This is the politics the book has tracked as *Lebensraum* in its newest dress, and the antifascist contribution to intersectional longtermism is the sober recognition beneath it: that the climate crisis will not automatically produce solidarity, that it can as easily produce its opposite, and that the cooperative future is therefore something to be organized and defended against an adversary already organizing, not a destination the crisis will deliver on its own.

Saito's degrowth is also where this tradition must do its most careful work, because degrowth is the clearest case of a project susceptible to reactionary capture. The reduction of material throughput is emancipatory when it is organized through collective ownership and an anti-eugenic commitment to provisioning everyone, but it is also eco-austerity that licenses triage when those commitments are stripped away, leaving only the bare imperative of less, to be borne by whoever has least power to refuse it. This is exactly the degrowth that Chapter 7 watched the regenerative-sovereignty regime absorb. The fork has a name in the tradition's own literature, the choice between ecosocialism and ecofascism [69]. That is why antifascism occupies a special place among the five traditions. Its defining content is opposition to fascism, and a vocabulary whose entire content is opposition to fascism is the one warrant in the repertoire that cannot be captured by the projects this book opposes, because admitting their content does not capture the position but converts it into its opposite. The antifascist long term is solidarity sustained across generations against an adversary that is itself generational, and its account of what is owed is the refusal to let any future serve as the alibi for present hierarchy.

Quantitative Egalitarianism and Solidarity Infrastructuring

The five resistance traditions summarized above supply the resources for a constructive program for imagining alternative long terms. It has two halves. *Quantitative egalitarianism* is the methodological counterpart to quantitative chauvinism: not the abandonment of number but its repurposing, the rebuilding of projection, disaggregation, decomposition, optimization, and shock analysis around the commitments the traditions have developed, so that probabilistic forecasts replace single official curves, environmental-justice screening replaces burden construction, and capability-based and capacity-based frameworks replace the optimization of demographic variables against a concealed objective [70]. *Solidarity infrastructuring* is the institutional-material counterpart to the *Lebensraum* imaginary: the building of the durable institutions, the cooperatives and care webs and sanctuary networks and data-sovereignty regimes, through which the traditions' commitments are carried across generations.

A reader who has followed the book this far is entitled to raise an objection: a program called "quantitative egalitarianism" proposes to answer quantitative chauvinism

with better numbers but a substantial body of scholarship holds that there are no better numbers to be had—the harm was never in the quality of the quantification but in the act of quantifying living relations at all. James Scott’s account of high-modernist legibility, encountered in Chapter 6, is also an indictment that reaches the present chapter: the violence of legibility is not that the state measures badly but that it measures, flattening the dense particularity of social life into the synoptic categories administration requires is the precondition of the coercion that so often follows [71]. Max Liboiron presses the argument: the threshold concept that organizes environmental governance, the determination of how much a body of water or a body of people can assimilate before harm registers, is not a neutral measurement that bad actors misuse but a colonial relation in mathematical form, and a more equitable threshold is still a threshold [72]. Elizabeth Popp Berman completes the indictment: the economic style of reasoning did not defeat the justice claims of American liberalism in open argument but dissolved them, by making efficiency the idiom in which all serious policy had to be conducted, so that reformers who adopted it to win a hearing found they had lost the vocabulary in which their commitments could be stated [73]. The objection these three assemble is sharp: a book that ends by proposing to quantify more justly has reabsorbed the disease it spent seven chapters diagnosing.

The chapter concedes most of this. Scott is right that legibility is the precondition of administrative coercion, and the book has shown the point in every era. Liboiron is right that some quantifications cannot be reformed but only refused: the refusal of the triage frame this book advances does not propose a “fairer” survival-probability score, because a score that decides who is worth rescuing is colonial whether or not its coefficients are balanced. Berman is right that the idiom can swallow the commitments of those who adopt it. If quantitative egalitarianism meant building a more equitable version of each instrument the book condemned, the objection would be fatal.

The problem the book has named is *jurisdiction*: the claim that a number has standing to *close* a political question, to convert who belongs here, whose fertility counts, and who is rescued from contestable matters of public judgment into settled findings that only experts may revise. Quantitative chauvinism is that confusion of jurisdictions, the elevation of a metric from a piece of evidence that informs a judgment into a verdict that forecloses it. Quantitative egalitarianism is the demotion of the number from verdict to evidence. It does not promise a fairer count that settles the question; it strips the count of its standing to settle the question at all, returning the judgment to the people the judgment governs. This is why Scott’s indictment, taken to its conclusion, does not touch the program: legibility grants the administrative monopoly, and the egalitarian counter-method refuses the monopoly, not the measurement, by making the categories, the assumptions, and the uncertainty contestable by the populations the numbers describe. A projection presented as a fan of weighted scenarios

with its classificatory choices labeled and renegotiable is still a quantification, but it has surrendered the jurisdiction the single official crossover date claimed. The number has stopped ruling and started reporting.

The strongest evidence that this is possible rather than wishful comes from the very communities whose critique the objection assembles. Indigenous data sovereignty is not a refusal of data; it is data under refused jurisdiction. The CARE Principles do not ask whether a count is accurate. They ask who holds the authority to control it, who benefits, and to whom it is accountable, and they place those questions prior to the technical question of measurement [74]. Te Hiku Media counts the Māori language meticulously and keeps the jurisdiction over that count with the communities the language belongs to, which is precisely the arrangement the threshold concept Li-boiron indicts was built to prevent. These communities have built the working model of quantification stripped of sovereignty, and the program is the generalization of what they already practice, not a humane gloss on the apparatus they refuse.

Solidarity infrastructuring is held to the same standard, and the five traditions supply its evidence. They are not theoretical resources to be activated from outside; they are operating infrastructures with documented intergenerational transmission, the Black church and the agrarian network, the ceremony and the data-sovereignty institution, the care web and the disability archive, the consciousness-raising group and the abortion fund, the cooperative federation and the antifascist documentation project. Across them, durability has required five practices: pedagogy that transmits across generations, material capacity sufficient to survive sustained attack, counter-archives that document the tradition on its own terms, institutional reproduction across changes in leadership and funding, and explicit anti-capture vigilance, the deliberate attention to the defining content that distinguishes a tradition from its capturable surface. That last practice is not optional, because every vocabulary the program deploys is a vocabulary the supremacist movements also covet, and the capture-resistance is structural rather than moral. Indigenous sovereignty without the standing of specific nations toward specific places is incoherent; Black abolition without anti-plantation politics is incoherent; disability justice without the indispensability of all is incoherent; antifascism is not capturable by fascism. The resistance follows from the definitions, not from the virtue of the definers, and that is why it can be built to last. Quantitative egalitarianism and solidarity infrastructuring are not blueprints. They are commitments and practices, drawn from traditions that have sustained them across generations, for the intergenerational work the long term requires.

The Diagnostic Answered

The book's diagnostic ran across five eras, and the program just described answers it without rehearsing it. Three structural features made *Lebensraum* portable: the con-

version of a demographic question into an ecological-spatial answer, the granting of demographic governance the authority of ecological science, and the disavowal of the historical name. Those features survived 1945, and the arrangement that combines them has been reassembled continuously: as the overpopulation panic of the Test Tube era, the immigration restrictionism of the Quarantine, the replacement anxiety of the Swarm, the pronatalism of the Portfolio, and the climate triage that Chapter 7 projected for the Evacuation. Each rehabilitation disavowed the name while preserving the arrangement, and each drew on whatever constructive vocabulary the moment supplied, until they converge in the contemporary synthesis the book has called regenerative sovereignty: *Lebensraum* under managed rebranding, deploying captured vocabulary as cover for the same spatial-biological governance the concept has always specified.

The traditions are the answer because they are what regenerative sovereignty captures. The regenerative agriculture it borrows was developed against the plantation; the sovereignty it invokes was developed against empire; the rootedness, the stewardship, the care, and the duty to future generations it performs were developed by communities for whom those words named obligations to specific people and places. Capture works by detaching the vocabulary from that underlying content, retaining the affective surface while evacuating the politics. Held to their original commitments, the traditions refuse the capture by definition, and that is the whole of the book's constructive claim: the resources adequate to the moment already exist, in the keeping of the communities the moment most endangers, and the work is to fund, defend, and transmit them rather than to invent a program from first principles.

This is why quantitative chauvinism and *Lebensraum* imaginaries are best understood, finally, as longtermist projects, exactly as Chapter 7 began to argue. They reproduce intergenerationally; they think in centuries; they claim the future to license the present. A resistance pitched at any shorter scale will lose to them. Intersectional longtermism is the resistance pitched at their scale, drawn from the only traditions that have sustained the long term against erasure and won the right to speak about it. It does not promise the future. It refuses to trade the present for one.

Conclusion

The chapter's analysis has operated at the level of concept and tradition. The rehabilitations of *Lebensraum* are not abstractions; they take place in specific institutions, policies, and movements, and the five traditions of surviving and prevailing are present in specific communities, organizations, and places. Boulder, Colorado, is one site among many at which the rehabilitation of *Lebensraum* through captured regenerative-sovereignty vocabulary is operating in real time, and at which the traditions are present in specific institutional and community form. The book began there. The next chapter returns there.

The coda does not propose a model for replicable application; it tests the chapter's conceptual apparatus at the specific site where the book's diagnostic work began. Boulder is a settler-colonial city on Arapaho and Cheyenne land whose environmental virtue has rested on infrastructural arrangements that displaced the peoples whose territorial claims the city still occupies. It is also a city whose growth-management regime, climate-action infrastructure, and environmental governance carry both the rehabilitative operations this book has documented and the foundational resistances it has named. Both are present. Both are operating. What intersectional longtermism looks like when it is brought to ground at a particular valley, under a particular deep-time horizon, at the place where the arithmetic of concern was first taught, is the work of the coda that follows.

9 Boulder Again

TWO VERY DIFFERENT KINDS OF OBITUARIES CIRCULATED after Albert Bartlett's death in September 2013. The first was carried in CU Boulder's institutional channels and the surviving paper of record, the *Daily Camera*: a remembrance of a beloved physics professor, a tireless public educator, a civic leader on Boulder's growth management questions, and the man who had delivered his lecture on the arithmetic of population and energy 1,742 times in service of public understanding [1]. The second appeared in *The Social Contract*, the intellectual organ of John Tanton's white nationalist anti-immigrant project, which remembered Bartlett as a heroic ally whose mathematical authority had supplied the restrictionist movement with one of its most effective pedagogical instruments.

Whether Bartlett was an ecofascist is not a particularly interesting question. While his extensive body of writing and deep affiliations certainly point to his importance within an unabashedly reactionary movement devoted to coercive exclusion with a fig leaf of conservation, some might rise to Bartlett's defense to say the white supremacists Bartlett collaborated with for decades did not reflect his personal politics as a "good Democrat" in mid-century Boulder. Adjudicating the behaviors and beliefs of a man who died over a decade ago would settle nothing about the need to confront what the movement Bartlett legitimized, how it has metastasized into American politics since 2016, and the predictable templates that will allow it to continue to be rehabilitated through the polycrises of the long heat [2].

Let us instead indulge in an alternative thought experiment that reflexively parrots the grandiose sociobiological appeals that seem to travel so well alongside *Lebensraum* imaginaries. This register of biological metaphors and their naturalistic fallacies are examples of portability that only ever seems to operate in the direction of justifying projects of coercion and dominance. Reversing the direction of making naturalistic fallacies to support intersectional perspectives is meant to elicit the unease and criticism that somehow escapes the other direction. Put another way, what Bartlett got wrong about bacterial growth mirrors what he got wrong about human capacities for cooper-

ation revealed by the resistance traditions of Chapter 8. What his thought experiment ignores about how real bacteria behave also revealingly mirrors what his *Lebensraum* imaginaries strip out at the macro scale of imagining humanity in the long term.

The five resistance traditions Chapter 8 developed are the macro counter-case to the macro simplification, just as actual bacterial complexity is the micro counter-case to Bartlett's bottle. The synthesis the chapter then develops carries through the climate polycrisis of the long heat that awaits us all. The resistance traditions are revisited here as macro human-scale analogies of biological processes, meant as provocations within the naturalistic fallacies of the sociobiological registers of *Lebensraum* imaginaries, not as reductions or endorsements. Rest assured the unease the reader feels with making these provocative equivalences will be critically unpacked.

The Bottle: What the Thought Experiment Assumes

To review, Bartlett's "bacteria in a bottle" thought experiment invited the audience to imagine a single bacterium is placed in a bottle at eleven o'clock in the morning. The population doubles every minute and by noon the bottle is full. The audience is invited to follow the arithmetic. At one minute before noon, the bottle is half full. At two minutes before noon, the bottle is one quarter full. At five minutes before noon, the bottle is less than 5% full and the bacteria experience the resources as abundant and unending. Exponential growth in a bounded environment produces inevitable carrying capacity exhaustion, and the perception of abundance immediately preceding exhaustion is the cognitive failure that prevents communities from recognizing the limits of the bottle they inhabit [3].

Bartlett's "greatest shortcoming" thought experiment assumes a great deal. The bottle is closed. The medium is uniform. The bacteria are incapable of communication, adaptation, or behavioral change. The resources are simple and exhaustible. The growth is only exponential with a fixed doubling time. There is no immune system, no environmental feedback, no genetic recombination, no symbiotic relationship, no quorum sensing, no biofilm formation, no horizontal gene transfer, no diauxic shift, no stress response, and no memory. Bartlett's bacteria are abstract counters whose only relevant property is their ability to double, and the bottle is an abstract container whose only relevant property is its finite capacity. Bartlett's thought experiment is mathematically rigorous and biologically false at every step.

Four important properties of real bacterial populations stripped from the bottle organize the biology that the thought experiment refuses. *Feedback* is the property by which bacterial populations respond to their own outputs. Bacteria detect accumulated waste products, depleted nutrients, and chemical signals from their neighbors, and they adjust metabolism, growth rate, and gene expression accordingly. *Coordination* is the property by which bacterial populations act collectively. Quorum sensing, first described in the 1970s and now understood as ubiquitous across the bacterial domain, allows

bacteria to detect population density through the accumulation of signaling molecules and to switch behaviors collectively when threshold concentrations are reached. The collective behaviors include biofilm formation, sporulation, the suppression of further growth, and the coordinated expression of metabolic pathways that individual cells could not sustain alone. *Memory* is the property by which bacterial populations encode information about past conditions and transmit the encoding across generations. CRISPR-Cas systems are bacterial adaptive immune systems: bacteria encode in their genomes short sequences derived from past phage attacks, transmit the sequences to subsequent generations, and use the sequences to recognize and destroy future infections by the same or similar pathogens [4]. CRISPR-Cas is, literally, immune memory: encoded recognition of past threats, transmitted across generations, capable of rejecting future encounters with recognized pathogens. *Survival* is the property by which bacterial populations persist through conditions that would otherwise eliminate them. Persister cells go dormant under stress and reactivate when conditions return to permissive states. Spores survive for decades or centuries in suspended animation. Biofilms protect inhabitants from antibiotics, predators, and environmental extremes. Horizontal gene transfer allows the acquisition of survival traits from neighboring populations, including across species lines, at rates that exceed conventional inheritance by orders of magnitude.

Together, feedback, coordination, memory, and survival are the actual properties of bacterial life produced across billions of years of exposure to the pressures Bartlett presumes to have discovered in 1969. The actual organisms in the actual bottles do not require the kinds of external control Bartlett's implications demand to govern their growth; their feedback, coordination, memory, and survival mechanisms handle the pressure through cooperation and recombination at scales his thought experiment cannot register. The question Bartlett's lecture poses, "what will impose the constraint the arithmetic predicts?", is the wrong question. The behaviors of actual organisms mirror many of the strategies we observe in human relations. The thought experiment's demand for external control is an artifact of the simplification, not a feature of any empirically grounded biology.

The Analogy: From Micro to Macro

What Bartlett got wrong at the micro level of his thought experiment also describes what he got wrong at the macro level, and the latter-day reanimations of *Lebensraum* imaginaries require us to ignore the same things. Just as Bartlett's bacteria in a bottle ignore fundamental bacterial mechanisms to justify coercive external interventions, *Lebensraum* imaginaries' commitments to purity and control demand coercive interventions only by ignoring the long history of possibilities within human communities for cooperating and surviving through pressure.

Human communities respond to their own conditions through political processes, economic adjustment, cultural negotiation, and institutional adaptation. Human communities have coordination: they organize through federations, mutual aid networks, political confederations, and intentional cooperation across community boundaries. Human communities have memory: they encode the recognition of past patterns in archives, oral traditions, organizing practices, and embodied knowledges, and they transmit the encoding across generations through institutions and intergenerational relationships. Human communities migrate, federate, recombine, hybridize, and reorganize under stress through mechanisms developed and refined across millennia of exposure to climate shifts, political pressure, and demographic disruption. The rhetorical registers common to sociobiology, behavioral genetics, and evolutionary psychology that accompany *Lebensraum* imaginaries are curiously resistant to grappling with these biological counter-metaphors.

The pairings that follow take the five traditions Chapter 8 developed on their own terms and set each, as provocation, beside a biological mechanism. Black ecologies recognize and remember patterns of disposability across centuries; the CRISPR-Cas systems within bacteria encode recognition of past pathogens into their own genomes. Indigenous environmental thought sustains the long-memory coordination through which communities reorganize their relationships to place; bacterial quorum sensing reads collective density through chemical signaling and triggers population-wide reorganization. Disability justice organizes mutual aid as the practice through which communities of different bodies sustain one another across need; mycorrhizal networks connect plants of different species through underground fungal partners, transferring nutrients to stressed individuals and signaling alarm across the network. Feminist political ecology has developed strategies for recognizing calls to subjugate bodily autonomy to political projects; bacterial populations exercise metabolic autonomy through diauxic shifts that let them handle conditions no single substrate could sustain. Antifascist environmental thought sustains the federated cooperation that resists authoritarian consolidation across generations; bacteria under stress organize into biofilm communities whose coordinated behaviors include resource sharing, division of labor, and defense against external threats. Bacteria outside of Bartlett's 1,742 tidy thought experiments know many inconvenient truths.

Lebensraum imaginaries share a demand for protecting the future through coercive intervention in the present. This is not a curious coincidence with Bartlett's thought experiment demanding paternalistic control over mindless bacteria. Both demands are artifacts of simplification in the name of portability that strips out the feedback, coordination, memory, and survival mechanisms found in real organisms from the microbiological to the macroinstitutional. Bartlett's thought experiment and its role within *Lebensraum* imaginaries both produce audiences that believe their situation requires the

control they have been taught to want. The actual organisms, at both scales, neither require nor benefit from the programs of control the imaginary demands. They are already products of long-term processes that allowed them to accumulate mechanisms for handling ecological pressure through accommodation, recombination, and inter-generational transmission rather than coercion and sacrifice.

Possibilities and Limits of the Biological Frame

This coda has paired the five resistance traditions with five biological mechanisms as deliberate provocations that invert the assumptions of sociobiological and other eugenic projects and, by now, should elicit some unease about the appropriateness of appealing to naturalistic fallacies to justify political projects. The implications of equating intersectional projects of resistance with biological mechanisms of survival carry both possibilities and risks that demand critique and reflection. What does this sociobiological register of resistance traditions reveal and obscure when confronting longtermist projects? At least three possibilities reveal themselves.

The first is temporal. Resistance traditions operate at the scale of centuries while biological mechanisms operate at the scale of millions of years. Setting the two scales alongside one another supplies an analytical frame in which long-term thinking does not belong only to longtermist projects allied with *Lebensraum* imaginaries, and invites us to examine what survives across deep time and how. The five resistance traditions perform this work at the human scale and the biological frame extends it into the ecological and geological scales implied by the long heat of the Anthropocene. The comparison supplies analytical frames for thinking about transmission, encoding, and recognition that should not be unilaterally ceded to *Lebensraum* imaginaries.

The second possibility is structural. The micro-macro isomorphism the chapter has been developing compares what Bartlett's bottle strips out with what *Lebensraum* imaginaries strip out: feedback, coordination, memory, survival. The simplification is identical at both scales, and it has a name the book has used from the first chapter forward: quantitative chauvinism, the will to let a single simplified quantity stand in for the political and biological complexity it suppresses. The bottle and the architecture are the same kind of analytical mistake performed at different temporal and spatial scales, and naming the isomorphism is what allows the chapter to recognize that the macro counter-case does at the human scale what the micro counter-case does at the microbial and ecological scales.

The third possibility is rhetorical. The biological frame disrupts the audience the architecture has been producing. Bartlett's lecture teaches audiences to want control of exponential processes; *Lebensraum* imaginaries tap into audiences' desire for conservation and preservation. The biological frame teaches that communities that recognize themselves as assemblages have a deep biological and historical precedent for the recombinations that ensure their survival.

The risks open in their own directions. The first risk is biological determinism. Treating the resistance traditions as biological inheritances rather than as politically chosen practices would commit the analytical error the book has spent eight chapters tracing: naturalizing political claims by encoding them as ecological facts. CRISPR-Cas systems are not chosen by bacteria. Black ecologies, Indigenous conservation, disability justice, feminist political ecology, and antifascist environmental thought are chosen, sustained, and transmitted by communities that have done the active work of carrying them across generations under conditions designed to extinguish them. The biological analogy frames recognition of what the political work resembles at deeper temporal scales, but it does not found or substitute for that work.

The second risk is organicism. Treating human communities as organisms carries historical baggage the book has been documenting throughout. Population biology, social Darwinism, organic statism, the carrying capacity literature whose lineage earlier chapters traced, and the *Lebensraum* tradition itself all derive their authority from various forms of the community-as-organism analogy. The chapter must not commit the same analytical move under a different name. Ecologies are not bottles. Human communities are not bacterial populations. Mutual aid is not mycorrhizal networking. Communities are political associations whose institutional forms emerge from contested choices about how to organize life together, and the choices are not constrained by biological analogies. The biological mechanisms are pedagogically useful because they show what Bartlett's analytical mistake *looks like* when applied to actual biology, and what feedback, coordination, memory, and survival look like as properties of complex systems. They are not pedagogically useful as templates for how communities should be organized.

The third risk is metaphorical drift. Provocations have a way of becoming conclusions, and analogies have a way of becoming identities. The chapter has been explicit about the provocational status of the pairings, and the explicitness is itself a discipline against the drift. But the drift can occur silently across the analogy's rhetoric, and the chapter has to remain vigilant against the possibility that what the metaphor offers as recognition of structural similarity becomes obscuration of political specificity. Bacteria do not have institutions, archives, presses, schools, or organizing networks. They do not vote, develop languages, or defend civil rights. The transmission of CRISPR-Cas across bacterial generations is automatic genetic inheritance; the transmission of the resistance traditions across human generations is active institutional labor performed by communities under conditions designed to actively destroy and erase them.

What the metaphor enables is a final provocation in the service of the intersectional longtermism Chapter 8 develops. That chapter named long-term thinking drawn from communities whose continued existence has required it, and set it against the effective-altruist longtermism that promises distant futures on the grounds that present coer-

cion is the price of securing them. The biological frame is one resource among the many intersectional longtermism draws on, and the most hazardous of them, offered as provocation and held under the vigilance the preceding pages have insisted on. It sits alongside the seven-generation framings of Indigenous environmental thought, the centuries-of-memory of Black ecologies, the intergenerational care of disability justice, the reproductive futurity of feminist political ecology, and the pattern recognition of antifascist organizing: the traditions whose institutional and intergenerational labor the analogy cannot replace and must not be mistaken for. The pluralism is the point. The biology is a provocation within it, not its foundation.

Recombinatory Futures

The historical and archaeological record offers a range of trajectories communities have followed through sudden shifts. At one end of the spectrum sit the collapses that produce catastrophic violence and the dispersal of communities through coordination failures, federation breakdowns, and the convergence of climate stress with political pressure. The Late Bronze Age Collapse around 1200 BCE is the canonical complex-systems case, and Chapter 8 used it as the foil for what breakdown without transmission produces [5]. The Sand Creek Massacre of 1864 is an under-appreciated instance more local to Boulder, in which climatic stress, settler-colonial pressure, and the bounded community premises that inspired generations of *Lebensraum*'s acolytes converged on the eradication of communities as a "common sense" response.

At the other end of the spectrum sit the cooperative reorganizations Chapter 8 examined in detail: the Ancestral Puebloan dispersal from Mesa Verde across the late-thirteenth-century megadrought, and the Pueblo Revolt of 1680 that the dispersal made possible four centuries later, which that chapter developed as evidence of persistence through erasure rather than as a tale of collapse. Between these endpoints sit communities that absorbed arrivals through cooperation, communities that lost some members through resettlement, and communities that survived in modified form through hybridization and recombination.

Boulder's recombinatory future, like that of every other human settlement on the planet in the early twenty-first century, will occupy some position on this spectrum. The position will be determined by which memories are transmitted through the poly-crisis decades ahead. The *Lebensraum* project's commitments to purity and constancy push communities toward the worst end of the spectrum, because the imaginary refuses the recombination the long heat will force and produces coercive interventions when the refusal fails under stress. The resistance traditions' immune memory pushes communities toward the best end of the spectrum, because the traditions encode the recognition of what coercive interventions produce and supply the feedback, coordination, memory, and survival mechanisms that make cooperative recombination institution-

ally possible. Both ends require active institutional and intergenerational transmission. The relevant analytical question about Boulder's future is not whether the future will involve recombination, but which recombination, organized through which inheritance, occupying which position on the spectrum the historical and archaeological record supplies.

The Cheyenne and Arapaho whose summer encampments preceded Boulder's founding by less than a century were themselves recent arrivals to the Front Range. Both nations had moved from Great Lakes and Missouri River homelands across the seventeenth and eighteenth centuries, responding to colonial pressure from the east and to the ecological opportunity opened by horse adoption after the seventeenth-century Pueblo Revolt released Spanish horses into Plains tribal networks [6]. The Plains life-way the white settlers encountered in the 1850s was a recombined formation less than a century and a half old. It was built on the integration of horse-mediated mobility with bison-economy subsistence and articulated through political confederations that had themselves recombined out of earlier homelands and earlier formations. The community Boulder displaced was not a static and eternal occupant of an unchanging territory. They were a reorganizing community whose own arrival had been a recombination response to climate, colonialism, and technological diffusion.

The Cheyenne language and the Arapaho language are both Algonquian, indicating their relatively recent origins in the upper Midwest and Great Lakes regions where the Algonquian language family had developed across millennia. The Plains adaptation involved the substantial reorganization of seasonal practice, settlement pattern, social organization, ceremonial calendar, and material culture: the abandonment of horticultural villages in favor of mobile camps, the development of the travois and the tipi as the material expressions of the new mobility, the elaboration of warrior societies whose function had been substantially different in the Great Lakes context, and the integration of trade and intermarriage relationships with other recently arrived Plains peoples (the Lakota, the Comanche, the Kiowa) whose own recombination histories paralleled the Cheyenne and Arapaho experience. The Plains tribal world of the early nineteenth century was a region of intensive cultural exchange, hybridization, and political innovation, and the bounded-community premise the architecture would later impose on the region was false and useless. The communities Boulder's white settlers encountered had already been making recombination their primary mode of cultural and political life for the better part of two centuries before the settlers arrived.

The recombination ran deeper still. The Plains tribal migrations of the seventeenth and eighteenth centuries unfolded across a continent that the post-1492 pandemic had already substantially depopulated. The Great Lakes and Missouri River communities whose ancestral homelands the Cheyenne and Arapaho left had experienced the pandemic shock generations earlier, and the demographic and political vacuum the pan-

demic left across what is now the central United States was part of the ecological opportunity the horse-mediated Plains adaptation took up [7]. The horse's arrival on the Plains, following its release from the Spanish colonial settlements after 1680, intersected with a continent whose Indigenous population had already been reduced by an order of magnitude across the preceding two centuries.

The Plains tribal recombination of the early modern period was therefore not only a response to colonial pressure from the east and ecological opportunity in the west. It was also a post-pandemic political formation of survivors and refugees, organized by communities whose collective memory of a catastrophic depopulation event would certainly be fresh and whose institutional capacities were those of survivors' communities that had passed through the most severe demographic catastrophe in the hemisphere's history. The Cheyenne and Arapaho summer encampments in Boulder Valley in the early nineteenth century were not the first or only encampments of communities that had always been at the Front Range. They were the recent encampments of communities whose presence on the Plains was itself the result of a recombination conducted under post-pandemic conditions across the entire continent.

The bison ecology of the Plains had been transformed by two decades of drought and intensified by an unapologetically genocidal project of over-hunting by the 1860s. The climatic and ecological conditions for Plains tribal autonomy had been deteriorating throughout the period. The colonial pressure from the gold rush and the Civil War's logistical demands converged on the Cheyenne and Arapaho communities whose seasonal use of Boulder Valley and the broader Front Range had been their sustenance for generations. On November 29, 1864, Colorado militia under Colonel John Chivington attacked the camp of Black Kettle and White Antelope along Sand Creek in southeastern Colorado, killing approximately 230 Cheyenne and Arapaho people, predominantly women, children, and elders, in a camp that had been promised federal protection by the same federal government whose military Chivington commanded [8].

The massacre was not climate genocide in any contemporary technical sense. It was a pogrom by white settlers against communities of survivors whose climatic and ecological conditions had been making their political autonomy increasingly precarious. Climate stress produces political conditions under which the *Lebensraum* imaginaries' purity-and-constancy commitments become violent, and the frontline of that violence is reliably the body of the person the architecture has marked as disposable, a pattern Peter Schwartzstein has documented across the contemporary world's climate-stressed peripheries, where drought and dislocation are metabolized into ethnic and sectarian killing by political actors who find the demographic argument ready to hand [9]. The 1865 congressional investigations into Sand Creek produced extensive documentation of the events, the political conditions that produced them, and the institutional fail-

ures that permitted them, and the documentary record has been continuously available across the century and a half since [10].

Sand Creek was not the deepest violence Lebensraum imaginaries would produce. America's settler-colonial project across the nineteenth century included massacres at Bear River (1863), Washita (1868), Marias (1870), Wounded Knee (1890), and dozens of others; forced removals across the continent (the Trail of Tears in 1838, the Long Walk of the Navajo in 1864, the Nez Perce flight in 1877); and the treaty violations and consciously genocidal erasures whose cumulative effect was to reduce the continent's Indigenous population by an estimated ninety percent across the four centuries from initial contact through the early twentieth century.

The Cheyenne and Arapaho camping at Sand Creek were the same Cheyenne and Arapaho whose seasonal use of Boulder Valley had been their sustenance for 150 years. The institutional pressure that produced the massacre also produced the founding of Boulder as a city in the same decade. The architecture this book has traced is not separable from the violence it produced when its commitments to bounded community encountered the climatic and political conditions of the 1860s Front Range.

The Indigenous communities attacked at Sand Creek were already survivors. The historiography of the past several decades has revised earlier estimates of the pre-contact population of the Americas substantially upward, and revised the post-contact mortality substantially upward in percentage terms [11]. The pandemics that followed European first contact, principally smallpox alongside measles, influenza, typhus, and the cumulative pathogens against which Indigenous populations had no acquired immunity, killed approximately ninety percent of the continent's Indigenous inhabitants across the first century and a half after 1492. Charles Mann's synthesis of the revised historiography drew on the demographic work of Henry Dobyns, William Denevan, and Russell Thornton, and recent paleoclimate evidence has documented that the post-pandemic depopulation was severe enough to produce a measurable drop in atmospheric carbon dioxide and to contribute to the Little Ice Age cooling. Communities like the Cheyenne and Arapaho who historically camped at Sand Creek were only five to seven generations removed from the most severe demographic catastrophe in the hemisphere's history. The slaughter at Sand Creek targeted survivors whose continued political and cultural life into the nineteenth century was itself the achievement of survivors of one of the most extreme stress tests imaginable.

The white settler community that displaced the Cheyenne and Arapaho was itself a recombined arrival. The settlers arriving in the 1850s Front Range were drawn from across the eastern United States, from Europe, and from earlier Western American mining frontiers. Boulder's earliest settlers were already a mixture of New Englanders, Pennsylvanians, Midwestern farmers, immigrant tradespeople, and refugees from the failed mining camps of the California and Nevada rushes. The Dust Bowl migration

of the 1930s added another layer of recombination. Climate-driven displacement from the southern Plains brought new populations to Colorado's Front Range, and Boulder absorbed migrants whose own communities had failed to persist in their previous locations. Post-war government investments in scientific research infrastructure like NIST, NOAA, and Rocky Flats precipitated a third wave of arrivals bearing slide rules while wearing hiking boots.

None of these arrivals respected any imaginary of constancy that *Lebensraum* imaginaries would later actively enact. The community whose character the comprehensive plan was charged with preserving in 1977 was the product of three previous waves of recombination within the city's institutional lifetime. The grasslands the settlers broke to the plow were themselves a conquered and reorganized ecology, the prairie sea whose ruin and partial redemption Dave Hage and Josephine Marcotty have traced across the American interior [12]. The historical record of the Front Range, from the Folsom hunters of the post-Younger Dryas period through the Apishapa and the early Plains horticulturalists through the Athapaskan and Algonquian arrivals to the colonial and post-colonial settlement waves, is a record of continuous recombination conducted under climatic, ecological, and political pressures of various kinds. The settler community's pretense to have established a stable bounded community in Boulder Valley was the institutional denial of the recombination that had constituted the valley's human history. David Nye's account of America's conflicted landscapes names the deeper mechanism: the same valley sustains incompatible stories at once, wilderness and frontier and homeland and resource, and which story a community tells itself determines which arrivals it can recognize as belonging [13].

Climate genocide can happen here because it has already happened here. The commitments of *Lebensraum* imaginaries to bounded community and homogeneous composition produced, under the specific climatic and political conditions of the 1860s Front Range, the systematic killing of the people incompatible with the architecture's premises. It is not difficult to imagine how the same architecture, under the climatic and political conditions the long heat will bring, can produce similar convergences again. The targets will be different: the climate-displaced migrants whose arrival the regenerative-sovereignty framework refuses; the populations whose presence in refugia the participatory-adaptation framework cannot accommodate; the communities whose vulnerability the triage discourse treats as collateral. The refusal of *Lebensraum* imaginaries to accommodate recombination, combined with the climatic stress that forces it, produces violence as the resolution. Catastrophic positions on the spectrum are not abstract historical possibilities but have compelling local precedents. The architecture that produced them before is likely to produce them again. This recognition is what immune memory equips communities to refuse.

The other end of the spectrum has its local precedent too, and it is the one Chapter 8 developed at length. The Pueblo nations that dispersed from Mesa Verde and federated in 1680 are recombined across nine centuries, carrying forward the relationships and recognitions that made coordinated refusal possible under the compounded stresses of drought, colonization, and pandemic. The Boulder of 2200 will be similarly recombined. The species that arrive will join those that persist; the human populations that the climate polycrisis displaces will join those that remain in place; the institutional forms that survive will be the forms the recombination has tolerated. The question is which commitments Boulder carries forward through the recombination. The *Lebensraum* imaginary's commitments to purity and constancy refuse the recombination and produce coerced violence when the refusal fails. The resistance traditions' commitments to egalitarianism and cooperation accept the recombination as the actual condition of communities, organize it through mutual aid and federation, and produce the cooperative outcomes the Pueblo political tradition exemplifies. Neither path is automatic. The polycrisis Boulder, like every other human settlement on the planet in the early twenty-first century, will inherit whatever has been transmitted to it.

Lessons for Deep Time

The Flatirons are slabs of ancient seafloor tilted vertically during the Laramide orogeny and exposed by erosion as the orange sandstone monoliths that frame the western edge of Boulder Valley. They have stood there for sixty-five million years. The basin they frame has supported many communities across that duration. The Pliocene Warm Period of about 3.3 to 3.0 million years ago corresponds to roughly two degrees of warming above pre-industrial baselines and is the closest paleoclimate analog for the lower end of the long heat's projected range. Forests of this era stood at higher elevations and latitudes than today, sea level was substantially higher than the present, and species composition was broadly recognizable but reorganized into patterns the Holocene would not have produced [14]. The Mid-Miocene Climatic Optimum of about seventeen to fifteen million years ago corresponds to roughly four degrees of warming and is the closest analog for the upper end of the projected range. Subtropical species reached temperate latitudes during the Miocene optimum, the C₄ grasslands that currently define the Great Plains had not yet established, and the regional ecology of what would become Colorado was substantially different from anything the Holocene has produced.

To name the range as a range is already to make a choice the book has spent eight chapters interrogating. The forward-looking scenarios that organize climate planning, the Shared Socioeconomic Pathways, sort the long heat into a low road of rapid decarbonization and cooperative development, a middle road that extends present trends, and a high road of fossil-fueled rivalry, and they attach to each a temperature and a

likelihood [15]. The book has shown what such numbers do when they are granted jurisdiction, when a single projected pathway becomes the future the model has discovered and forecloses the political choice it actually encodes. The coda invokes the scenarios in the only register consistent with the argument of Chapter 8: as evidence, not verdict, their assumptions visible and contestable, a fan of weighted possibilities rather than a destiny. The difference between the low road and the high road is roughly the difference between the Pliocene and the Miocene, and it is not a difference the modeling will settle. It is a difference the present will choose, and the scenarios are useful precisely insofar as they keep the choice open rather than closing it.

The basin has been hotter than the long heat will make it. The basin will support the long heat's recombined community. The speed at which the recombination is being forced, however, is what makes the contemporary case unprecedented. The paleoclimate analogs document basin hospitality but they do not document the rates at which contemporary species and communities can move to take up the conditions the polycrisis is producing. The Pliocene's warming unfolded across hundreds of thousands of years; the Mid-Miocene's transition operated on similar deep-time timescales; the contemporary polycrisis compresses warming of comparable magnitudes into the span of decades or centuries. Species that could have migrated across the Pliocene's slow transition cannot migrate across the polycrisis's compressed transition. Communities that could have reorganized across the Miocene's gradual warming may not be able to reorganize across the long heat's accelerated arrival. The chapter's analytical body has been showing what outcomes are available across the historical spectrum; the deep-time frame is showing that the spectrum itself is being compressed by the polycrisis into temporal windows the historical record may not be able to inform.

The water will move first, because it always does. The Front Range is a place that has always lived on water it does not have, importing snowmelt through tunnels from the western slope and drawing the rest from a snowpack that functions as the region's largest reservoir. The long heat shrinks that reservoir and changes its timing: more winter precipitation falls as rain, the snow that does fall melts earlier, and the runoff that the whole apparatus of ditch and reservoir and water right was built to capture arrives in a pulse the apparatus was not designed to hold. To the east, the Ogallala aquifer that underwrites the High Plains is being drawn down toward exhaustion, the slow emptying of a fossil water that the irrigation economy spent in a single century, and Lucas Bessire's account of that draining describes the particular grief of a community watching the ground beneath its future go dry [16]. Boulder Creek itself, the thread the city is strung along, will run differently: lower and warmer through the late-summer months, and then, when the rain does come, higher and faster than the channel and its floodplain have been managed to expect. The 2013 flood was a rehearsal. James Scott's last book, in praise of floods, argued that the river's overflowing is not a failure of the

river but the source of the life along it, and that the long project of disciplining rivers into channels has been a project of refusing precisely the recombination a floodplain performs [17]. The long heat will overrule the discipline.

The cottonwood gallery forests along Boulder Creek are dispersing now, in the sense that their species composition is shifting under the stresses the long heat is producing. Plains cottonwood and narrowleaf cottonwood depend on overbank flooding for regeneration, and the flooding regime is shifting as snowpack declines and precipitation patterns change. The cohorts establishing in the 2050s and 2060s will be sparser than the cohorts that established in the twentieth century, and the species that establish in their place will increasingly be drought-tolerant arrivals from elsewhere along the Front Range and from across the south. Tamarisk and Russian olive press from downstream, and the gallery forest visible from the bike paths in 2055 will be the same shape it was in 2025 and a different forest in its species composition.

The American pika, the cold-adapted talus-dwelling lagomorph that has inhabited Boulder's higher elevations for millennia, is dispersing upslope at a rate the talus cannot match, and the lower-elevation populations are contracting toward extirpation [18]. The dispersals do not respect property lines or political boundaries. They respond to temperature, moisture, and the availability of the conditions the organisms' biology requires to survive. The Pueblo dispersal across the late thirteenth century that Chapter 8 examined, and the species dispersals across Boulder Valley in the early twenty-first, both refuse the *Lebensraum* premise of a bounded community. Both are immune-memory resources for the recombined assemblage, and both occupy a recognizable position on the spectrum of outcomes that await us in the long heat. The position the dispersals open is not collapse but reorganization, with the receiving communities and the receiving ecologies absorbing the arrivals through cooperative mechanisms whose institutional and biological forms the book has now spent two chapters describing.

Piñon-juniper woodlands are expanding upslope into former ponderosa zones, in some places replacing the species that had defined Boulder's foothill ecology for centuries. Southern grass species are moving into the prairie transition zones east of the city, carried by climate-shifted dispersal patterns and accelerated by the disturbance regimes that drought and fire have produced. Novel community assemblages with no historical precedent in Boulder Valley are establishing in disturbed zones and along the urban-wildland interface. The arrivals are not invasions but responses.

Fire is the other agent of the reorganization, and it has stopped waiting for summer. On the penultimate day of 2021, a grass fire driven by hurricane-force downslope winds ran out of the foothills into the suburban subdivisions between Boulder and Denver and destroyed more than a thousand homes in an afternoon, the most destructive fire in Colorado's history and a fire that burned, pointedly, not in remote forest but in graded lots and cul-de-sacs in the dead of winter [19]. The Marshall Fire was a lesson

in the new regime: the grassland that the settlers broke and the suburb that replaced it are both fuel, the fire season is now the year, and the wildland-urban interface the planners drew is a line the fire does not read. Fire is not the enemy of this landscape; the grasslands and the ponderosa both evolved with it and require it. What the long heat changes is the timing, the intensity, and the proximity, so that a process the ecology has always metabolized becomes, where it meets the built environment, a recurring catastrophe the built environment has not learned to expect.

What all of this asks of the people who manage the valley's ecologies is a reckoning that the language of restoration is only beginning to absorb. Ecological restoration, as Laura Martin's history of the field shows, was always a project of deciding which past to restore a place to, and the long heat removes the baseline the decision depended on: there is no Holocene Boulder Valley to return to, because the climate that produced it is gone [20]. The restorationist's primer now teaches assisted migration and novel-ecosystem management as much as it teaches the recovery of historical reference conditions [21], and the rewilding literature has shifted from restoring lost assemblages toward cultivating the capacity of an ecology to reorganize itself, to recombine into something viable that no one specified in advance [22]. Peter Alagona's account of the accidental ecosystem, the dense and surprising wildlife of American cities, suggests that the recombined Boulder may be wilder than the preserved one, its open-space buffers and irrigated corridors and warming creek assembling communities of coyote and bear and heron that the 1959 Blue Line never imagined [23]. And beneath all of it runs the mycorrhizal substrate Merlin Sheldrake describes, the fungal networks that connect the roots of the gallery forest and the foothill woodland and move water and warning between them, the literal underground of the recombination, indifferent to the property lines drawn above [24].

A central feature in the architecture of *Lebensraum* projects is to refuse arrivals. The ecological fallout from the long heat will eventually overrule that refusal at the biological scale, as the pika and the cottonwood and the piñon are already demonstrating. The harder question is the human one. The long heat will produce human migration on a scale the modern world has not governed, and the Front Range, with its relative elevation, its relative water, and its relative distance from the coasts and the worst of the heat, will be a place people move toward. The literature that has begun to map the next American migration, Jake Bittle's account of the communities already on the move and Abrahm Lustgarten's projections of where the uprooting leads, places the interior West and the Front Range among the regions that receive [25]. Gaia Vince has argued that the century's defining project will be the planned movement of people toward the latitudes and elevations that remain habitable, though her top-down framing of managed migration carries its own jurisdictional hazards, the same hazards the book has tracked wherever a population becomes a quantity to be allocated [26].

Here the coda makes its turn, and it is the turn the whole book has been building toward. The nativist question, the question *Lebensraum* has always asked, is who belongs here, and it answers with a bounded community and a refused arrival. The question the long heat actually poses is the inverse: given that the arrivals are coming, given that the biological refusal has already failed and the human refusal will fail after it, what does the receiving community owe the people who arrive? David Wallace-Wells's catalog of the uninhabitable earth performs the catastrophist version of the future, in which the arrivals are a threat and the refuge a fortress [27]; but the recombinatory account the book has assembled reverses the charge. Boulder's habitability is not a fixed quantity to be defended against the arrivals who would exhaust it. It is a relationship to be extended or refused, and the question is not how many the valley can hold but which inheritance the valley carries forward, the inheritance that meets the arrival with the lifeboat and the gunwale, or the inheritance that meets it with the mutual-aid network and the federation. Carrying capacity was always the wrong question. The right question is which community Boulder chooses to become as the people arrive.

The boreal toad is regionally extinct by 2200 under the conditions of most medium-term projections of two degrees of warming. The lower-elevation aspen clones have largely failed. The greenback cutthroat trout is concentrated in headwater refugia, isolated from its previously connected populations. The American pika is gone from all but the highest peaks. The failures of species are not violence in any moral sense, but their human correlates often produce profound institutional disruption and galling violence. *Lebensraum* has spent a century and a quarter administering refusal at the political scale that the biological scale cannot administer, and the convergence of biological and political refusal is what makes the long heat dangerous beyond its biological consequences.

Four lessons converge from the chapter's movements. The first lesson is that recombination is the actual condition of communities, and that the architecture's imaginary of purity and constancy nostalgizes a condition that has never existed. Bartlett's bottle stripped out the recombination because the model required organisms incapable of it. Real bacteria recombine constantly; real human communities recombine constantly; the architecture's purity-and-constancy commitments are biologically and historically incoherent. The Pueblo communities, the Front Range settler community, the species composition of Boulder Valley: all of these are recombined assemblages whose persistence has been the persistence of relationships rather than of substances. The recombination the long heat will force is not a deviation from a previous static state. It is the continuation of a process that has been continuous across the basin's entire human and ecological history.

The second lesson is that climate genocide can happen here because it has happened here. The Sand Creek Massacre is not an unfortunate pogrom in our history but a local

precedent for what the architecture's premises produce when they encounter climatic stress and political pressure simultaneously. Pandemic-scale demographic collapse has happened in the Americas, and the communities that survived it produced the political traditions Chapter 8 drew on. The resistance traditions are not abstract theoretical resources assembled from comfortable conditions. They are the encoded memory of communities that have already passed through the most extreme demographic catastrophe in continental history, and the institutional capacities they carry forward are the proof-tested capacities of pandemic-survivor communities whose continued political life into the twenty-first century is itself the demonstration of what cooperative recombination can do under the most severe possible stress. Without the immune-memory work the resistance traditions transmit, the architecture's premises will produce the same convergence again, against different targets, in conditions the polycrisis is now producing.

The third lesson is that the traditions Chapter 8 developed, which this chapter has been reading through the figure of immune memory, are the actual resource for cooperative recombination, and that their transmission is the work of the decades immediately ahead. Immune memory is not the preservation of the present; it is the recognition of patterns the architecture has produced before and the rejection of them when they reappear under new vocabulary. What gets preserved through the long heat is not contemporary Boulder's specific institutional configuration but the analytical commitments and embodied knowledges that sustain the traditions. Schools, presses, archives, organizing networks, mutual aid structures, intergenerational mentorship, the embodied knowledges carried by elders and renewed by younger practitioners: none of these arrive without resources, and none survive sustained pressure without active support. The traditions have been built and maintained without the resources the dominant institutions have appropriated, and their preservation through the long heat will depend on whether the polycrisis-stressed Boulder of the coming decades recognizes the traditions as the immune system the city's own future will require. None of this is guaranteed. The institutional work is the practice. The practice is the immune memory in action.

The fourth lesson is the realism question. The *Lebensraum* tradition has always claimed to be realistic. Its triage discourse, its bounded-community premise, its lifeboat ethics, its carrying-capacity calculations, and its participatory-adaptation frameworks all present themselves as the realistic responses to scarce resources and demographic pressure. The claim is unrealistic about biology, history, and human social life at both scales. Real bacteria do not require Bartlett's control; their feedback mechanisms evolved across billions of years to handle exponential pressure through accommodation and recombination. Real human communities do not require the architecture's coercive interventions; their political traditions across millennia handle climate shifts through migration, federation,

and cooperative recombination. The appropriate target for coercive intervention in the present on behalf of the future is to stop the climate-changing emissions immediately, but that goal is inconvenient for late neoliberal capitalism. So as we make the severity of the long heat worse, the realism of migration, recombination, egalitarianism, and cooperation becomes more urgent, not less. The chapter rejects the *Lebensraum* architecture's claim to realism at both scales and asserts the counter-realism the immune-memory traditions transmit. Quantitative chauvinism has always supplied that false realism its alibi: the simplified number that makes coercion look like arithmetic and accommodation look like sentiment. The counter-realism reverses the charge. It is the accommodation, not the control, that the evidence at both scales actually supports.

It is not irresponsible to think about the present's responsibilities to the future. Longtermists are predictably late to realizing that the intersectional long term of the past might have some lessons about securing the existence of their people and a future for their children. [28] The work of mobilizing paleoclimate analogs, archaeological precedents, the centuries-long memory of dispersed and reorganized communities, and the billions-of-years evolutionary scale of bacterial complexity is exactly the kind of intersectional long-term thinking the moment requires.

Not all forms of long-term thinking are responsible. The *Lebensraum* tradition's claim to be thinking about the future has always been a claim to justify coercion in the present, and the futurity the tradition invokes has always been a future the present's coerced victims cannot consent to. A reader should be skeptical of any framework that promises a *Lebensraum* future on the grounds that present chauvinist coercion is the only realistic path to securing it. The future cannot answer for the present's violence that produces dispersals, arrivals, and recombination. The intersectional long term the immune-memory traditions transmit is a different kind of long-term thinking. It treats the past and the future as multiple and the present as the moment when the choice among possible recombinations is made. Boulder is where the arithmetic of concern was first taught, and Boulder is where the choice, like the bottle, is still open.

Notes

Notes for Chapter 1

1. [Olmsted, 1910](#)
2. [?](#), 4
3. [Colorado Encyclopedia](#)
4. [Olmsted, 1910](#), 65
5. [?](#)
6. [Goldberg, 1981](#)
7. [Ngai, 2004](#)
8. [Corson, 1997](#)
9. [Corson, 1997](#)
10. [?](#), 394–95
11. [Corson, 1997](#)
12. [Corson, 1997](#)
13. [Sargent, 2023](#)
14. [PLAN-Boulder County](#)
15. [?](#)
16. [?](#)
17. [Porter, 1995](#)
18. [Kolankiewicz, 2013](#); [Elbel, 2013](#)
19. [Robertson, 2012](#)
20. [Connelly, 2010](#)

21. Normandin and Valles, 2015
22. Bracke and Aguilar, 2023
23. Porter, 1995
24. Griffin, 2018

Notes for Chapter 2

1. Allen, 1986
2. Allen, 1986
3. Davenport, 1911
4. Laughlin, 1922
5. Lombardo, 2008
6. Stern, 2005; Kline, 2001
7. Gosney and Popenoe, 1929
8. Grant, 1916
9. Spiro, 2009
10. Hodgson, 1991
11. Ramsden, 2002
12. Ramsden, 2003
13. Ramsden, 2009
14. Osborn, 1940
15. Sear, 2021
16. Smith, 1980
17. Ratzel, 1901
18. ?
19. Abrahamsson, 2013
20. Marklund, 2015
21. Chiantera-Stutte, 2023
22. Chiantera-Stutte, 2023
23. Allen, 2013
24. Spiro, 2009
25. Rollins, 1999
26. Allen, 2013

27. [Turner, 1893](#)
28. [Roosevelt, 1889](#)
29. [Madley, 2016](#)
30. [Grant, 1916](#)
31. [Kakel, 2011](#)
32. [Westermann, 2016](#)
33. [Wolfe, 2006](#)
34. [Dunbar-Ortiz, 2014](#)
35. [Murphy, 1997](#)
36. [Murphy, 1997](#)
37. [Barnes and Minca, 2013](#); [Elden, 2006](#)
38. [Griffin, 2018](#)
39. [Campion, 2023](#)
40. [Conversi, 2024](#)
41. [Hughes et al., 2022](#)
42. [Simpson and Cheever, 2025](#)
43. [Taylor, 2002](#)
44. [Jasanoff and Kim, 2015](#)
45. [Hendriks et al., 2025](#)
46. [Connelly, 2010](#)
47. [Connelly, 2010](#)
48. [Connelly, 2010](#)
49. [Mass, 1974](#)
50. [Greenhalgh, 2012](#)
51. [Vogt, 1948](#)
52. [McCann, 2016](#)
53. [McCann, 2016](#)
54. [Sasser, 2014](#); [Dyett and Thomas, 2019](#)
55. [Meadows, 1974](#)
56. [Porter, 1995](#)
57. [Hardin, 1968](#)
58. [Hardin, 1974](#)

59. Paddock and Paddock, 1967
60. Chavez-Garcia, 2023
61. Robertson, 2012; Connelly, 2010
62. Notestein, 1970
63. Davis, 1973
64. Ryder, 1973
65. Mamdani, 1976
66. Park, 1974
67. Hofsten, 1975
68. Chamberlain, 1971
69. Mccann, 2009
70. Dyett and Thomas, 2019
71. Hartmann, 2016
72. Porter, 1995

Notes for Chapter 3

1. <https://www.colorado.edu/asmagazine/2016/12/06/physics-profs-home-man-himself-now-icon>
2. [Bartlett, 2006a](#)
3. [Bartlett, 1976](#)
4. [Porter, 1995](#)
5. [Zwietering et al., 1990](#)
6. ?
7. [Porter, 1995](#)
8. [Desrosières, 1998](#)
9. Annual growth-rate estimates derived from the Maddison Project Database 2023 (world, pre-1950) and UN World Population Prospects 2024 (1950 onward); U.S. counts and projections from the U.S. Census Bureau (2020 apportionment series, 1910–2020; 2023 National Population Projections); computation in `notebooks/ch02_bartlett_growth_models.ipynb`.
10. [Bartlett, 1978](#)
11. [United Nations Department of Economic and Social Affairs, Population Division, 1969](#)
12. [Desrosières, 1998](#)
13. Gompertz's original 1825 derivation ([Gompertz, 1825](#)) was a mortality function, but the curve's S-shape with asymptotic saturation had been imported into population and growth modeling well before Bartlett's lecture debut.
14. Fits and projections in `notebooks/ch02_bartlett_growth_models.ipynb`; U.S. series from the Maddison Project Database 2023, UN World Population Prospects 2024, and the U.S. Census Bureau (decennial counts and 2023 projection).
15. [Notestein, 1945](#)
16. [Coale and Hoover, 1958](#)
17. [Kirk, 1996](#)
18. [United Nations, Department of Economic and Social Affairs, Population Division, 2024](#)
19. [Bartlett and Lytwak, 1995b](#)
20. [United Nations, Department of Economic and Social Affairs, Population Division, 2024](#); [Riahi et al., 2017](#)
21. [Venuti, 1995](#)
22. [Bartlett, 1995](#)
23. [Bartlett, 1994](#)
24. [Bartlett, 2004a](#)
25. [Nachman, 2005](#)

26. [Bartlett, 1998](#)
27. [Bartlett, 1996](#)
28. [Bartlett, 2004b](#)
29. [Sayre, 2008](#)
30. [?](#)
31. [Bhatia, 2003](#)
32. [Coole, 2021](#)
33. [Bartlett and Lytwak, 1995b](#)
34. [Bartlett and Lytwak, 1995a](#)
35. [Daily et al., 1995](#)
36. [Lytwak and Bartlett, 1995](#)
37. [Bartlett and Lytwak, 1995b](#)
38. [Daily et al., 1995](#)
39. [Lytwak and Bartlett, 1995](#)
40. [Lytwak and Bartlett, 1995](#)
41. [?](#)
42. [Lytwak and Bartlett, 1995](#)
43. [Normandin and Valles, 2015](#); [Elcioglu, 2025](#)
44. [Kolankiewicz, 2013](#)
45. [Elbel, 2013](#)
46. [Dyett and Thomas, 2019](#)
47. [Bartlett, 2010](#)

Notes for Chapter 4

1. Connelly, 2010
2. United Nations, 1994
3. Hartmann, 2016
4. Hendrixson et al., 2020a
5. Ehrlich, 1972; Meadows, 1974
6. Hartmann, 2016
7. Connelly, 2010
8. Sasser, 2014
9. Normandin and Valles, 2015
10. Coole, 2021
11. Connelly, 2010; Hartmann, 2016
12. Daily et al., 1995
13. Coole, 2021
14. Porter, 1995
15. Normandin and Valles, 2015
16. Ehrlich, 1972
17. Normandin and Valles, 2015; , 2002
18. Normandin and Valles, 2015
19. Hultgren, 2015
20. Binkowski, 2022
21. Tucker, 1994
22. Beirich, 2008;
23. Beirich, 2008
24. Bhatia, 2003
25. Beck and Kolankiewicz, 2001
26. Bartlett and Lytwak, 1995a
27. Normandin and Valles, 2015; Sohoni, 2006
28. Porter, 1995
29. Sohoni, 2017
30. Barrett and Kurzman, 2004
31. Elcioglu, 2025

32. [Beirich, 2007](#)
33. [Huddle, 1993a](#)
34. [Fix and Passel, 1994](#)
35. [Sohoni, 2017](#)
36. [Desrosières, 1998](#)
37. [Smith and Edmonston, 1997](#)
38. [Sohoni, 2017](#)
39. [Porter, 1995](#)
40. [Power, 1997](#)
41. [Strathern, 2000](#)
42. [Bowker and Star, 1999](#)
43. [Sohoni, 2017](#)
44. [Bonilla-Silva, 2003](#)
45. [Mayorga-Gallo, 2015](#)
46. [Huang, 2008](#)
47. [Bhatia, 2003](#)
48. [Hultgren, 2015](#)
49. [Beirich, 2008](#)
50. [Fairchild, 1991](#)
51. [Benjamin, 2019](#)
52. [Bowker and Star, 1999](#)
53. [Zuckerman, 1999](#)
54. [Elcioglu, 2025](#); [King, 2008](#)
55. [Barringer, 2004](#)
56. [Normandin and Valles, 2015](#)
57. [Proctor and Schiebinger, 2008](#)
58. [Oreskes and Conway, 2011](#)
59. [Arora-Jonsson and Wahlström, 2023](#)
60. [Herrnstein and Murray, 1994](#)
61. [Tucker, 1994](#)
62. [Sohoni, 2017](#)
63. [Huddle, 1993b](#)

64. [Huddle, 1993b](#)
65. Aggregate omitted-category figures and per-line dollar values from the National Research Council's reconstruction in [National Research Council \(1997\)](#), Tables 6.1–6.6; replication tables in `notebooks/ch03_immigration_impact_statement.ipynb`.
66. [National Research Council, 1997](#)
67. Range derived from the National Academies' 2017 update ([National Academies of Sciences, Engineering, and Medicine, 2017](#)), Tables 8-1 through 8-12, which replicate the longitudinal framework with post-1997 data; see also ([Peri, 2012](#)) for the labor-market effects component.
68. Crossover horizons are sensitive to the immigrant cohort's average age at arrival, education, and country of origin; see ([National Academies of Sciences, Engineering, and Medicine, 2017](#)), Chapter 9.
69. [National Research Council, 1997](#)
70. Lifetime decomposition methodology in ([National Research Council, 1997](#)), Tables 6.4–6.6, with updated post-2010 estimates in ([National Academies of Sciences, Engineering, and Medicine, 2017](#)), Table 8-12.
71. [Smith and Edmonston, 1997](#)
72. [Power, 1997](#)
73. [Abernethy, 1997](#)
74. [MacDonald, 2016](#)
75. [Brimelow, 1995](#)
76. [Beirich, 2008](#)
77. [Beirich, 2007](#)
78. [Beirich, 2008](#)
79. [Sohoni, 2022](#)
80. [Hogan et al., 2014–2015](#)
81. [Farr, 2021](#); [Craig et al., 2018](#)

Notes for Chapter 5

1. ?
2. ?
3. Bowker and Star, 1999
4. ?
5. Frey, 2008
6. Sohoni, 2022
7. Craig and Richeson, 2014
8. Porter, 1995
9. Desrosières, 1998
10. Bowker and Star, 1999
11. ?
12. Porter, 1995
13. Sohoni, 2022
14. ?
15. Alba, 2020
16. Preston et al., 2001
17. Bowker and Star, 1999
18. Alba, 2020
19. ?
20. Craig and Richeson, 2014
21. Noble, 2018
22. Benjamin, 2019
23. Data & Society Research Institute, 2022
24. Weiner and Teitelbaum, 2001
25. Frey, 2008
26. Frey, 2018
27. Sohoni, 2017
28. ?
29. Richards et al., 2022
30. Ekman, 2022
31. Camus, 2011

32. ?
33. [Hughes et al., 2022](#)
34. [Stanley, 2020](#)
35. [Rueda, 2020](#)
36. ?
37. [Stanley, 2020](#)
38. [Biehl and Staudenmaier, 1995](#)
39. [Thomas and Gosink, 2021](#)
40. [Conversi, 2024](#)
41. [Ahmann and Oguz, 2025](#)
42. ?
43. ?
44. [Griffin, 2018](#)
45. ?
46. [Williams, 2025](#)
47. [Henderson et al., 2025](#)
48. [Simpson and Cheever, 2025](#)
49. [Bracke and Aguilar, 2023](#)
50. [Ortman and Hogan, 2009](#)
51. Sensitivity-analysis tables in `notebooks/ch04_majority_minority_crossover.ipynb`, cell `analyze_definition_sensitivity`; methodology consistent with ([Alba, 2018](#)).
52. [Alba, 2018, 2020](#)
53. [Levy and Myers, 2018](#)
54. 2020 multiracial-population growth figures from Census Bureau decennial release tables P1 and P2; Bureau retirement of the majority-minority framing in 2023 in ?.
55. ?
56. ?
57. ?
58. [Alba, 2020](#)
59. [Myers and Levy, 2018](#)
60. [Myers and Levy, 2018](#)
61. [King, 2002](#)
62. ?

Notes for Chapter 6

1. ?
2. ?
3. Porter, 1995
4. ?
5. Porter, 1995
6. Greenhalgh, 2012
7. Hartmann, 2016
8. Power, 1997
9. Murillo, 2025a
10. Murillo, 2025b
11. Data & Society Research Institute, 2022
12. Desrosières, 1998
13. Bracke and Aguilar, 2023
14. Bartlett, 2010
15. ?
16. Vesoulis, 2023
17. ?
18. King, 2002
19. Bracke and Aguilar, 2023
20. ?
21. King, 2002
22. Vesoulis, 2023
23. ?
24. Merchant, 2025
25. Martuscelli, 2023
26. Perry and Grubbs, 2025
27. Perry et al., 2022
28. Perry et al., 2022
29. Schultz, 2019
30. Hendrixson et al., 2020a
31. Bhatia et al., 2020

32. [Coole, 2013](#)
33. [Greenhalgh, 2012](#)
34. [Roberts, 1997](#)
35. [Huang, 2008](#)
36. [Dyett and Thomas, 2019](#)
37. [Younis, 2026](#)
38. ?
39. [Berman, 2022](#)
40. [Schultz, 2019](#)
41. [Porter, 1995](#)
42. [Power, 1997](#)
43. [Scott, 1998](#)
44. [Eubanks, 2018](#)
45. [Greenhalgh, 2012](#)
46. [Power, 1997](#)
47. [Data & Society Research Institute, 2022](#)
48. [Desrosières, 1998](#)
49. [Keilman, 2020](#)
50. ?
51. [Schultz, 2019](#)
52. ?
53. [Porter, 1995](#)
54. [Beckert, 2013](#)
55. [Esguerra, 2019](#)
56. [Bracke and Aguilar, 2023](#)
57. [Dyett and Thomas, 2019](#)
58. [Murillo, 2025a](#)
59. [Henderson et al., 2025](#)
60. [Bhatia and Hendrixson, 2026](#)
61. [Hartmann, 2016](#)
62. [Connelly, 2010](#)
63. [Gebru and Torres, 2024](#)

64. Range based on Guttmacher Institute estimates of unintended-pregnancy carry-to-term rates and the post-*Dobbs* differential in state-level abortion access; computation in `notebooks/ch05_portfolio_stress` cell `analyze_abortion_immigration`.
65. [Balinski and Young, 1982](#)
66. [Balinski and Young, 1982](#)
67. Forecast-error envelopes derived from comparison of past Census Bureau projection vintages (NP1992, NP2000, NP2008) against actual decennial counts; computation in `notebooks/ch05_portfolio_stress` cell `analyze_forecast_errors`.
68. [Greenhalgh, 2012](#)
69. [Porter, 1995](#)
70. [Eubanks, 2018](#)
71. [Hartmann, 2016](#)
72. [Data & Society Research Institute, 2022](#)
73. [Hendrixson et al., 2020a](#)
74. [Sen, 1997](#)
75. [Fung, 2015](#)
76. [Taylor, 2017a](#)
77. [Baldwin et al., 2019](#)
78. [Siders, 2019](#)

Notes for Chapter 7

1. ?
2. [Meadows, 1974](#)
3. ?
4. [Homer-Dixon, 1999](#)
5. ?
6. ?
7. ?
8. ?
9. [Riahi et al., 2017](#)
10. ?
11. ?
12. [Lawrence et al., 2024](#)
13. [Rockström et al., 2009](#)
14. [Folke, 2006](#)
15. [Renn, 2008](#)
16. ?
17. [Kemp, 2025](#); [Turchin, 2025](#)
18. [Scranton, 2015](#)
19. [Schwartzstein, 2024](#)
20. [Bettini, 2013, 2017](#); [Bettini et al., 2017](#)
21. [Andreucci and Zografos, 2022](#)
22. [Turner and Bailey, 2022](#)
23. [Hultgren, 2012, 2015](#)
24. [Bartlett, 2004b](#)
25. [Bartlett, 2006b](#)
26. [Bartlett, 2010](#)
27. [Homer-Dixon, 1999](#)
28. [Lenton et al., 2008](#)
29. [Porter, 1995](#)
30. [Power, 1997](#)
31. [Muiderman et al., 2020](#); [Granjou et al., 2017](#); [Esguerra, 2019](#)

32. [Esguerra, 2019](#)
33. [Eubanks, 2018; Noble, 2018; Benjamin, 2019](#)
34. [Elish, 2019](#)
35. [Eubanks, 2018; Noble, 2018; Benjamin, 2019](#)
36. [Eubanks, 2018](#)
37. [Benjamin, 2019](#)
38. [Noble, 2018](#)
39. [Pasquale, 2015](#)
40. [Birhane, 2021](#)
41. [Selbst et al., 2019](#)
42. [Molnar, 2024](#)
43. [Crawford, 2021](#)
44. [MacAskill, 2022; Ord, 2020](#)
45. [Gebru and Torres, 2024](#)
46. [Malm and Carton, 2024](#)
47. [Turner and Bailey, 2022](#)
48. [Hardin, 1974](#)
49. [Næss, 2004](#)
50. [Biehl and Staudenmaier, 1995](#)
51. [Næss, 2004](#)
52. [Robinson, 2020](#)
53. [Haraway, 2016](#)
54. [Gordon, 1997](#)
55. [Muiderman et al., 2020; Granjou et al., 2017](#)
56. [Beckert, 2013](#)
57. [Garforth, 2019](#)
58. [Gebru and Torres, 2024](#)
59. [Eubanks, 2018; Noble, 2018; Benjamin, 2019](#)
60. [Ramsden, 2002](#)

Notes for Chapter 8

1. Something
2. Carton and Malm, 2026
3. Ghosh, 2016
4. Rich, 2019
5. MacAskill, 2022; Ord, 2020
6. Gebru and Torres, 2024
7. Stern, 2007; Nordhaus, 1992
8. Riahi et al., 2017
9. Malm and Carton, 2024
10. Hansen, 2009
11. Wang, 2025
12. Hawken, 2021
13. Martin, 2022
14. Brand, 1999
15. Weiss, 1989
16. Lent, 2021
17. Wallace-Wells, 2019; Scranton, 2015
18. Kemp, 2025; Turchin, 2025
19. Woods, 1998
20. McKittrick, 2013
21. Roane and Hosbey, 2019
22. Butler, 1993
23. Womack, 2013
24. Gilmore, 2007, 2022
25. Hartman, 2007
26. Sharpe, 2016
27. Bullard, 1990; Pellow and Park, 2017
28. Whyte et al., 2022
29. Kimmerer, 2013
30. Watts, 2013
31. Estes, 2019

32. [Carroll et al., 2020](#); [Lewis, 2020](#)
33. [Diamond, 2005](#)
34. [Whyte et al., 2022](#)
35. [Koch et al., 2019](#)
36. [Knaut, 1995](#)
37. [Cline, 2014](#)
38. [Gudynas, 2011](#)
39. [Mhlambi, 2020](#)
40. [Escobar, 2018](#)
41. [Sins Invalid, 2019](#)
42. [Pellow and Park, 2017](#)
43. [Kafer, 2013](#)
44. [Piepzna-Samarasinha, 2018](#)
45. [Wong, 2020](#)
46. [Clare, 2017](#)
47. [Taylor, 2017b](#)
48. [Hamraie, 2017](#)
49. [Federici, 2004](#)
50. [Ross and Solinger, 2017](#)
51. [Hartmann, 2016](#); [Hendrixson et al., 2020b](#)
52. [Merchant, 1980](#)
53. [Shiva, 1988](#)
54. [Mies, 1986](#)
55. [Plumwood, 1993](#)
56. [Fraser, 2022](#)
57. [Bhattacharya, 2017](#)
58. [Tronto, 2013](#)
59. [Haraway, 1988](#)
60. [Collins, 1990](#); [Crenshaw, 1991](#)
61. [Moore, 2016](#); [Johnson and Wilkinson, 2020](#)
62. [Bray, 2017](#)
63. [Scholz and Schneider, 2016](#)

64. [Spade, 2020](#); [Solnit, 2009](#)
65. [Moore, 2015](#)
66. [Saito, 2023, 2017, 2024](#)
67. [Bookchin, 1982](#)
68. [Malm and Collective, 2021](#)
69. [Harris, 2022](#); [Tilley and Ajl, 2023](#)
70. [Lee and Tuljapurkar, 1994](#); [Carroll et al., 2020](#); [Sen, 1999](#)
71. [Scott, 1998](#)
72. [Liboiron, 2021](#)
73. [Berman, 2022](#)
74. [Carroll et al., 2020](#)

Notes for Chapter 9

1. Byars, 2013
2. Carton and Malm, 2026
3. Bartlett, 1976
4. Barrangou et al., 2007
5. Cline, 2014
6. ?
7. Koch et al., 2019
8. Kelman, 2013
9. Schwartzstein, 2024
10. Kelman, 2013
11. Mann, 2005; Koch et al., 2019
12. Hage and Marcotty, 2025
13. Nye, 2021
14. Burke et al., 2018
15. Riahi et al., 2017
16. Bessire, 2021
17. Scott, 2025
18. Beever et al., 2003
19. , 2021
20. Martin, 2022
21. Holl, 2020
22. Jepson and Blythe, 2020; Monbiot, 2013
23. Alagona, 2022
24. Sheldrake, 2020
25. Bittle, 2023; Lustgarten, 2024
26. Vince, 2022
27. Wallace-Wells, 2019
28. This is a conscious reference to white supremacists' Fourteen Words.

Bibliography

- Abernethy, Virginia Deane. 1997. Editorial: Political Science. *Population and Environment* 19 (2): 113–117.
- Abrahamsson, C. 2013. On the genealogy of Lebensraum. *Geographica Helvetica* 68 (1): 37–44.
- Ahmann, Chloe, and Zeynep Oguz. 2025. Collection introduction: Everyday and emergent ecofascisms. *Environment and Planning D: Society and Space*.
- Alagona, Peter S. 2022. *The Accidental Ecosystem: People and Wildlife in American Cities*. Berkeley: University of California Press.
- Alba, Richard. 2018. What Majority-Minority Society? A Critical Analysis of the Census Bureau's Projections of America's Demographic Future. *Socius: Sociological Research for a Dynamic World* 4: 1–10. doi:10.1177/2378023118796932.
- Alba, Richard. 2020. *The Great Demographic Illusion: Majority, Minority, and the Expanding American Mainstream*. Princeton, NJ: Princeton University Press.
- Allen, Garland E. 1986. The Eugenics Record Office at Cold Spring Harbor, 1910–1940: An Essay in Institutional History. *Osiris* 2: 225–264.
- Allen, Garland E. 2013. "Culling the Herd": Eugenics and the Conservation Movement in the United States, 1900–1940. *Journal of the History of Biology* 46 (1): 31–72.
- Andreucci, Diego, and Christos Zografos. 2022. Between improvement and sacrifice: Othering and the (bio)political ecology of climate change. *Political Geography* 92: 102512.
- Arora-Jonsson, Seema, and Nora Wahlström. 2023. Unraveling the production of ignorance in climate policymaking: The imperative of a decolonial feminist intervention for transformation. *Environmental Science & Policy* 149: 103564.
- Baldwin, Andrew, Christiane Fröhlich, and Delf Rothe. 2019. From climate migration to anthropocene mobilities: Shifting the debate. *Mobilities* 14 (3): 289–297.
- Balinski, Michel L., and H. Peyton Young. 1982. *Fair Representation: Meeting the Ideal of One Man, One Vote*. New Haven, CT: Yale University Press.
- Barnes, Trevor J., and Claudio Minca. 2013. Nazi Spatial Theory: The Dark Geographies of Carl Schmitt and Walter Christaller. *Annals of the Association of American Geographers* 103 (3): 669–687.

- Barrangou, Rodolphe, Christophe Fremaux, Hélène Deveau, Melissa Richards, Patrick Boyaval, Sylvain Moineau, Dennis A. Romero, and Philippe Horvath. 2007. CRISPR provides acquired resistance against viruses in prokaryotes. *Science* 315 (5819): 1709–1712.
- Barrett, Deborah, and Charles Kurzman. 2004. Globalizing Social Movement Theory: The Case of Eugenics. *Theory and Society* 33 (5): 487–527.
- Barringer, Felicity. 2004. Bitter Division For Sierra Club On Immigration. *The New York Times*. Chap. U.S..
- Bartlett, Albert A. 1976. The exponential function—Part 1. *The Physics Teacher* 14 (7): 393–401.
- Bartlett, Albert A. 1978. Forgotten fundamentals of the energy crisis. *American Journal of Physics* 46 (9): 876–888.
- Bartlett, Albert A. 1994. Reflections on sustainability, population growth, and the environment. *Population and Environment* 16 (1): 5–35.
- Bartlett, Albert A. 1995. Arithmetic, Population and Energy: The Impossibility of Exponential Growth on a Finite Planet.
- Bartlett, Albert A. 1996. The Exponential Function, XI: The New Flat Earth Society. *The Physics Teacher* 34 (6): 342–343.
- Bartlett, Albert A. 1998. Malthus Marginalized. *The Social Contract*.
- Bartlett, Albert A. 2004a. Exponential recollections, 1st edn. In *The Essential Exponential! For The Future of Our Planet*. Lincoln, Nebraska, United States: Center for Science, Mathematics & Computer Education.
- Bartlett, Albert A. 2004b. Thoughts on Long-Term Energy Supplies: Scientists and the Silent Lie. *Physics Today* 57 (7): 53–55.
- Bartlett, Albert A. 2006a. Albert Bartlett's Interview. Voices of the Manhattan Project, Atomic Heritage Foundation and National Museum of Nuclear Science & History. Oral history; interview by Cindy Kelly. Year approximate.
- Bartlett, Albert A. 2006b. Scientific American and the Silent Lie. *The Physics Teacher* 44 (9): 623–624.
- Bartlett, Albert A. 2010. Al Bartlett on Population, Energy, and Sustainability. YouTube, <https://www.youtube.com/watch?v=F8ZJCtL6bPs>. TENTATIVE CITATION — to be definitively finalized at a later stage.
- Bartlett, Albert A., and Edward P. Lytwak. 1995a. Population Stability and Immigration. *The Social Contract* 6.
- Bartlett, Albert A., and Edward P. Lytwak. 1995b. Zero growth of the population of the United States. *Population and Environment* 16 (5): 415–428.
- Beck, Roy, and Leon Kolankiewicz. 2001. Forsaking Fundamentals, Technical report, Center for Immigration Studies.
- Beckert, Jens. 2013. Imagined futures: Fictional expectations in the economy. *Theory and Society* 42 (3): 219–240.
- Beever, Erik A., Peter F. Brussard, and Joel Berger. 2003. Patterns of apparent extirpation among isolated populations of pikas (*Ochotona princeps*) in the Great Basin. *Journal of Mammalogy* 84 (1): 37–54.

- Beirich, Heidi. 2007. Federation for American Immigration Reform's Hate Filled Track Record.
- Beirich, Heidi. 2008. John Tanton's Private Papers Expose More Than 20 Years of Hate.
- Benjamin, Ruha. 2019. *Race After Technology: Abolitionist Tools for the New Jim Code*, 1st edition edn. Medford, MA: Polity.
- Berman, Elizabeth Popp. 2022. *Thinking like an Economist: How Efficiency Replaced Equality in U.S. Public Policy*. Princeton, NJ: Princeton University Press.
- Bessire, Lucas. 2021. *Running Out: In Search of Water on the High Plains*. Princeton, NJ: Princeton University Press.
- Bettini, Giovanni. 2013. Climate Barbarians at the Gate? A critique of apocalyptic narratives on 'climate refugees'. *Geoforum* 45: 63–72.
- Bettini, Giovanni. 2017. Archaeologies of the future: Tracing the lineage of contemporary discourses on the climate–migration nexus. In *Environmental History of Modern Migrations*. Routledge.
- Bettini, Giovanni, Sarah Louise Nash, and Giovanna Gioli. 2017. One step forward, two steps back? The fading contours of (in)justice in competing discourses on climate migration. *The Geographical Journal* 183 (4): 348–358.
- Bhatia, Rajani. 2003. Green or Brown? White Nativist Environmental Movements. In *Home-Grown Hate*, 194–214. Routledge.
- Bhatia, Rajani, and Anne Hendrixson. 2026. Kindling green hate through eco- and demographic anxiety. *Environmental Politics* 35 (1): 129–149.
- Bhatia, Rajani, Jade S. Sasser, Diana Ojeda, Anne Hendrixson, Sarojini Nadimpally, and Ellen E. Foley. 2020. A feminist exploration of 'populationism': Engaging contemporary forms of population control. *Gender, Place & Culture* 27 (3): 333–350.
- Bhattacharya, Tithi, ed. 2017. *Social Reproduction Theory: Remapping Class, Recentering Oppression*. London: Pluto Press.
- Biehl, Janet, and Peter Staudenmaier. 1995. *Ecofascism: Lessons from the German Experience*. Edinburgh: AK Press.
- Binkowski, Brooke. 2022. Eugenics, Border Wars & Population Control: The Tanton Network. *Unicorn Riot*.
- Birhane, Abeba. 2021. Algorithmic injustice: A relational ethics approach. *Patterns* 2 (2): 100205.
- Bittle, Jake. 2023. *The Great Displacement: Climate Change and the Next American Migration*. New York: Simon & Schuster.
- Bonilla-Silva, Eduardo. 2003. *Racism without Racists: Color-Blind Racism and the Persistence of Racial Inequality in the United States*. Lanham, MD: Rowman & Littlefield.
- Bookchin, Murray. 1982. *The Ecology of Freedom: The Emergence and Dissolution of Hierarchy*. Palo Alto, CA: Cheshire Books.
- Bowker, Geoffrey C., and Susan Leigh Star. 1999. *Sorting things out: Classification and Its Consequences*. Cambridge, MA: MIT Press.
- Bracke, Sarah, and Luis Manuel Hernández Aguilar, eds. 2023. *The Politics of Replacement: Demographic Fears, Conspiracy Theories, and Race Wars*. London: Routledge.

- Brand, Stewart. 1999. *The Clock of the Long Now: Time and Responsibility*. New York: Basic Books.
- Bray, Mark. 2017. *Antifa: The Anti-Fascist Handbook*. Brooklyn, NY: Melville House.
- Brimelow, Peter. 1995. *Alien Nation: Common Sense about America's Immigration Disaster*. New York: Random House.
- Bullard, Robert D. 1990. *Dumping in Dixie: Race, Class, and Environmental Quality*. Boulder, CO: Westview Press.
- Burke, Kevin D., John W. Williams, Mark A. Chandler, Alan M. Haywood, Daniel J. Lunt, and Bette L. Otto-Bliesner. 2018. Pliocene and Eocene provide best analogs for near-future climates. *Proceedings of the National Academy of Sciences* 115 (52): 13288–13293.
- Butler, Octavia E. 1993. *Parable of the Sower*. New York: Four Walls Eight Windows.
- Byars, Mitchell. 2013. Boulder honors 'Big Al': Friends, family celebrate life of CU-Boulder professor Al Bartlett. *Boulder Daily Camera*.
- Campion, Kristy. 2023. Defining Ecofascism: Historical Foundations and Contemporary Interpretations in the Extreme Right. *Terrorism and Political Violence* 35 (4): 926–944.
- Camus, Renaud. 2011. *Le Grand Remplacement*. Neuilly-sur-Seine: David Reinharc. AUTHOR REVIEW: French-language original; verify publisher and exact pagination. English-language scholarly discussions usually cite the French edition.
- Carroll, Stephanie Russo, Ibrahim Garba, Oscar L. Figueroa-Rodríguez, Jarita Holbrook, Raymond Lovett, Simeon Materechera, Meghan Parsons, Kay Raseroka, Desi Rodriguez-Lonebear, Robyn Rowe, Rodrigo Sara, Jennifer D. Walker, Jane Anderson, and Maui Hudson. 2020. The CARE Principles for Indigenous Data Governance. *Data Science Journal* 19 (1): 43.
- Carton, Wim, and Andreas Malm. 2026. *The Long Heat: Climate Politics When It's Too Late*. London: Verso. Year to confirm; source of the term “the long heat”.
- Chamberlain, William M. 1971. Population Control—the Legal Approach to a Biological Imperative. *Ecology Law Quarterly* 1 (1): 143–172.
- Chavez-Garcia, Mirosława. 2023. *Architects of Hate: Eugenists, Conservationists, and the Racist Origins of the Immigration Debate*. Oakland, CA: University of California Press.
- Chiantera-Stutte, Patricia. 2023. The State as a “Form of Life” and the Space as Leistungsraum: The reception of Ratzel in the First and Second World Wars. *Geographica Helvetica* 78. doi:10.5194/gh-78-29-2023.
- Clare, Eli. 2017. *Brilliant Imperfection: Grappling with Cure*. Durham, NC: Duke University Press.
- Cline, Eric H. 2014. *1177 B.C.: The Year Civilization Collapsed*. Princeton, NJ: Princeton University Press.
- Coale, Ansley J., and Edgar M. Hoover. 1958. *Population Growth and Economic Development in Low-Income Countries: A Case Study of India's Prospects*. Princeton, NJ: Princeton University Press.
- Collins, Patricia Hill. 1990. *Black Feminist Thought: Knowledge, Consciousness, and the Politics of Empowerment*. New York: Routledge.
- Colorado Encyclopedia. Colorado Chautauqua. <https://coloradoencyclopedia.org/article/colorado-chautauqua>. AUTHOR REVIEW: web encyclopedia entry cited in ch01 (Boulder Chautauqua history). Verify URL, retrieval date, exact title, and named author/editor.

- Connelly, Matthew. 2010. *Fatal Misconception: The Struggle to Control World Population*. Harvard.
- Conversi, Daniele. 2024. Eco-fascism: An oxymoron? Far-right nationalism, history, and the climate emergency. *Frontiers in Human Dynamics* 6.
- Coole, Diana. 2013. Too many bodies? The return and disavowal of the population question. *Environmental Politics* 22 (2): 195–215.
- Coole, Diana. 2021. The Toxification of Population Discourse. A Genealogical Study. *The Journal of Development Studies* 57 (9): 1454–1469.
- Corson, Christopher. 1997. Boulder's First Zoning Ordinance. Seminar paper, URP 6635, University of Colorado; Carnegie Branch Library for Local History, Boulder. First name and pagination to be confirmed..
- Craig, Maureen A., and Jennifer A. Richeson. 2014. On the Precipice of a “Majority–Minority” America: Perceived Status Threat From the Racial Demographic Shift Affects White Americans’ Political Ideology. *Psychological Science* 25 (6): 1189–1197.
- Craig, Maureen A., Julian M. Rucker, and Jennifer A. Richeson. 2018. Racial and Political Dynamics of an Approaching “Majority–Minority” United States. *The ANNALS of the American Academy of Political and Social Science* 677 (1): 204–214.
- Crawford, Kate. 2021. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. New Haven, CT: Yale University Press.
- Crenshaw, Kimberlé. 1991. Mapping the Margins: Intersectionality, Identity Politics, and Violence against Women of Color. *Stanford Law Review* 43 (6): 1241–1299.
- Daily, Gretchen C., Anne H. Ehrlich, and Paul R. Ehrlich. 1995. Response to Bartlett and Lytwak (1995): Population and immigration policy in the United States. *Population and Environment* 16 (6): 521–526.
- Data & Society Research Institute. 2022. Powerful Numbers: A Primer on Statistics for Democracy, Technical report, Data & Society Research Institute, New York.
- Davenport, Charles Benedict. 1911. *Heredity in Relation to Eugenics*. New York: Henry Holt and Company.
- Davis, Kingsley. 1973. Zero Population Growth: The Goal and the Means. *Daedalus* 102 (4): 15–30.
- Desrosières, Alain. 1998. *The Politics of Large Numbers: A History of Statistical Reasoning*. Harvard University Press.
- Diamond, Jared. 2005. *Collapse: How Societies Choose to Fail or Succeed*. New York: Viking.
- Dunbar-Ortiz, Roxanne. 2014. *An Indigenous Peoples’ History of the United States*. Boston, MA: Beacon Press.
- Dyett, Jordan, and Cassidy Thomas. 2019. Overpopulation Discourse: Patriarchy, Racism, and the Specter of Ecofascism. *Perspectives on Global Development and Technology* 18 (1–2): 205–224.
- Ehrlich, Paul R. 1972. *The Population Bomb*. Sierra Club.
- Ekman, Mattias. 2022. The great replacement: Strategic mainstreaming of far-right conspiracy claims. *Convergence: The International Journal of Research into New Media Technologies* 28 (4): 1127–1143.

- Elbel, Fred. 2013. The Social Contract – A Tribute to Al Bartlett (1923–2013). *The Social Contract* 24 (1): 42–43.
- Elcioglu, Emine Fidan. 2025. Seeding the divide: John Tanton, the Sierra Club and the struggle over US environmentalism. *Race & Class*.
- Elden, Stuart. 2006. National Socialism and the politics of calculation. *Social & Cultural Geography* 7 (5): 753–769.
- Elish, Madeleine Clare. 2019. Moral Crumple Zones: Cautionary Tales in Human-Robot Interaction. *Engaging Science, Technology, and Society* 5: 40–60. doi:10.17351/ests2019.260.
- Escobar, Arturo. 2018. *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*. Durham, NC: Duke University Press.
- Esguerra, Alejandro. 2019. Future objects: Tracing the socio-material politics of anticipation. *Sustainability Science* 14 (4): 963–971.
- Estes, Nick. 2019. *Our History Is the Future: Standing Rock versus the Dakota Access Pipeline, and the Long Tradition of Indigenous Resistance*. London: Verso.
- Eubanks, Virginia. 2018. *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*. New York: St. Martin's Press.
- Fairchild, Halford H. 1991. Scientific Racism: The Cloak of Objectivity. *Journal of Social Issues* 47 (3): 101–115.
- Farr, Brittany. 2021. A Demographic Moral Panic: Fears of a Majority-Minority Future and the Depreciating Value of Whiteness.
- Federici, Silvia. 2004. *Caliban and the Witch: Women, the Body and Primitive Accumulation*. New York: Autonomedia.
- Fix, Michael, and Jeffrey S. Passel. 1994. Immigration and Immigrants: Setting the Record Straight, Technical report, Urban Institute, Washington, DC. Metadata from standard bibliographic knowledge; not indexed in OpenAlex. Verify before publication..
- Folke, Carl. 2006. Resilience: The emergence of a perspective for social-ecological systems analyses. *Global Environmental Change* 16 (3): 253–267. doi:10.1016/j.gloenvcha.2006.04.002.
- Fraser, Nancy. 2022. *Cannibal Capitalism: How Our System Is Devouring Democracy, Care, and the Planet*. London: Verso.
- Frey, William H. 2008. America's Regional Demographics in the '00s Decade: The Role of Seniors, Boomers and New Minorities Washington, DC: Brookings Institution Metropolitan Policy Program.
- Frey, William H. 2018. *Diversity Explosion: How New Racial Demographics Are Remaking America*, Revised edn. Washington, DC: Brookings Institution Press.
- Fung, Archon. 2015. Putting the Public Back into Governance: The Challenges of Citizen Participation and Its Future. *Public Administration Review* 75 (4): 513–522.
- Garforth, Lisa. 2019. Environmental Futures, Now and Then: Crisis, Systems Modeling, and Speculative Fiction. *Osiris* 34 (1): 238–257.
- Gebru, Timnit, and Émile P. Torres. 2024. The TESCREAL bundle: Eugenics and the promise of utopia through artificial general intelligence. *First Monday*.

- Ghosh, Amitav. 2016. *The Great Derangement: Climate Change and the Unthinkable*. Chicago: University of Chicago Press.
- Gilmore, Ruth Wilson. 2007. *Golden Gulag: Prisons, Surplus, Crisis, and Opposition in Globalizing California*. Berkeley: University of California Press.
- Gilmore, Ruth Wilson. 2022. *Abolition Geography: Essays Towards Liberation*. London: Verso.
- Goldberg, Robert Alan. 1981. *Hooded Empire: The Ku Klux Klan in Colorado*. Urbana: University of Illinois Press.
- Gompertz, Benjamin. 1825. On the Nature of the Function Expressive of the Law of Human Mortality, and on a New Mode of Determining the Value of Life Contingencies. *Philosophical Transactions of the Royal Society of London* 115: 513–583. doi:10.1098/rstl.1825.0026.
- Gordon, Avery F. 1997. *Ghostly Matters: Haunting and the Sociological Imagination*. Minneapolis: University of Minnesota Press.
- Gosney, E. S., and Paul Popenoe. 1929. *Sterilization for Human Betterment: A Summary of Results of 6,000 Operations in California, 1909–1929*. New York: Macmillan.
- Granjou, Céline, Jeremy Walker, and Juan Francisco Salazar. 2017. The politics of anticipation: On knowing and governing environmental futures. *Futures* 92: 5–11.
- Grant, Madison. 1916. *The Passing of the Great Race; or, The Racial Basis of European History*. New York: Charles Scribner's Sons.
- Greenhalgh, Susan. 2012. On the Crafting of Population Knowledge. *Population and Development Review* 38 (1): 121–131.
- Griffin, Roger. 2018. *Fascism*. John Wiley & Sons.
- Gudynas, Eduardo. 2011. Buen Vivir: Today's tomorrow. *Development* 54 (4): 441–447.
- Hage, Dave, and Josephine Marcotty. 2025. *Sea of Grass: The Conquest, Ruin, and Redemption of Nature on the American Prairie*. New York: Random House.
- Hamraie, Aimi. 2017. *Building Access: Universal Design and the Politics of Disability*. Minneapolis: University of Minnesota Press.
- Hansen, James. 2009. *Storms of My Grandchildren: The Truth about the Coming Climate Catastrophe and Our Last Chance to Save Humanity*. New York: Bloomsbury.
- Haraway, Donna. 1988. Situated knowledges: The science question in feminism and the privilege of partial perspective. *Feminist Studies* 14 (3): 575–599.
- Haraway, Donna J. 2016. *Staying with the Trouble: Making Kin in the Chthulucene*. Durham, NC: Duke University Press.
- Hardin, Garrett. 1968. The tragedy of the commons. *Science* 162 (3859): 1243–1248.
- Hardin, Garrett. 1974. Lifeboat Ethics: The Case Against Helping the Poor. *Psychology Today* 8 (4): 38–43. AUTHOR REVIEW: Psychology Today issue/page metadata not confirmed via OpenAlex (Psychology Today is not indexed). The closely related “Living on a Lifeboat,” *BioScience* 24(10):561–568 (DOI 10.2307/1296629) is sometimes cited interchangeably.
- Harris, Jerry. 2022. Our Future: Ecosocialism or Ecofascism. *The International Journal of Interdisciplinary Global Studies* 17 (2): 35–47.

- Hartman, Saidiya. 2007. *Lose Your Mother: A Journey Along the Atlantic Slave Route*. New York: Farrar, Straus and Giroux.
- Hartmann, Betsy. 2016. *Reproductive Rights and Wrongs: The Global Politics of Population Control*. Haymarket Books.
- Hawken, Paul. 2021. *Regeneration: Ending the Climate Crisis in One Generation*. New York: Penguin Books.
- Henderson, Jane, Bruno Seraphin, April Anson, Alex Menrisky, Shane Hall, and Cassie Galentine. 2025. How Ecofascism Creeps. *American Quarterly* 77 (1): 181–188.
- Hendriks, Abe, Kamilla Karhunmaa, and Pierre Delvenne. 2025. Shaping the future: A conceptual review of sociotechnical imaginaries. *Futures* 170: 103607.
- Hendrixson, Anne, Diana Ojeda, Jade S. Sasser, Sarojini Nadimpally, Ellen E. Foley, and Rajani Bhatia. 2020a. Confronting populationism: Feminist challenges to population control in an era of climate change. *Gender, Place & Culture* 27 (3): 307–315.
- Hendrixson, Anne, Diana Ojeda, Jade S. Sasser, Sarojini Nadimpally, Ellen E. Foley, and Rajani Bhatia. 2020b. Confronting populationism: Feminist challenges to population control in an era of climate change. *Gender, Place & Culture* 27 (3): 307–315.
- Herrnstein, Richard J., and Charles Murray. 1994. *The Bell Curve: Intelligence and Class Structure in American Life*. New York: Free Press.
- Hodgson, Dennis. 1991. The Ideological Origins of the Population Association of America. *Population and Development Review* 17 (1): 1–34.
- Hofsten, Erland. 1975. Population Growth—Menace to What? *International Journal of Health Services* 5 (3): 417–424.
- Hogan, Howard, Jennifer M. Ortman, and Sandra L. Colby. 2014–2015. Projecting Diversity: The Methods, Results, Assumptions and Limitations of the U.S. Census Bureau's Population Projections. *West Virginia Law Review* 117 (3): 1047–1080.
- Holl, Karen D. 2020. *Primer of Ecological Restoration*. Washington, DC: Island Press.
- Homer-Dixon, Thomas F. 1999. *Environment, Scarcity, and Violence*. Princeton, NJ: Princeton University Press.
- Huang, Priscilla. 2008. Anchor Babies, Over-Breeders, and the Population Bomb: The Reemergence of Nativism and Population Control in Anti-Immigration Policies. *Harvard Law and Policy Review* 2.
- Huddle, Donald. 1993a. The Costs of Immigration, Technical report, Carrying Capacity Network, Washington, DC.
- Huddle, Donald L. 1993b. The Costs of Immigrants to American Taxpayers Washington, DC: Carrying Capacity Network. Reissued and updated in subsequent CCN bulletins; circulated as the foundational document for the FAIR-network impact-statement genre..
- Hughes, Brian, Dave Jones, and Amarnath Amarasingam. 2022. Ecofascism: An Examination of the Far-Right/Ecology Nexus in the Online Space. *Terrorism and Political Violence* 34 (5): 997–1023.
- Hultgren, John. 2012. Natural Exceptions to Green Sovereignty? American Environmentalism and the "Immigration Problem". *Alternatives: Global, Local, Political* 37 (4): 300–316.

- Hultgren, John. 2015. *Border Walls Gone Green: Nature and Anti-immigrant Politics in America*. U of Minnesota Press.
- Jasanoff, Sheila, and Sang-Hyun Kim, eds. 2015. *Dreamscapes of Modernity: Sociotechnical Imaginaries and the Fabrication of Power*. Chicago: University of Chicago Press.
- Jepson, Paul, and Cain Blythe. 2020. *Rewilding: The Radical New Science of Ecological Recovery*. London: Icon Books.
- Johnson, Ayana Elizabeth, and Katharine K. Wilkinson, eds. 2020. *All We Can Save: Truth, Courage, and Solutions for the Climate Crisis*. New York: One World.
- Kafer, Alison. 2013. *Feminist, Queer, Crip*. Bloomington: Indiana University Press.
- Kakel, Carroll P. 2011. *The American West and the Nazi East: A Comparative and Interpretive Perspective*. New York: Palgrave Macmillan.
- Keilman, Nico. 2020. Uncertainty in Population Forecasts for the Twenty-First Century. *Annual Review of Resource Economics* 12: 449–470.
- Kelman, Ari. 2013. *A Misplaced Massacre: Struggling over the Memory of Sand Creek*. Cambridge, MA: Harvard University Press.
- Kemp, Luke. 2025. *Goliath's Curse: The History and Future of Societal Collapse*. New York: Viking.
- Kimmerer, Robin Wall. 2013. *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge, and the Teachings of Plants*. Minneapolis: Milkweed Editions.
- King, Leslie. 2002. Demographic trends, pronatalism, and nationalist ideologies in the late twentieth century. *Ethnic and Racial Studies* 25 (3): 367–389.
- King, Leslie. 2008. Ideology, Strategy and Conflict in a Social Movement Organization: The Sierra Club Immigration Wars. *Mobilization: An International Quarterly* 13 (1): 45–61.
- Kirk, Dudley. 1996. Demographic Transition Theory. *Population Studies* 50 (3): 361–387.
- Kline, Wendy. 2001. *Building a Better Race: Gender, Sexuality, and Eugenics from the Turn of the Century to the Baby Boom*. Berkeley, CA: University of California Press.
- Knaut, Andrew L. 1995. *The Pueblo Revolt of 1680: Conquest and Resistance in Seventeenth-Century New Mexico*. Norman: University of Oklahoma Press.
- Koch, Alexander, Chris Brierley, Mark M. Maslin, and Simon L. Lewis. 2019. Earth system impacts of the european arrival and Great Dying in the Americas after 1492. *Quaternary Science Reviews* 207: 13–36.
- Kolankiewicz, Leon. 2013. Remembering Dr. Al Bartlett. *The Social Contract* 24 (1): 36–41.
- Laughlin, Harry H. 1922. *Eugenical Sterilization in the United States*. Chicago: Psychopathic Laboratory of the Municipal Court of Chicago.
- Lawrence, Michael, Thomas Homer-Dixon, Scott Janzwood, Johan Rockström, Ortwin Renn, and Jonathan F. Donges. 2024. Global polycrisis: The causal mechanisms of crisis entanglement. *Global Sustainability* 7: 6. doi:10.1017/sus.2024.1.
- Lee, Ronald D., and Shripad Tuljapurkar. 1994. Stochastic Population Forecasts for the United States: Beyond High, Medium, and Low. *Journal of the American Statistical Association* 89 (428): 1175–1189.

- Lent, Jeremy. 2021. *The Web of Meaning: Integrating Science and Traditional Wisdom to Find Our Place in the Universe*. Gabriola Island, BC: New Society Publishers.
- Lenton, Timothy M., Hermann Held, Elmar Kriegler, Jim W. Hall, Wolfgang Lucht, Stefan Rahmstorf, and Hans Joachim Schellnhuber. 2008. Tipping elements in the Earth's climate system. *Proceedings of the National Academy of Sciences* 105 (6): 1786–1793. doi:10.1073/pnas.0705414105.
- Levy, Morris E., and Dowell Myers. 2018. Reconstructing the Racial Demographic Frame: Public Reaction to Census Bureau Projections of a Majority-Minority America. *Public Opinion Quarterly* 82 (4): 743–758. doi:10.1093/poq/nfy052.
- Liboiron, Max. 2021. *Pollution Is Colonialism*. Durham, NC: Duke University Press. doi:10.1215/9781478021445.
- Lombardo, Paul A. 2008. *Three Generations, No Imbeciles: Eugenics, the Supreme Court, and Buck v. Bell*. Baltimore, MD: Johns Hopkins University Press.
- Lustgarten, Abrahm. 2024. *On the Move: The Overheating Earth and the Uprooting of America*. New York: Farrar, Straus and Giroux.
- Lytwak, Edward P., and Albert A. Bartlett. 1995. Rejoinder to daily, Ehrlich, and Ehrlich: Immigration and population policy in the United States. *Population and Environment* 16 (6): 527–537.
- MacAskill, William. 2022. *What we Owe the Future*. New York: Basic Books.
- MacDonald, Kevin B. 2016. Kevin MacDonald on the David Duke Show.
- Madley, Benjamin. 2016. *An American Genocide: The United States and the California Indian Catastrophe, 1846–1873*. New Haven, CT: Yale University Press.
- Malm, Andreas, and Wim Carton. 2024. *Overshoot: How the World Surrendered to Climate Breakdown*. London: Verso.
- Malm, Andreas, and The Zetkin Collective. 2021. *White Skin, Black Fuel: On the Danger of Fossil Fascism*. Verso Books.
- Mamdani, Mahmood. 1976. The Ideology of Population Control. *Economic and Political Weekly* 11 (31/33): 1141–1148.
- Mann, Charles C. 2005. *1491: New Revelations of the Americas Before Columbus*. New York: Alfred A. Knopf.
- Marklund, Carl. 2015. The Return of Geopolitics in the Era of Soft Power: Rereading Rudolf Kjellén on Geopolitical Imaginary and Competitive Identity. *Geopolitics* 20 (2): 248–266.
- Martin, Laura J. 2022. *Wild by Design: The Rise of Ecological Restoration*. Cambridge, MA: Harvard University Press.
- Martuscelli, Carlo. 2023. The populist right wants you to make more babies. The question is how. *Politico*.
- Mass, Bonnie. 1974. An Historical Sketch of the American Population Control Movement. *International Journal of Health Services* 4 (4): 651–676.
- Mayorga-Gallo, Sarah. 2015. Racism Without Racists. In *The Wiley Blackwell Encyclopedia of Race, Ethnicity, and Nationalism*, 1–2. John Wiley & Sons, Ltd.
- McCann, Carole. 2016. *Figuring the Population Bomb*. University of Washington Press.

- Mccann, Carole R. 2009. Malthusian Men and Demographic Transitions: A Case Study of Hegemonic Masculinity in Mid-Twentieth-Century Population Theory. *Frontiers: A Journal of Women Studies* 30 (1): 142–171.
- McKittrick, Katherine. 2013. Plantation Futures. *Small Axe* 17 (3): 1–15.
- Meadows, Donella H. 1974. *The Limits to Growth; a Report for the Club of Rome's Project on the Predicament of Mankind*. New American Library.
- Merchant, Carolyn. 1980. *The Death of Nature: Women, Ecology, and the Scientific Revolution*. San Francisco: Harper & Row.
- Merchant, Emily Klancher. 2025. Reproducing Intelligence: Eugenics and Behavior Genetics Past and Present, 1st edn. In *DNA, Race, and Reproduction*, eds. Emily Klancher Merchant and Meaghan O'Keefe, 120–150. University of California Press.
- Mhlambi, Sabelo. 2020. From Rationality to Relationality: Ubuntu as an Ethical and Human Rights Framework for Artificial Intelligence Governance, Technical Report 2020-009, Carr Center for Human Rights Policy, Harvard Kennedy School, Cambridge, MA.
- Mies, Maria. 1986. *Patriarchy and Accumulation on a World Scale: Women in the International Division of Labour*. London: Zed Books.
- Molnar, Petra. 2024. *The Walls Have Eyes: Surviving Migration in the Age of Artificial Intelligence*. New York: The New Press.
- Monbiot, George. 2013. *Feral: Rewilding the Land, the Sea, and Human Life*. London: Allen Lane.
- Moore, Jason W. 2015. *Capitalism in the Web of Life: Ecology and the Accumulation of Capital*. London: Verso.
- Moore, Kathleen Dean. 2016. *Great Tide Rising: Towards Clarity and Moral Courage in a Time of Planetary Change*. Berkeley, CA: Counterpoint.
- Muiderman, Karlijn, Aarti Gupta, Joost Vervoort, and Frank Biermann. 2020. Four approaches to anticipatory climate governance: Different conceptions of the future and implications for the present. *WIREs Climate Change* 11 (6): 673.
- Murillo, Lina-Maria. 2025a. Abortion and Demographic Fears in the Heartland. *Middle West Review* 11 (2): 105–117.
- Murillo, Lina-Maria. 2025b. Fears of white population decline: The politics of race, gender, reproductive justice, and white supremacy in the US. *Journal of Gender Studies* 34 (5): 716–731.
- Murphy, David Thomas. 1997. *The Heroic Earth: Geopolitical Thought in Weimar Germany, 1918–1933*. Kent, OH: Kent State University Press.
- Myers, Dowell, and Morris Levy. 2018. Racial Population Projections and Reactions to Alternative News Accounts of Growing Diversity. *The ANNALS of the American Academy of Political and Social Science* 677 (1): 215–228.
- Nachman, Paul. 2005. Albert A. Bartlett, PhD – An Interview. *The Social Contract* 16 (1).
- Næss, Petter. 2004. Live and let die: The tragedy of Hardin's social Darwinism. *Journal of Environmental Policy & Planning* 6 (1): 19–34.
- National Academies of Sciences, Engineering, and Medicine. 2017. *The Economic and Fiscal Consequences of Immigration*. Washington, DC: National Academies Press. doi:10.17226/23550.

- National Research Council. 1997. *The New Americans: Economic, Demographic, and Fiscal Effects of Immigration*. Washington, DC: National Academies Press. doi:10.17226/5779.
- Ngai, Mae M. 2004. *Impossible Subjects: Illegal Aliens and the Making of Modern America*. Princeton, NJ: Princeton University Press.
- Noble, Safiya Umoja. 2018. *Algorithms of Oppression: How Search Engines Reinforce Racism*. New York: NYU Press.
- Nordhaus, William D. 1992. An Optimal Transition Path for Controlling Greenhouse Gases. *Science* 258 (5086): 1315–1319. doi:10.1126/science.258.5086.1315.
- Normandin, Sebastian, and Sean A. Valles. 2015. How a network of conservationists and population control activists created the contemporary US anti-immigration movement. *Endeavour* 39 (2): 95–105.
- Notestein, Frank W. 1945. Population — The Long View. In *Food for the World*, ed. Theodore W. Schultz, 36–57. Chicago: University of Chicago Press.
- Notestein, Frank W. 1970. Zero Population Growth. *Population Index* 36 (4): 444–452.
- Nye, David E. 2021. *Conflicted American Landscapes*. Cambridge, MA: MIT Press.
- Olmsted, Frederick Law. 1910. The Improvement of Boulder, Colorado: Report to the City Improvement Association, Technical report, Olmsted Brothers, Brookline, MA. AUTHOR REVIEW: archival report not indexed in OpenAlex; verify exact title, author attribution (firm vs. individual), and pagination against the Boulder Carnegie Library or Olmsted Archives copy.
- Ord, Toby. 2020. *The Precipice: Existential Risk and the Future of Humanity*. New York: Hachette Books.
- Oreskes, Naomi, and Erik M. Conway. 2011. *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Global Warming*. Bloomsbury Publishing USA.
- Ortman, Jennifer M., and Howard Hogan. 2009. Methodology and Assumptions for the 2009 National Population Projections of the United States. Washington, DC: U.S. Census Bureau.
- Osborn, Frederick. 1940. *Preface to Eugenics*. New York: Harper & Brothers.
- Paddock, William, and Paul Paddock. 1967. *Famine—1975! America's Decision: Who Will Survive?* Boston: Little, Brown.
- Park, Robert M. 1974. Not Better Lives, Just Fewer People: The Ideology of Population Control. *International Journal of Health Services* 4 (4): 691–700.
- Pasquale, Frank. 2015. *The Black Box Society: The Secret Algorithms that Control Money and Information*. Cambridge, MA: Harvard University Press.
- Pellow, David Naguib, and Lisa Sun-Hee Park. 2017. Nativist politics and environmental privilege: Ecological and cultural conflicts concerning Latin American migration to the United States. In *Environmental History of Modern Migrations*. Routledge.
- Peri, Giovanni. 2012. The Effect of Immigration on Productivity: Evidence from U.S. States. *Review of Economics and Statistics* 94 (1): 348–358. doi:10.1162/REST_a_00137.
- Perry, Samuel L., and Joshua B. Grubbs. 2025. Make One for the Team: Culture Wars and Group-Serving Pronatalism. *PS: Political Science & Politics* 58 (1): 13–22.

Perry, Samuel L., Elizabeth E. McElroy, Landon Schnabel, and Joshua B. Grubbs. 2022. Fill the Earth and Subdue It: Christian Nationalism, Ethno-Religious Threat, and Nationalist Pronatalism. *Sociological Forum* 37 (4): 995–1017.

Piepzna-Samarasinha, Leah Lakshmi. 2018. *Care Work: Dreaming Disability Justice*. Vancouver: Arsenal Pulp Press.

PLAN-Boulder County. Timeline of PLAN-Boulder County and Boulder Growth Management. AUTHOR REVIEW: web source cited in ch01. Verify URL, retrieval date, and exact title; PLAN-Boulder County publishes a timeline of Boulder's growth-management history.

Plumwood, Val. 1993. *Feminism and the Mastery of Nature*. London: Routledge.

Porter, Theodore M. 1995. *Trust in Numbers: The Pursuit of Objectivity in Science and Public Life*. Princeton, NJ: Princeton University Press.

Power, Michael. 1997. *The Audit Society: Rituals of Verification*. Oxford: Oxford University Press.

Preston, Samuel H., Patrick Heuveline, and Michel Guillot. 2001. *Demography: Measuring and Modeling Population Processes*. Oxford: Blackwell.

Proctor, Robert, and Londa L. Schiebinger. 2008. *Agnotology: The Making and Unmaking of Ignorance*. Stanford University Press.

Ramsden, Edmund. 2002. Carving up Population Science: Eugenics, Demography and the Controversy over the 'Biological Law' of Population Growth. *Social Studies of Science* 32 (5/6): 857–899.

Ramsden, Edmund. 2003. Social Demography and Eugenics in the Interwar United States. *Population and Development Review* 29 (4): 547–593.

Ramsden, Edmund. 2009. Confronting the Stigma of Eugenics: Genetics, Demography and the Problems of Population. *Social Studies of Science* 39 (6): 853–884.

Ratzel, Friedrich. 1901. Lebensraum: A Biogeographical Study. *Journal of Historical Geography* 61: 59–80. English translation by Tul'si Bhambry published in *Journal of Historical Geography* 61 (2018): 59–80; original German essay 1901.

Renn, Ortwin. 2008. *Risk Governance: Coping with Uncertainty in a Complex World*. London: Earthscan.

Riahi, Keywan, Detlef P. van Vuuren, Elmar Kriegler, Jae Edmonds, Brian C. O'Neill, et al.. 2017. The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global Environmental Change* 42: 153–168. doi:10.1016/j.gloenvcha.2016.05.009.

Rich, Nathaniel. 2019. *Losing Earth: A Recent History*. New York: MCD/Farrar, Straus and Giroux.

Richards, Imogen, Callum Jones, and Gearóid Brinn. 2022. Eco-Fascism Online: Conceptualizing Far-Right Actors' Response to Climate Change on Stormfront. *Studies in Conflict & Terrorism* 0 (0): 1–27.

Roane, J. T., and Justin Hosbey. 2019. Mapping Black Ecologies. *Current Research in Digital History* 2.

Roberts, Dorothy E. 1997. *Killing the Black Body: Race, Reproduction, and the Meaning of Liberty*. New York: Pantheon Books.

- Robertson, Thomas. 2012. *The Malthusian Moment: Global Population Growth and the Birth of American Environmentalism*. New Brunswick, UNITED STATES: Rutgers University Press.
- Robinson, Kim Stanley. 2020. *The Ministry for the Future*. New York: Orbit.
- Rockström, Johan, Will Steffen, Kevin Noone, Åsa Persson, F. Stuart Chapin, Eric F. Lambin, Timothy M. Lenton, Marten Scheffer, Carl Folke, Hans Joachim Schellnhuber, Björn Nykvist, Cynthia A. de Wit, Terry Hughes, Sander van der Leeuw, Henning Rodhe, Sverker Sörlin, Peter K. Snyder, Robert Costanza, Uno Svedin, Malin Falkenmark, Louise Karlberg, Robert W. Corell, Victoria J. Fabry, James Hansen, Brian Walker, Diana Liverman, Katherine Richardson, Paul Crutzen, and Jonathan A. Foley. 2009. A Safe Operating Space for Humanity. *Nature* 461 (7263): 472–475. doi:10.1038/461472a.
- Rollins, William H. 1999. Imperial Shades of Green: Conservation and Environmental Chauvinism in the German Colonial Project. *German Studies Review* 22 (2): 187.
- Roosevelt, Theodore. 1889. *The Winning of the West*. New York: G. P. Putnam's Sons. 4 vols., 1889–1896.
- Ross, Loretta, and Rickie Solinger. 2017. *Reproductive Justice: An Introduction*. Oakland, CA: University of California Press.
- Rueda, Daniel. 2020. Neoecofascism: The Example of the United States. *Journal for the Study of Radicalism* 14 (2): 95–126.
- Ryder, Norman B. 1973. Two Cheers for ZPG. *Daedalus* 102 (4): 45–62.
- Saito, Kohei. 2017. *Karl Marx's Ecosocialism: Capital, Nature, and the Unfinished Critique of Political Economy*. New York: Monthly Review Press.
- Saito, Kohei. 2023. *Marx in the Anthropocene: Towards the Idea of Degrowth Communism*. Cambridge: Cambridge University Press.
- Saito, Kohei. 2024. *Slow Down: The Degrowth Manifesto*. New York: Astra House.
- Sargent, Andrew. 2023. Boulder is for People: Zoning Reform and the Fight for Affordable Housing. *University of Colorado Law Review* 94: 857. Author first name to be confirmed..
- Sasser, Jade. 2014. From Darkness into Light: Race, Population, and Environmental Advocacy. *Antipode* 46 (5): 1240–1257.
- Sayre, Nathan F. 2008. The Genesis, History, and Limits of Carrying Capacity. *Annals of the Association of American Geographers* 98 (1): 120–134. doi:10.1080/00045600701734356.
- Scholz, Trebor, and Nathan Schneider, eds. 2016. *Ours to Hack and to Own: The Rise of Platform Cooperativism*. New York: OR Books.
- Schultz, Susanne. 2019. Demographic futurity: How statistical assumption politics shape immigration policy rationales in Germany. *Environment and Planning D: Society and Space* 37 (4): 644–662.
- Schwartzstein, Peter. 2024. *The Heat and the Fury: On the Frontlines of Climate Violence*. Washington, DC: Island Press.
- Scott, James C. 1998. *Seeing like a State: How Certain Schemes to Improve the Human Condition Have Failed*. New Haven: Yale University Press.
- Scott, James C. 2025. *In Praise of Floods: The Untamed River and the Life It Brings*. New Haven, CT: Yale University Press.

- Scranton, Roy. 2015. *Learning to Die in the Anthropocene: Reflections on the End of a Civilization*. San Francisco: City Lights Books.
- Sear, Rebecca. 2021. Demography and the rise, apparent fall, and resurgence of eugenics. *Population Studies* 75: 201–220.
- Selbst, Andrew D., Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and Abstraction in Sociotechnical Systems. In *Proceedings of the conference on fairness, accountability, and transparency (fat* '19)*, 59–68. doi:10.1145/3287560.3287598.
- Sen, Amartya. 1997. Population Policy: Authoritarianism versus Cooperation. *Journal of Population Economics* 10 (1): 3–22.
- Sen, Amartya. 1999. *Development as Freedom*. New York: Knopf.
- Sharpe, Christina. 2016. *In the Wake: On Blackness and Being*. Durham, NC: Duke University Press.
- Sheldrake, Merlin. 2020. *Entangled Life: How Fungi Make Our Worlds, Change Our Minds & Shape Our Futures*. New York: Random House.
- Shiva, Vandana. 1988. *Staying Alive: Women, Ecology and Development*. London: Zed Books.
- Siders, A. R. 2019. Managed Retreat in the United States. *One Earth* 1 (2): 216–225.
- Simpson, Sid, and Kate Cheever. 2025. It Was Always Blood and Soil: Ecofascism and the Racial Capitalocene. *Antipode* 57 (3): 1126–1147.
- Sins Invalid. 2019. *Skin, Tooth, and Bone: The Basis of Movement Is Our People*, 2nd edn. Berkeley, CA: Sins Invalid.
- Smith, James P., and Barry Edmonston, eds. 1997. *The New Americans: Economic, Demographic, and Fiscal Effects of Immigration*. Washington, DC: National Academy Press. National Research Council, Panel on the Demographic and Economic Impacts of Immigration. Metadata from standard bibliographic knowledge; verify before publication..
- Smith, Woodruff D. 1980. Friedrich Ratzel and the Origins of Lebensraum. *German Studies Review* 3 (1): 51–68.
- Sohoni, Deenesh. 2006. The ‘Immigrant Problem’: Modern-Day Nativism on the Web. *Current Sociology* 54 (6): 827–850.
- Sohoni, Deenesh. 2017. Restrictionist Discourse by the Numbers: The Framing of the Demographic Impacts of Immigration. *Social Problems* 64 (4): 476–496.
- Sohoni, Deenesh. 2022. The Coming Majority-Minority State?: Media Coverage of U.S. Census Projections, Demographic Threat, and the Construction of Racial Boundaries. *The Sociological Quarterly* 63 (1): 94–113.
- Solnit, Rebecca. 2009. *A Paradise Built in Hell: The Extraordinary Communities That Arise in Disaster*. New York: Viking Penguin.
- Spade, Dean. 2020. *Mutual Aid: Building Solidarity during this Crisis (and the Next)*. London: Verso.
- Spiro, Jonathan. 2009. *Defending the Master Race: Conservation, Eugenics, and the Legacy of Madison Grant*. UPNE.
- Stanley, Jason. 2020. *How Fascism Works: The Politics of Us and Them*. Random House Publishing Group.

- Stern, Alexandra Minna. 2005. *Eugenic Nation: Faults and Frontiers of Better Breeding in Modern America*. Berkeley, CA: University of California Press.
- Stern, Nicholas. 2007. *The Economics of Climate Change: The Stern Review*. Cambridge, UK: Cambridge University Press.
- Strathern, Marilyn, ed. 2000. *Audit Cultures: Anthropological Studies in Accountability, Ethics and the Academy*. London: Routledge.
- Taylor, Charles. 2002. Modern Social Imaginaries. *Public Culture* 14 (1): 91–124.
- Taylor, Linnet. 2017a. What is data justice? The case for connecting digital rights and freedoms globally. *Big Data & Society* 4 (2): 205395171773633.
- Taylor, Sunaura. 2017b. *Beasts of Burden: Animal and Disability Liberation*. New York: The New Press.
- Thomas, Cassidy, and Elhom Gosink. 2021. At the Intersection of Eco-Crises, Eco-Anxiety, and Political Turbulence: A Primer on Twenty-First Century Ecofascism. *Perspectives on Global Development and Technology* 20 (1-2): 30–54.
- Tilley, Lisa, and Max Ajl. 2023. Eco-socialism will be anti-eugenic or it will be nothing: Towards equal exchange and the end of population. *Politics* 43 (2): 201–218.
- Tronto, Joan C. 2013. *Caring Democracy: Markets, Equality, and Justice*. New York: New York University Press.
- Tucker, William H. 1994. *The Science and Politics of Racial Research*. University of Illinois Press.
- Turchin, Peter. 2025. *The Great Holocene Transformation: What Complexity Science Tells Us about the Evolution of Complex Societies*. New York: Penguin Press. Publisher and year to confirm.
- Turner, Frederick Jackson. 1893. The Significance of the Frontier in American History. In *Annual Report of the American Historical Association for the Year 1893*, 197–227. Washington, DC: Government Printing Office.
- Turner, Joe, and Dan Bailey. 2022. ‘Ecobordering’: Casting immigration control as environmental protection. *Environmental Politics* 31 (1): 110–131.
- United Nations. 1994. Programme of Action of the International Conference on Population and Development, Cairo, 5–13 September 1994. United Nations document A/CONF.171/13. AUTHOR REVIEW: official UN ICPD Programme of Action; verify document number and URL.
- United Nations Department of Economic and Social Affairs, Population Division. 1969. *World Population Prospects as Assessed in 1968*. New York: United Nations.
- United Nations, Department of Economic and Social Affairs, Population Division. 2024. *World Population Prospects 2024*, Technical report, United Nations, New York. Data source for global and regional fertility and growth-rate trends..
- Venuti, Lawrence. 1995. *The Translator’s Invisibility: A History of Translation*. London: Routledge.
- Vesoulis, Abby. 2023. The GOP is flirting with this Hungarian autocrat’s generous—and exclusionary—family benefits. *Mother Jones*. Chap. Politics.
- Vince, Gaia. 2022. *Nomad Century: How Climate Migration Will Reshape Our World*. New York: Flatiron Books.

- Vogt, William. 1948. *Road to Survival*. New York: William Sloane Associates.
- Wallace-Wells, David. 2019. *The Uninhabitable Earth: Life After Warming*. New York: Tim Dugan Books.
- Wang, Dan. 2025. *Breakneck: China's Quest to Engineer the Future*. New York: W. W. Norton.
- Watts, Vanessa. 2013. Indigenous place-thought and agency amongst humans and non-humans (First Woman and Sky Woman go on a european world tour!). *Decolonization: Indigeneity, Education & Society* 2 (1): 20–34.
- Weiner, Myron, and Michael S. Teitelbaum. 2001. *Political Demography, Demographic Engineering*. New York: Berghahn Books.
- Weiss, Edith Brown. 1989. In *Fairness to Future Generations: International Law, Common Patrimony, and Intergenerational Equity*. Dobbs Ferry, NY: Transnational Publishers.
- Westermann, Edward B. 2016. *Hitler's Ostkrieg and the Indian Wars: Comparing Genocide and Conquest*. Norman, OK: University of Oklahoma Press.
- Whyte, Jennifer, Alice Comi, and Luigi Mosca. 2022. Making futures that matter: Future making, online working and organizing remotely. *Organization Theory* 3 (1): 26317877211069138.
- Williams, Casey A. 2025. Settler Ecofascism, Fossil Capitalism, and Democratic Crisis. *Environmental Communication* 0 (0): 1–15.
- Wolfe, Patrick. 2006. Settler Colonialism and the Elimination of the Native. *Journal of Genocide Research* 8 (4): 387–409. doi:10.1080/14623520601056240.
- Womack, Ytasha L. 2013. *Afrofuturism: The World of Black Sci-Fi and Fantasy Culture*. Chicago: Lawrence Hill Books.
- Wong, Alice, ed. 2020. *Disability Visibility: First-Person Stories from the Twenty-First Century*. New York: Vintage Books.
- Woods, Clyde. 1998. *Development Arrested: The Blues and Plantation Power in the Mississippi Delta*. London: Verso.
- Younis, Musab. 2026. Lineages of a conspiracy: The 'great replacement' and demography. *Race & Class*.
- Zuckerman, Ben. 1999. The Sierra Club Immigration Debate: National Implications. *Population and Environment* 20 (5): 401–412.
- Zwietering, M. H., I. Jongenburger, F. M. Rombouts, and K. van 't Riet. 1990. Modeling of the Bacterial Growth Curve. *Applied and Environmental Microbiology* 56 (6): 1875–1881. doi:10.1128/aem.56.6.1875-1881.1990.
- John Tanton Papers, 1960–2007 - University of Michigan Bentley Historical Library - University of Michigan Finding Aids. <https://findingaids.lib.umich.edu/catalog/umich-bhl-861056#using-these-materials>.
2002. John Tanton is the Mastermind Behind the Organized Anti-Immigration Movement. *Southern Poverty Law Center*.
2020. Indigenous Protocol and Artificial Intelligence Position Paper, Technical report, Indigenous AI Working Group / Concordia University, Honolulu.
2021. Marshall Fire, Boulder County, Colorado, December 30, 2021. AUTHOR REVIEW: news/governmental source describing the December 30, 2021 wildfire that destroyed over 1,000 homes in Louisville and Superior, CO. Author should specify primary news source or Boulder County after-action report.

